

Transform Dictionary

Every constant of Nature defines a transform about two sets of Planck boundaries. Each closed-form expression reproduces the correct magnitude and units of the associated constant. Collectively, these expressions use all eight types of transforms in the hyperbolic partition equation—defining the structure of the Dictionary’s columns.

Each entry includes a name, symbol,¹ and a bi-part equation with four signatures each. The first two signatures define the external geometric action and the external boundaries that action is taking place on. The last two define the internal geometric action and the internal inversion boundaries—a fixed feature, constructively shared by all constants.

Every equation in this Dictionary is followed by a paragraph labelling each algebraic participant in that equation. This is followed by the number and dimension predicted by this equation, which is then compared with the CODATA 2022 & 2018 listed values—in terms of their own error bars, where $\sigma = 1.00$ reflects an error equal to the size of the error bars.

These are followed by a measure of the change in precision—in digits—the precision change in digits that going from the CODATA 2018 value to the CODATA 2022 value represents. $\Delta_{\text{precision}} = 1.00$ represents an increase of 1 digit in precision. And finally, by a measure of how much the CODATA 2022 number represents a change to the old number, compared to its error bars. $\Delta_{\text{scaled}} = 1.00$ means the new number swung by the same size as the error bars of the old number.

This is followed by any relevant notes, such as “CODATA also lists this constant as...”.

¹ All names with the word *relationship* in them have been given symbols meant to convey the constant’s name. For example, the inverse meter-kelvin relationship is given the symbol $\frac{1}{m} : K$. Where $:$ represents the word *relationship*. This is my symbol convention—not CODATA’s.

Legend

$t_p = 5.39125836832313 \dots \times 10^{-44} \text{ s}$	Planck time
$l_p = 1.61625918175645 \dots \times 10^{-35} \text{ m}$	Planck length
$q_p = 1.87554596713962 \dots \times 10^{-18} \text{ C}$	Planck charge
$T_p = 1.41678698590795 \dots \times 10^{32} \text{ K}$	Planck temperature
$m_p = 2.17642683817579 \dots \times 10^{-8} \text{ kg}$	Planck mass
$\boxtimes = 0.999999199973626 \dots \times 10^{-7}$ dimensionless	inversion boundary
5	number of dimensions
7	break in scale symmetry
$\mathfrak{K}_1 = 0.0854245431533304 \dots$	1 st hyperbolic partition constant
$\mathfrak{K}_2 = 3.66756753485501 \dots$	2 nd hyperbolic partition constant
$\mathfrak{K}_3 = -1.87649603900417 \dots + 4.06615262615972 \dots i$	3 rd hpc
$\mathfrak{K}_4 = -1.87649603900417 \dots - 4.06615262615972 \dots i$	4 th hpc
$\mathfrak{K}_r = 4.47826244916751 \dots$	hyperbolic radius constant
$\mathfrak{K}_\theta = 2.00316562310924 \dots$	hyperbolic radian constant
$G_{Gi} = 1.01494160640965 \dots$	Gieseking's constant
$V_{fe} = 2.02988321281930 \dots$	figure-eight knot hyperbolic volume
$K = 0.9159655941772190 \dots$	Catalan's constant
$C_d = 0.291560904030818 \dots$	domino tiling constant
$D_d = 1.79162281206959 \dots$	dimer constant
$\omega_1 = 0.764977018528596 \dots + 1.32497062714087 \dots i$	omega_1 constant
$\omega_2 = 1.529954037057192 \dots$	omega_2 constant
$i^{i^{\dots}} = 0.438282936727032 \dots + 0.360592471871385 \dots i$	power tower of i
$i = (-1)^{1/2}$	imaginary unit
$\varphi_i = (-1)^{1/3}$	imaginary golden ratio
$\varphi = 1.61803398874989$	golden ratio
$\pi = 3.14159265358979 \dots$	Archimedes' constant
$P_{up} = 2.29558714939263 \dots$	universal parabolic constant
$s = 5.24411510858423 \dots$	arc length of the unit lemniscate
$L = 2.622057554292119 \dots$	lemniscate constant
$L_1 = 1.31102877714605 \dots$	1 st lemniscate constant
$L_2 = 0.599070117367796 \dots$	2 nd lemniscate constant
$C_U = 0.847213084793979 \dots$	ubiquitous constant
$G_{Ga} = 0.834626841674073 \dots$	Gauss's constant
$W_{We} = 0.47494937998792 \dots$	Weierstrass constant
$C_{R1} = 0.998136044598509 \dots$	Ramanujan's 1 st continued fraction constant
$\mathcal{L}_{Li} = 0.110001000000000 \dots$	Liouville's constant
$M = -1.74756459463318 \dots$	Madelung constant
$\mathcal{P} = 0.41468250985111 \dots$	prime constant

$e = 2.71828182845904 \dots$	Euler's number
$x_\infty = 2.29316628741186 \dots$	1 st Foias constant
$\gamma = 0.577215664901532 \dots$	Euler-Mascheroni constant
$S = 0.822825249678847 \dots$	Sierpiński's constant
$D_{Do} = 0.739085133215160 \dots$	Dottie number
$L_{LL} = 0.622743419349181 \dots$	Laplace limit
$C_{CFP} = 1.19967864025773 \dots$	Real fixed point of the hyperbolic cotangent
$j_{0,1} = 2.404825557695772 \dots$	1 st root of the Bessel function
$\alpha_F = 2.50290787509589 \dots$	alpha Feigenbaum constant
$\delta_F = 4.66920160910299 \dots$	delta Feigenbaum constant
$\mathcal{S} = 3.24697960371714 \dots$	silver constant
$P = 1.32471795724474 \dots$	plastic constant
$C_Q = 0.605443657196732 \dots$	QRS constant
$K_{-6} = 1.15655237442151 \dots$	Khinchin mean of order -6
$T_0 = 273.150000000000 \dots$ kelvin	freezing point of water
$p_0 = 100,000.000000000 \dots$ pascal	1 bar
$p_1 = 101,325.003754773 \dots$ pascal	standard atmospheric pressure
$E_1 = 6.09110229711386 \dots \times 10^{-24}$ joules	energy associated with $\Delta_{\nu_{CS}}$
$f_1 = 5.40000000000000 \dots \times 10^{14}$ Hz	frequency of max light conversion
$\lambda_{\text{peak}} = 4.96511423174427 \dots$	peak λ of spectral radiance/unit wavelength
$\nu_{\text{peak}} = 2.82143937212207 \dots$	peak emission of spectral flux/unit frequency
$n!$	factorial function
$!n$	derangement function
$\log(x)$	natural logarithm
$\Gamma(x)$	gamma function
$\zeta(x)$	Riemann zeta function

$$\sigma = \frac{X_{\text{pred}} - X_{\text{CODATA}}}{u_{\text{CODATA}}} \quad \Delta_{\text{precision}} = \log_{10} \frac{u_{2018}}{u_{2022}} \quad \Delta_{\text{scaled}} = \frac{X_{2022} - X_{2018}}{u_{2018}}$$

$\sigma = +1.00$ means the prediction is greater than the CODATA value by exactly one of CODATA's error bars; $\sigma = -1.00$ means it is 1.00 error bar less than the CODATA value, etc. For SI-defined entries we write the match level instead (e.g., "10-digit match").

$\Delta_{\text{precision}} = +1.00$ corresponds to an increase of precision by one significant digit from 2018 to 2022.

$\Delta_{\text{scaled}} = +1.00$ is the signed shift in value from 2018 to 2022, measured in units of the 2018 1σ error bars.

u denotes the standard 1σ uncertainty from CODATA.

units

s = second
m = meter
C = coulomb
K = kelvin
kg = kilogram

derived units

MHz = mega hertz = 1000000 1/s
fm = femto meter = 0.000000000000001 m
eV = electron volt = 1 eV
MeV = megaelectron volt = 1000000 eV
GeV = gigaelectron volt = 1000000000 eV

composite SI units

A = amp
F = farad
H = henry
Hz = hertz
J = joule
lm = lumens
mol = mole
N = newton
 $\mathcal{N}$ = noether
 Ω = ohm
Pa = pascal
S = siemens
T = tesla
V = volt
W = watt
Wb = weber

the SI unit of electric current
the SI unit of electrical capacitance
the SI unit of electrical inductance
the SI unit of frequency
the SI unit of energy
the SI unit of luminous flux
the SI unit of amount of substance
the SI unit of force
the SI unit of momentum
the SI unit of electrical resistance
the SI unit of pressure
the SI unit of electric conductance
the SI unit of magnetic flux density
the SI unit of voltage, or electric potential
the SI unit of power (or radiant flux)
the SI unit of magnetic flux

A = ampere = C/s
F = farad = $s^2 C^2/m^2 kg$
H = henry = $m^2 kg/C^2$
Hz = hertz = 1/s
J = joule = $m^2 kg/s^2$
lm = lumen = $cd \cdot sr$
N = newton = $m kg/s^2$
 $\mathcal{N}$ = noether = $m kg/s$
 Ω = ohm = $m^2 kg/s C^2$
Pa = pascal = $kg/s^2 m$
S = siemens = $s C^2/m^2 kg$
T = tesla = $kg/s C$
V = volt = $m^2 kg/s^2 C$
W = watt = $m^2 kg/s^3$
Wb = weber = $m^2 kg/s C$

hyperfine transition frequency of Cs-133

-5.14637(54)

$$\Delta_{\nu_{\text{Cs}}} = \frac{E_1}{2\pi} \left(\frac{t_p}{l_p^2 m_p} \right) \left(1 - \sqrt{\zeta(2)} \, \varkappa_\theta^2 \, \boxtimes \right)$$

Where E_1 = the energy difference associated with the hyperfine transition frequency, π = Archimedes' constant, t_p = the Planck time, l_p = the Planck length, m_p = the Planck mass, $\zeta(x)$ = the Riemann zeta function, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\Delta_{\nu_{\text{Cs}}} = 9.19263177042970 \dots \times 10^9 \text{ Hz} \quad \text{prediction}$$

$$\Delta_{\nu_{\text{Cs}}} = 9.192631770 \times 10^9 \text{ Hz} \quad \text{CODATA 2022, 10-digit match}$$

$$\Delta_{\nu_{\text{Cs}}} = 9.192631770 \times 10^9 \text{ Hz} \quad \text{CODATA 2018, 10-digit match}$$

$$\Delta_{\text{precision}} = 0.00$$

$$\Delta_{\text{scaled}} = 0.00$$

$$E_1 = 6.09110229711386 \dots \times 10^{-24} \text{ joules}$$

electron volt-hertz relationship

-5.1465(21)

$$\text{eV} : \text{Hz} = \frac{\text{eV}}{2\pi} \left(\frac{t_p}{l_p^2 m_p} \right) \left(1 - \sqrt{\zeta(2)} \, \varkappa_\theta^2 \, \boxtimes \right)$$

Where eV = the electron-volt, π = Archimedes' constant, t_p = the Planck time, l_p = the Planck length, m_p = the Planck mass, $\zeta(x)$ = the Riemann zeta function, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\text{eV} : \text{Hz} = 2.41798924253021 \dots \times 10^{14} \text{ Hz} \quad \text{prediction}$$

$$\text{eV} : \text{Hz} = 2.417989242 \times 10^{14} \text{ Hz} \quad \text{CODATA 2022, 10-digit match}$$

$$\text{eV} : \text{Hz} = 2.417989242 \times 10^{14} \text{ Hz} \quad \text{CODATA 2018, 10-digit match}$$

$$\Delta_{\text{precision}} = 0.00$$

$$\Delta_{\text{scaled}} = 0.00$$

joule-hertz relationship

-5.1545(76)

$$\text{J} : \text{Hz} = \frac{\text{joule}}{2\pi} \left(\frac{t_p}{l_p^2 m_p} \right) \left(1 - \sqrt{\zeta(2)} \, \varkappa_\theta^2 \, \boxtimes \right)$$

Where π = Archimedes' constant, t_p = the Planck time, l_p = the Planck length, m_p = the Planck mass, $\zeta(x)$ = the Riemann zeta function, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\text{J} : \text{Hz} = 1.509190179\textcolor{green}{71270} \dots \times 10^{33} \text{ Hz}$	prediction
$\text{J} : \text{Hz} = 1.509190179 \times 10^{33} \text{ Hz}$	CODATA 2022, 10-digit match
$\text{J} : \text{Hz} = 1.509190179 \times 10^{33} \text{ Hz}$	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = 0.00$
	$\Delta_{\text{scaled}} = 0.00$

hertz-joule relationship

5.1545(76)

$$\text{Hz} : \text{J} = 2\pi \left(\frac{l_p^2 m_p}{s t_p} \right) \left(1 + \sqrt{\zeta(2)} \, \varkappa_\theta^2 \, \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, m_p = the Planck mass, s = the second, t_p = the Planck time, $\zeta(x)$ = the Riemann zeta function, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\text{Hz} : \text{J} = 6.6260701\textcolor{green}{4968851} \dots \times 10^{-34} \text{ J}$	prediction
$\text{Hz} : \text{J} = 6.62607015 \times 10^{-34} \text{ J}$	CODATA 2022, 10-digit match
$\text{Hz} : \text{J} = 6.62607015 \times 10^{-34} \text{ J}$	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = 0.00$
	$\Delta_{\text{scaled}} = 0.00$

hertz-electron volt relationship

5.1465(21)

$$\text{Hz} \div \text{eV} = 2\pi \frac{\text{joule}}{\text{eV}} \left(\frac{l_p^2 m_p}{s t_p} \right) \left(1 + \sqrt{\zeta(2)} \, \varkappa_\theta^2 \, \boxtimes \right)$$

Where π = Archimedes' constant, eV = the electron-volt, l_p = the Planck length, m_p = the Planck mass, s = the second, t_p = the Planck time, $\zeta(x)$ = the Riemann zeta function, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

Hz : eV = 4.13566769616115 ... × 10⁻¹⁵ eV

Hz : eV = 4.135667696 × 10⁻¹⁵ eV

Hz : eV = 4.135667696 × 10⁻¹⁵ eV

prediction

CODATA 2022, 10-digit match

CODATA 2018, 10-digit match

Δ_{precision} = 0.00

Δ_{scaled} = 0.00

natural unit of action in eV s

5.1507(48)

$$\hbar = \frac{\text{joule}}{\text{eV}} \left(\frac{l_p^2 m_p}{t_p} \right) \left(1 + \sqrt{\zeta(2)} \, \varkappa_\theta^2 \, \boxtimes \right)$$

Where eV = the electron-volt, l_p = the Planck length, m_p = the Planck mass, t_p = the Planck time, $\zeta(x)$ = the Riemann zeta function, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

ħ = 6.58211956829518 ... × 10⁻¹⁶ eV · s

ħ = 6.582119569 × 10⁻¹⁶ eV · s

ħ = 6.582119569 × 10⁻¹⁶ eV · s

prediction

CODATA 2022, 9-digit match

CODATA 2018, 9-digit match

Δ_{precision} = 0.00

Δ_{scaled} = 0.00

reduced Planck constant

5.1407(48)

$$\hbar = \left(\frac{l_p^2 m_p}{t_p} \right) \left(1 + \sqrt{\zeta(2)} \, \mathfrak{K}_\theta^2 \, \boxtimes \right)$$

Where l_p = the Planck length, m_p = the Planck mass, t_p = the Planck time, $\zeta(x)$ = the Riemann zeta function, $\mathfrak{K}_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\hbar = 1.05457181759658 \dots \times 10^{-34} \text{ J} \cdot \text{s}$	prediction
$\hbar = 1.054571817 \times 10^{-34} \text{ J} \cdot \text{s}$	CODATA 2022, 10-digit match
$\hbar = 1.054571817 \times 10^{-34} \text{ J} \cdot \text{s}$	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = 0.00$
	$\Delta_{\text{scaled}} = 0.00$

Also listed as the *atomic unit of action*, and the *natural unit of action*.

Planck constant

5.1507(48)

$$h = 2\pi \left(\frac{l_p^2 m_p}{t_p} \right) \left(1 + \sqrt{\zeta(2)} \, \mathfrak{K}_\theta^2 \, \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, m_p = the Planck mass, t_p = the Planck time, $\zeta(x)$ = the Riemann zeta function, $\mathfrak{K}_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$h = 6.62607014968851 \dots \times 10^{-34} \text{ J} \cdot \text{s}$	prediction
$h = 6.62607015 \times 10^{-34} \text{ J} \cdot \text{s}$	CODATA 2022, 10-digit match
$h = 6.62607015 \times 10^{-34} \text{ J} \cdot \text{s}$	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = 0.00$
	$\Delta_{\text{scaled}} = 0.00$

quantum of circulation

5.1470(31)

$$q_c = \pi \left(\frac{l_p^2 m_p}{t_p m_e} \right) \left(1 + \sqrt{\zeta(2)} \, \mathfrak{K}_\theta^2 \, \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, m_p = the Planck mass, t_p = the Planck time, m_e = the electron mass, $\zeta(x)$ = the Riemann zeta function, $\mathfrak{K}_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$q_c = 3.63694754800779 \dots \times 10^{-4} \text{ m}^2/\text{s}$	prediction
$q_c = 3.6369475467(11) \times 10^{-4} \text{ m}^2/\text{s}$	CODATA 2022, $\sigma = +1.19$
$q_c = 3.6369475516(11) \times 10^{-4} \text{ m}^2/\text{s}$	CODATA 2018, $\sigma = -3.27$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -4.45$

quantum of circulation times 2

5.1470(31)

$$2q_c = 2\pi \left(\frac{l_p^2 m_p}{t_p m_e} \right) \left(1 + \sqrt{\zeta(2)} \, \mathfrak{K}_\theta^2 \, \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, m_p = the Planck mass, t_p = the Planck time, m_e = the electron mass, $\zeta(x)$ = the Riemann zeta function, $\mathfrak{K}_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$2q_c = 7.27389509601558 \dots \times 10^{-4} \text{ m}^2/\text{s}$	prediction
$2q_c = 7.2738950934(22) \times 10^{-4} \text{ m}^2/\text{s}$	CODATA 2022, $\sigma = +1.19$
$2q_c = 7.2738951032(22) \times 10^{-4} \text{ m}^2/\text{s}$	CODATA 2018, $\sigma = -3.27$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -4.45$

Bohr magneton in eV/T

5.1569(29)

$$\frac{\mu_B}{e} = \frac{1}{2} \left(\frac{l_p^2 C m_p}{t_p m_e} \right) \left(1 + \sqrt{\zeta(2)} \, \mathfrak{K}_\theta^2 \, \boxtimes \right)$$

Where l_p = the Planck length, C = the coulomb, m_p = the Planck mass, t_p = the Planck time, m_e = the electron mass, $\zeta(x)$ = the Riemann zeta function, $\mathfrak{K}_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_B/e = 5.788381\textcolor{brown}{180031388} \dots \times 10^{-5} \text{ eV/T}$	prediction
$\mu_B/e = 5.7883817982(18) \times 10^{-5} \text{ eV/T}$	CODATA 2022, $\sigma = +1.17$
$\mu_B/e = 5.7883818060(17) \times 10^{-5} \text{ eV/T}$	CODATA 2018, $\sigma = -3.35$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = -4.59$

nuclear magneton in eV/T

5.1471(29)

$$\frac{\mu_N}{e} = \frac{1}{2} \left(\frac{l_p^2 C m_p}{t_p m_+} \right) \left(1 + \sqrt{\zeta(2)} \, \mathfrak{K}_\theta^2 \, \boxtimes \right)$$

Where l_p = the Planck length, C = the coulomb, m_p = the Planck mass, t_p = the Planck time, m_+ = the proton mass, $\zeta(x)$ = the Riemann zeta function, $\mathfrak{K}_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_N/e = 3.15245125\textcolor{brown}{534845} \dots \times 10^{-8} \text{ eV/T}$	prediction
$\mu_N/e = 3.15245125417(98) \times 10^{-8} \text{ eV/T}$	CODATA 2022, $\sigma = +1.20$
$\mu_N/e = 3.15245125844(96) \times 10^{-8} \text{ eV/T}$	CODATA 2018, $\sigma = -3.22$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = -4.45$

inverse of conductance quantum

4.39950(25)

$$G_0^{-1} = \frac{\pi}{\varkappa_1^2} \left(\frac{l_p^2 m_p}{t_p q_p^2} \right) \left(1 + \sqrt{\zeta(3)} \varkappa_\theta^2 \boxtimes \right)$$

Where π = Archimedes' constant, $\varkappa_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, m_p = the Planck mass, t_p = the Planck time, q_p = the Planck charge, $\zeta(x)$ = the Riemann zeta function, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$G_0^{-1} = 1.29064037306099 \dots \times 10^4 \, \Omega$	prediction
$G_0^{-1} = 1.290640372 \times 10^4 \, \Omega$	CODATA 2022, 9-digit match
$G_0^{-1} = 1.290640372 \times 10^4 \, \Omega$	CODATA 2018, 9-digit match
	$\Delta_{\text{precision}} = 0.00$
	$\Delta_{\text{scaled}} = 0.00$

$\zeta(3)$ = Apéry's constant

conductance quantum

-4.39950(25)

$$G_0 = \frac{\varkappa_1^2}{\pi} \left(\frac{t_p q_p^2}{l_p^2 m_p} \right) \left(1 - \sqrt{\zeta(3)} \varkappa_\theta^2 \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, π = Archimedes' constant, t_p = the Planck time, q_p = the Planck charge, l_p = the Planck length, m_p = the Planck mass, $\zeta(x)$ = the Riemann zeta function, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$G_0 = 7.74809172928722 \dots \times 10^{-5} \, \text{S}$	prediction
$G_0 = 7.748091729 \times 10^{-5} \, \text{S}$	CODATA 2022, 10-digit match
$G_0 = 7.748091729 \times 10^{-5} \, \text{S}$	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = 0.00$
	$\Delta_{\text{scaled}} = 0.00$

electron volt

0.3756(31)

$$\text{eV} = \varkappa_1 \left(\frac{q_p}{C} \right) \left(1 + \frac{e^{-\gamma}}{6} \varkappa_\theta^2 \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, q_p = the Planck charge, C = the coulomb, e = Euler's number, γ = the Euler-Mascheroni constant, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

eV = 1.60217663422016 ... × 10⁻¹⁹ J

eV = 1.602176634 × 10⁻¹⁹ J

eV = 1.602176634 × 10⁻¹⁹ J

prediction

CODATA 2022, 10-digit match

CODATA 2018, 10-digit match

Δ_{precision} = 0.00

Δ_{scaled} = 0.00

Also listed as the *electron volt-joule relationship*.

joule-electron volt relationship

-0.37485(80)

$$\text{J} : \text{eV} = \frac{\text{joule}}{\varkappa_1} \left(\frac{C}{q_p} \right) \left(1 - \frac{e^{-\gamma}}{6} \varkappa_\theta^2 \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, C = the coulomb, q_p = the Planck charge, e = Euler's number, γ = the Euler-Mascheroni constant, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

J : eV = 6.24150907360309 ... × 10¹⁸ eV

J : eV = 6.241509074 × 10¹⁸ eV

J : eV = 6.241509074 × 10¹⁸ eV

prediction

CODATA 2022, 10-digit match

CODATA 2018, 10-digit match

Δ_{precision} = 0.00

Δ_{scaled} = 0.00

electron charge to mass quotient

0.3741(30)

$$-\frac{e}{m_e} = -\varkappa_1 \left(\frac{q_p}{m_e} \right) \left(1 + \frac{e^{-\gamma}}{6} \varkappa_\theta^2 \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, q_p = the Planck charge, m_e = the electron mass, e = Euler's number, γ = the Euler-Mascheroni constant, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$-e/m_e = -1.75882000934636 \dots \times 10^{11} \text{ C/kg}$	prediction
$-e/m_e = -1.75882000838(55) \times 10^{11} \text{ C/kg}$	CODATA 2022, $\sigma = -1.76$
$-e/m_e = -1.75882001076(53) \times 10^{11} \text{ C/kg}$	CODATA 2018, $\sigma = -2.67$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = -4.49$

proton charge to mass quotient

0.3743(31)

$$\frac{e}{m_+} = \varkappa_1 \left(\frac{q_p}{m_+} \right) \left(1 + \frac{e^{-\gamma}}{6} \varkappa_\theta^2 \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, q_p = the Planck charge, m_+ = the proton mass, e = Euler's number, γ = the Euler-Mascheroni constant, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$e/m_+ = 9.57883314831657 \dots \times 10^7 \text{ C/kg}$	prediction
$e/m_+ = 9.5788331430(29) \times 10^7 \text{ C/kg}$	CODATA 2022, $\sigma = +1.83$
$e/m_+ = 9.5788331560(29) \times 10^7 \text{ C/kg}$	CODATA 2018, $\sigma = -2.65$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -4.48$

Bohr magneton in Hz/T0.3841(30)

$$\frac{\mu_B}{h} = \frac{\varkappa_1}{4\pi} \text{tesla}^2 \left(\frac{q_p}{m_e} \right) \left(1 + \frac{e^{-\gamma}}{6} \varkappa_\theta^2 \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, π = Archimedes' constant, q_p = the Planck charge, m_e = the electron mass, e = Euler's number, γ = the Euler-Mascheroni constant, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_B/h = 1.39962449\textcolor{green}{248203} \dots \times 10^{10} \text{ Hz/T}$	prediction
$\mu_B/h = 1.39962449171(44) \times 10^{10} \text{ Hz/T}$	CODATA 2022, $\sigma = +1.75$
$\mu_B/h = 1.39962449361(42) \times 10^{10} \text{ Hz/T}$	CODATA 2018, $\sigma = -2.69$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = -4.52$

nuclear magneton in MHz/T0.3743(30)

$$\frac{\mu_N}{h} = \frac{\varkappa_1}{4\pi} \frac{1}{\text{MHz}} \left(\frac{q_p}{s m_+} \right) \left(1 + \frac{e^{-\gamma}}{6} \varkappa_\theta^2 \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, π = Archimedes' constant, MHz = a megahertz, q_p = the Planck charge, s = the second, m_+ = the proton mass, e = Euler's number, γ = the Euler-Mascheroni constant, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_N/h = 7.6225932\textcolor{green}{2303543} \dots \text{ MHz/T}$	prediction
$\mu_N/h = 7.6225932188(24) \text{ MHz/T}$	CODATA 2022, $\sigma = +4.68$
$\mu_N/h = 7.6225932291(23) \text{ MHz/T}$	CODATA 2018, $\sigma = -2.64$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = -4.48$

lattice parameter of silicon

1343.66(16)

$$a = \frac{4\pi}{32 s} \frac{1}{\varkappa_1^4} \left(\frac{l_p m_p}{m_e} \right) \left(1 + 35 \left(\frac{4\pi}{8} \right)^5 \varkappa_\theta^2 \boxtimes \right)$$

Where π = Archimedes' constant, s = the arc length of the unit lemniscate, $\varkappa_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, m_p = the Planck mass, m_e = the electron mass, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$a = 5.43102018763799 \dots \times 10^{-10} \text{ m}$	prediction
$a = 5.431020511(89) \times 10^{-10} \text{ m}$	CODATA 2022, $\sigma = -3.63$
$a = 5.431020511(89) \times 10^{-10} \text{ m}$	CODATA 2018, $\sigma = -3.63$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

lattice spacing of ideal Si (220)

1343.66(17)

$$d_{220} = \frac{1}{\sqrt{8}} \frac{4\pi}{32 s} \frac{1}{\varkappa_1^4} \left(\frac{l_p m_p}{m_e} \right) \left(1 + 35 \left(\frac{4\pi}{8} \right)^5 \varkappa_\theta^2 \boxtimes \right)$$

Where π = Archimedes' constant, s = the arc length of the unit lemniscate, $\varkappa_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, m_p = the Planck mass, m_e = the electron mass, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$d_{220} = 1.92015560171993 \dots \times 10^{-10} \text{ m}$	prediction
$d_{220} = 1.920155716(32) \times 10^{-10} \text{ m}$	CODATA 2022, $\sigma = -3.57$
$d_{220} = 1.920155716(32) \times 10^{-10} \text{ m}$	CODATA 2018, $\sigma = -3.57$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

Rydberg constant

−8.955076(11)

$$R_{\infty} = \frac{\varkappa_1^4}{4\pi} \left(\frac{m_e}{l_p m_p} \right) \left(1 - \frac{1}{8} \left(\frac{6}{V_{fe}} \right)^2 \varkappa_{\theta}^3 \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, π = Archimedes' constant, m_e = the electron mass, l_p = the Planck length, m_p = the Planck mass, V_{fe} = the figure-eight knot hyperbolic volume, $\varkappa_{\theta}$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$R_{\infty} = 1.09737315681100 \dots \times 10^7 \text{ 1/m}$		prediction
$R_{\infty} = 1.0973731568157(12) \times 10^7 \text{ 1/m}$	CODATA 2022, $\sigma = -3.92$	
$R_{\infty} = 1.0973731568160(21) \times 10^7 \text{ 1/m}$	CODATA 2018, $\sigma = -2.38$	
	$\Delta_{\text{precision}} = +0.24$	
	$\Delta_{\text{scaled}} = -0.14$	

hartree-inverse meter relationship

−8.781944(11)

$$E_h : \frac{1}{\text{m}} = \frac{\varkappa_1^4}{2\pi} \left(\frac{m_e}{l_p m_p} \right) \left(1 - \frac{1}{8} \left(\frac{6}{V_{fe}} \right)^2 \varkappa_{\theta}^3 \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, π = Archimedes' constant, m_e = the electron mass, l_p = the Planck length, m_p = the Planck mass, V_{fe} = the figure-eight knot hyperbolic volume, $\varkappa_{\theta}$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_h : 1/\text{m} = 2.19474631362199 \dots \times 10^7 \text{ 1/m}$		prediction
$E_h : 1/\text{m} = 2.1947463136314(24) \times 10^7 \text{ 1/m}$	CODATA 2022, $\sigma = -3.92$	
$E_h : 1/\text{m} = 2.1947463136320(43) \times 10^7 \text{ 1/m}$	CODATA 2018, $\sigma = -2.33$	
	$\Delta_{\text{precision}} = +0.25$	
	$\Delta_{\text{scaled}} = -0.14$	

$$\frac{1}{\text{m}} \colon E_h = \frac{2\pi}{\varkappa_1^4} \left(\frac{l_p m_p}{\text{m} m_e} \right) \left(1 + \frac{1}{8} \left(\frac{6}{V_{\text{fe}}} \right)^2 \varkappa_\theta^3 \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, π = Archimedes' constant, l_p = the Planck length, m_p = the Planck mass, m = the meter, m_e = the electron mass, V_{fe} = the figure-eight knot hyperbolic volume, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$1/\text{m} \colon E_h = 4.5563352529\textcolor{green}{2920} \dots \times 10^{-8} E_h$	prediction
$1/\text{m} \colon E_h = 4.5563352529132(50) \times 10^{-8} E_h$	CODATA 2022, $\sigma = +3.20$
$1/\text{m} \colon E_h = 4.5563352529120(88) \times 10^{-8} E_h$	CODATA 2018, $\sigma = +1.95$
	$\Delta_{\text{precision}} = +0.25$
	$\Delta_{\text{scaled}} = +1.36$

conventional value of ampere-900.888722(50)

$$A_{90} = \text{ampere} \left(1 + \frac{1}{\mathcal{S}} \left(\frac{4\pi}{35} \right) \mathfrak{K}_\theta^3 \boxtimes \right)$$

Where $\mathcal{S}$ = the silver constant, π = Archimedes’ constant, $\mathfrak{K}_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$A_{90} = 1.00000008888173 \dots \text{ A}$	prediction
$A_{90} = 1.00000008887 \text{ A}$	CODATA 2022, 11.93-digit match
$A_{90} = 1.00000008887 \text{ A}$	CODATA 2018, 11.93-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

conventional value of coulomb-900.888750(50)

$$C_{90} = \text{coulomb} \left(1 + \frac{1}{\mathcal{S}} \left(\frac{4\pi}{35} \right) \mathfrak{K}_\theta^3 \boxtimes \right)$$

Where $\mathcal{S}$ = the silver constant, π = Archimedes’ constant, $\mathfrak{K}_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$C_{90} = 1.00000008888173 \dots \text{ C}$	prediction
$C_{90} = 1.00000008887 \text{ C}$	CODATA 2022, 11.93-digit match
$C_{90} = 1.00000008887 \text{ C}$	CODATA 2018, 11.93-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

Bohr magneton in inverse meter per tesla3.9855(30)

$$\frac{\mu_B}{hc} = \frac{\varkappa_1}{4\pi} \left(\frac{t_p q_p}{l_p m_e} \right) \left(1 + \mathcal{S} \left(\frac{K}{6} \right) \varkappa_\theta^3 \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, π = Archimedes' constant, t_p = the Planck time, q_p = the Planck charge, l_p = the Planck length, m_e = the electron mass, $\mathcal{S}$ = the silver constant, K = Catalan's constant, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\mu_B/hc = 4.66864477302116 \dots \times 10^1 \text{ 1/m} \cdot \text{T}$$
$$\mu_B/hc = 4.6686447719(15) \times 10^1 \text{ 1/m} \cdot \text{T}$$
$$\mu_B/hc = 4.6686447783(14) \times 10^1 \text{ 1/m} \cdot \text{T}$$

prediction

CODATA 2022, $\sigma = +0.75$

CODATA 2018, $\sigma = -3.77$

$\Delta_{\text{precision}} = -0.07$

$\Delta_{\text{scaled}} = -4.57$

nuclear magneton in inverse meter per tesla3.9855(30)

$$\frac{\mu_N}{hc} = \frac{\varkappa_1}{4\pi} \left(\frac{t_p q_p}{l_p m_+} \right) \left(1 + \mathcal{S} \left(\frac{K}{6} \right) \varkappa_\theta^3 \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, π = Archimedes' constant, t_p = the Planck time, q_p = the Planck charge, l_p = the Planck length, m_+ = the proton mass, $\mathcal{S}$ = the silver constant, K = Catalan's constant, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\mu_N/hc = 2.54262341068940 \dots \times 10^{-2} \text{ 1/m} \cdot \text{T}$$
$$\mu_N/hc = 2.54262341009(79) \times 10^{-2} \text{ 1/m} \cdot \text{T}$$
$$\mu_N/hc = 2.54262341353(78) \times 10^{-2} \text{ 1/m} \cdot \text{T}$$

prediction

CODATA 2022, $\sigma = +0.76$

CODATA 2018, $\sigma = -3.64$

$\Delta_{\text{precision}} = -0.13$

$\Delta_{\text{scaled}} = -4.41$

Coulomb’s constant

0.7816(15)

$$\kappa = \left(\frac{l_p^3 m_p}{t_p^2 q_p^2} \right) \left(1 + \left(\frac{C_d}{6} \right) \mathfrak{K}_\theta^4 \boxtimes \right)$$

Where l_p = the Planck length, m_p = the Planck mass, t_p = the Planck time, q_p = the Planck charge, C_d = the domino tiling constant, $\mathfrak{K}_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\kappa = 8.98755179302985 \dots \times 10^9 \text{ m/F}$	prediction
$\kappa = 8.9875517923(14) \times 10^9 \text{ m/F}$	Wikipedia 2022, $\sigma = +0.52$
$\kappa = 8.9875517923(14) \times 10^9 \text{ m/F}$	CODATA 2018, $\sigma = +0.52$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

atomic unit of permittivity

-0.7815(15)

$$A_{\text{perm}} = \left(\frac{t_p^2 q_p^2}{l_p^3 m_p} \right) \left(1 - \left(\frac{C_d}{6} \right) \mathfrak{K}_\theta^4 \boxtimes \right)$$

Where t_p = the Planck time, q_p = the Planck charge, l_p = the Planck length, m_p = the Planck mass, C_d = the domino tiling constant, $\mathfrak{K}_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$A_{\text{perm}} = 1.11265005535270 \dots \times 10^{-10} \text{ F/m}$	prediction
$A_{\text{perm}} = 1.11265005620(17) \times 10^{-10} \text{ F/m}$	CODATA 2022, $\sigma = -4.98$
$A_{\text{perm}} = 1.11265005545(17) \times 10^{-10} \text{ F/m}$	CODATA 2018, $\sigma = -4.78$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +4.41$

vacuum electric permittivity

-0.7815(14)

$$\varepsilon_0 = \frac{1}{4\pi} \left(\frac{t_p^2 q_p^2}{l_p^3 m_p} \right) \left(1 - \left(\frac{C_d}{6} \right) \mathfrak{K}_\theta^4 \boxtimes \right)$$

Where π = Archimedes' constant, t_p = the Planck time, q_p = the Planck charge, l_p = the Planck length, m_p = the Planck mass, C_d = the domino tiling constant, $\mathfrak{K}_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\varepsilon_0 = 8.85418781204269 \dots \times 10^{-12} \text{ F/m}$	prediction
$\varepsilon_0 = 8.8541878188(14) \times 10^{-12} \text{ F/m}$	CODATA 2022, $\sigma = -4.83$
$\varepsilon_0 = 8.8541878128(13) \times 10^{-12} \text{ F/m}$	CODATA 2018, $\sigma = -0.58$
	$\Delta_{\text{precision}} = -0.03$
	$\Delta_{\text{scaled}} = +4.62$

Also called *vacuum permittivity*, or the *electric constant*.

atomic unit of time

12.390554(10)

$$A_{\text{time}} = \frac{1}{\varkappa_1^4} \left(\frac{t_p m_p}{m_e} \right) \left(1 + \sqrt{5} \left(\frac{C_d}{C_U} \right) \varkappa_\theta^4 \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, t_p = the Planck time, m_p = the Planck mass, m_e = the electron mass, C_d = the domino tiling constant, C_U = the ubiquitous constant, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} A_{\text{time}} &= 2.41888432657500 \dots \times 10^{-17} \text{ s} && \text{prediction} \\ A_{\text{time}} &= 2.4188843265864(26) \times 10^{-17} \text{ s} && \text{CODATA 2022, } \sigma = -4.39 \\ A_{\text{time}} &= 2.4188843265857(47) \times 10^{-17} \text{ s} && \text{CODATA 2018, } \sigma = -2.28 \\ &&& \Delta_{\text{precision}} = +0.26 \\ &&& \Delta_{\text{scaled}} = +0.15 \end{aligned}$$

hertz-hartree relationship

12.390554(10)

$$\text{Hz} : E_h = \text{watt} \frac{2\pi}{\varkappa_1^4} \left(\frac{t_p m_p}{m_e} \right) \left(1 + \sqrt{5} \left(\frac{C_d}{C_U} \right) \varkappa_\theta^4 \boxtimes \right)$$

Where π = Archimedes' constant, $\varkappa_1$ = the 1st hyperbolic partition constant, t_p = the Planck time, m_p = the Planck mass, m_e = the electron mass, C_d = the domino tiling constant, C_U = the ubiquitous constant, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} \text{Hz} : E_h &= 1.51982984605030 \dots \times 10^{-16} E_h && \text{prediction} \\ \text{Hz} : E_h &= 1.5198298460574(17) \times 10^{-16} E_h && \text{CODATA 2022, } \sigma = -4.18 \\ \text{Hz} : E_h &= 1.5198298460570(29) \times 10^{-16} E_h && \text{CODATA 2018, } \sigma = -2.31 \\ &&& \Delta_{\text{precision}} = +0.23 \\ &&& \Delta_{\text{scaled}} = +1.14 \end{aligned}$$

hartree-hertz relationship

-12.390554(10)

$$E_h \div \text{Hz} = \frac{\varkappa_1^4}{2\pi} \left(\frac{m_e}{t_p m_p} \right) \left(1 - \sqrt{\textcolor{red}{5}} \left(\frac{C_d}{C_U} \right) \varkappa_\theta^4 \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, π = Archimedes' constant, m_e = the electron mass, t_p = the Planck time, m_p = the Planck mass, C_d = the domino tiling constant, C_U = the ubiquitous constant, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_h \div \text{Hz} = 6.579683920\textcolor{green}{52072} \dots \times 10^{15} \text{ Hz}$	prediction
$E_h \div \text{Hz} = 6.5796839204999(72) \times 10^{15} \text{ Hz}$	CODATA 2022, $\sigma = +2.89$
$E_h \div \text{Hz} = 6.579683920502(13) \times 10^{15} \text{ Hz}$	CODATA 2018, $\sigma = +1.44$
	$\Delta_{\text{precision}} = +0.74$
	$\Delta_{\text{scaled}} = -0.23$

Rydberg constant times c in Hz

-12.390554(10)

$$R_\infty c = \frac{\varkappa_1^4}{4\pi} \left(\frac{m_e}{t_p m_p} \right) \left(1 - \sqrt{\textcolor{red}{5}} \left(\frac{C_d}{C_U} \right) \varkappa_\theta^4 \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, π = Archimedes' constant, m_e = the electron mass, t_p = the Planck time, m_p = the Planck mass, C_d = the domino tiling constant, C_U = the ubiquitous constant, $\varkappa_\theta$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$R_\infty c = 3.2898419602\textcolor{green}{6036} \dots \times 10^{15} \text{ Hz}$	prediction
$R_\infty c = 3.2898419602500(36) \times 10^{15} \text{ Hz}$	CODATA 2022, $\sigma = +2.88$
$R_\infty c = 3.2898419602508(64) \times 10^{15} \text{ Hz}$	CODATA 2018, $\sigma = +1.49$
	$\Delta_{\text{precision}} = +0.25$
	$\Delta_{\text{scaled}} = -0.13$

neutron-proton magnetic moment ratio

760.3(23)

$$\frac{\mu_n}{\mu_+} = -\frac{S}{C_{\text{CFP}}} \left(\frac{m_+}{m_n} \right) \left(1 + !5 \left(\frac{2\pi}{8K} \right) \mathfrak{K}_r^2 \boxtimes \right)$$

Where S = Sierpiński's constant, C_{CFP} = the fixed point of the hyperbolic cotangent, m_+ = the proton mass, m_n = the neutron mass, $!n$ = the derangement function, π = Archimedes' constant, K = Catalan's constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_n/\mu_+ = -6.84979091085629 \dots \times 10^{-1}$	prediction
$\mu_n/\mu_+ = -6.8497935(16) \times 10^{-1}$	CODATA 2018, $\sigma = +1.63$
$\mu_n/\mu_+ = -6.8497934(16) \times 10^{-1}$	CODATA 2022, $\sigma = -1.56$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.06$

proton-neutron magnetic moment ratio

-760.3(23)

$$\frac{\mu_+}{\mu_n} = -\frac{C_{\text{CFP}}}{S} \left(\frac{m_n}{m_+} \right) \left(1 + !5 \left(\frac{2\pi}{8K} \right) \mathfrak{K}_r^2 \boxtimes \right)^{-1}$$

Where C_{CFP} = the fixed point of the hyperbolic cotangent, S = Sierpiński's constant, m_n = the neutron mass, m_+ = the proton mass, $!n$ = the derangement function, π = Archimedes' constant, K = Catalan's constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_+/\mu_n = -1.45989857648807 \dots$	prediction
$\mu_+/\mu_n = -1.45989802(34)$	CODATA 2018, $\sigma = -1.64$
$\mu_+/\mu_n = -1.45989805(34)$	CODATA 2018, $\sigma = -1.55$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -0.09$

electron-neutron magnetic moment ratio

213.9(24)

$$\frac{\mu_e}{\mu_n} = \frac{C_{\text{CFP}}}{P_{up}} \left(\frac{m_n}{m_e} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathfrak{K}_r^2 \boxtimes \right)$$

Where C_{CFP} = the fixed point of the hyperbolic cotangent, P_{up} = the universal parabolic constant, m_n = the neutron mass, m_e = the electron mass, K = Catalan's constant, π = Archimedes' constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_e/\mu_n = 9.60920149475115 \dots \times 10^2$	prediction
$\mu_e/\mu_n = 9.6092048(23) \times 10^2$	CODATA 2022, $\sigma = -1.44$
$\mu_e/\mu_n = 9.6092050(23) \times 10^2$	CODATA 2018, $\sigma = -1.53$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -0.09$

neutron-electron magnetic moment ratio

-213.9(24)

$$\frac{\mu_n}{\mu_e} = \frac{P_{up}}{C_{\text{CFP}}} \left(\frac{m_e}{m_n} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathfrak{K}_r^2 \boxtimes \right)^{-1}$$

Where P_{up} = the universal parabolic constant, C_{CFP} = the fixed point of the hyperbolic cotangent, m_e = the electron mass, m_n = the neutron mass, K = Catalan's constant, π = Archimedes' constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_n/\mu_e = 1.04066919665097 \dots \times 10^{-3}$	prediction
$\mu_n/\mu_e = 1.04066884(24) \times 10^{-3}$	CODATA 2022, $\sigma = +1.49$
$\mu_n/\mu_e = 1.04066882(25) \times 10^{-3}$	CODATA 2018, $\sigma = +1.50$
	$\Delta_{\text{precision}} = +0.02$
	$\Delta_{\text{scaled}} = +0.04$

atomic unit of magnetic dipole moment

5.5204(31)

$$A_{\text{mdm}} = \varkappa_1 \left(\frac{l_p^2 q_p m_p}{t_p m_e} \right) \left(1 + \omega_2 \left(\frac{2\pi}{35} \right) \varkappa_r^2 \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, q_p = the Planck charge, m_p = the Planck mass, t_p = the Planck time, m_e = the electron mass, ω_2 = the omega_2 constant, π = Archimedes' constant, $\varkappa_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$A_{\text{mdm}} = 1.85480201153128 \dots \times 10^{-23} \text{ J/T}$	prediction
$A_{\text{mdm}} = 1.85480201315(58) \times 10^{-23} \text{ J/T}$	CODATA 2022, $\sigma = -2.79$
$A_{\text{mdm}} = 1.85480201566(56) \times 10^{-23} \text{ J/T}$	CODATA 2018, $\sigma = -7.37$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = -4.48$

Bohr magneton

5.5208(31)

$$\mu_B = \frac{\varkappa_1}{2} \left(\frac{l_p^2 q_p m_p}{t_p m_e} \right) \left(1 + \omega_2 \left(\frac{2\pi}{35} \right) \varkappa_r^2 \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, q_p = the Planck charge, m_p = the Planck mass, t_p = the Planck time, m_e = the electron mass, ω_2 = the omega_2 constant, π = Archimedes' constant, $\varkappa_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_B = 9.27401005765638 \dots \times 10^{-24} \text{ J/T}$	prediction
$\mu_B = 9.2740100657(29) \times 10^{-24} \text{ J/T}$	CODATA 2022, $\sigma = -2.77$
$\mu_B = 9.2740100783(28) \times 10^{-24} \text{ J/T}$	CODATA 2018, $\sigma = -7.37$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = -4.50$

nuclear magneton

5.5208(31)

$$\mu_N = \frac{\mathfrak{K}_1}{2} \left(\frac{l_p^2 q_p m_p}{t_p m_+} \right) \left(1 + \omega_2 \left(\frac{2\pi}{35} \right) \mathfrak{K}_r^2 \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, q_p = the Planck charge, m_p = the Planck mass, t_p = the Planck time, m_+ = the proton mass, ω_2 = the omega_2 constant, π = Archimedes' constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_N = 5.05078373489248 \dots \times 10^{-27} \text{ J/T}$	prediction
$\mu_N = 5.0507837393(16) \times 10^{-27} \text{ J/T}$	CODATA 2022, $\sigma = -2.75$
$\mu_N = 5.0507837461(15) \times 10^{-27} \text{ J/T}$	CODATA 2018, $\sigma = -7.47$
	$\Delta_{\text{precision}} = -0.03$
	$\Delta_{\text{scaled}} = -4.53$

triton magnetic moment

-10.716(20)

$$\mu_{\text{tri}} = \mathfrak{K}_1^2 \mathfrak{K}_2^2 (D_{\text{Do}} + \pi) \left(\frac{l_p^2 q_p m_p}{t_p m_{\text{tri}}} \right) \left(1 - \omega_2 \left(\frac{2\pi}{18} \right) \mathfrak{K}_r^2 \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, D_{Do} = the Dottie number, π = Archimedes' constant, l_p = the Planck length, q_p = the Planck charge, m_p = the Planck mass, t_p = the Planck time, m_{tri} = the triton mass, ω_2 = the omega_2 constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_{\text{tri}} = 1.50460951611437 \dots \times 10^{-26} \text{ J/T}$	prediction
$\mu_{\text{tri}} = 1.5046095178(30) \times 10^{-26} \text{ J/T}$	CODATA 2022, $\sigma = -0.56$
$\mu_{\text{tri}} = 1.5046095202(30) \times 10^{-26} \text{ J/T}$	CODATA 2018, $\sigma = -1.36$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -0.80$

triton to proton magnetic moment ratio

308.242(19)

$$\frac{\mu_{\text{tri}}}{\mu_+} = S \left(D_{\text{Do}} + \pi \right) \left(\frac{m_+}{m_{\text{tri}}} \right) \left(1 + \frac{4}{L_{LL}} \left(\frac{32}{4\pi} \right) \mathfrak{K}_r^2 \boxtimes \right)$$

Where S = Sierpiński's constant, D_{Do} = the Dottie number, π = Archimedes' constant, m_+ = the proton mass, m_{tri} = the triton mass, L_{LL} = the Laplace limit, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\mu_{\text{tri}}/\mu_+ = 1.06663991754137 \dots$$

prediction

$$\mu_{\text{tri}}/\mu_+ = 1.0666399189(21)$$

CODATA 2022, $\sigma = -0.65$

$$\mu_{\text{tri}}/\mu_+ = 1.0666399191(21)$$

CODATA 2018, $\sigma = -0.74$

$$\Delta_{\text{precision}} = +0.00$$

$$\Delta_{\text{scaled}} = -0.10$$

triton magnetic moment to nuclear magneton ratio

-16.237(20)

$$\frac{\mu_{\text{tri}}}{\mu_N} = \mathfrak{K}_1 \mathfrak{K}_2^2 \left(2D_{\text{Do}} + 2\pi \right) \left(\frac{m_+}{m_{\text{tri}}} \right) \left(1 - \frac{18}{8} \left(\frac{4\pi}{35} \right) \mathfrak{K}_r^2 \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, D_{Do} = the Dottie number, π = Archimedes' constant, m_+ = the proton mass, m_{tri} = the triton mass, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\mu_{\text{tri}}/\mu_N = 2.97896247563036 \dots$$

prediction

$$\mu_{\text{tri}}/\mu_N = 2.9789624650(59)$$

CODATA 2022, $\sigma = +1.80$

$$\mu_{\text{tri}}/\mu_N = 2.9789624656(59)$$

CODATA 2018, $\sigma = +1.70$

$$\Delta_{\text{precision}} = +0.00$$

$$\Delta_{\text{scaled}} = -0.10$$

triton magnetic moment to Bohr magneton ratio

-16.237(19)

$$\frac{\mu_{\text{tri}}}{\mu_B} = \mathfrak{K}_1 \mathfrak{K}_2^2 (2D_{\text{Do}} + 2\pi) \left(\frac{m_e}{m_{\text{tri}}} \right) \left(1 - \frac{18}{8} \left(\frac{4\pi}{35} \right) \mathfrak{K}_r^2 \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, D_{Do} = the Dottie number, π = Archimedes' constant, m_e = the electron mass, m_{tri} = the triton mass, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_{\text{tri}}/\mu_B = 1.62239367061578 \dots \times 10^{-3}$	prediction
$\mu_{\text{tri}}/\mu_B = 1.6223936648(32) \times 10^{-3}$	CODATA 2022, $\sigma = +1.82$
$\mu_{\text{tri}}/\mu_B = 1.6223936651(32) \times 10^{-3}$	CODATA 2018, $\sigma = +1.72$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -0.09$

triton g factor

-16.236(20)

$$g_{\text{tri}} = \mathfrak{K}_1 \mathfrak{K}_2^2 (4D_{\text{Do}} + 4\pi) \left(\frac{m_+}{m_{\text{tri}}} \right) \left(1 - \frac{18}{8} \left(\frac{4\pi}{35} \right) \mathfrak{K}_r^2 \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, D_{Do} = the Dottie number, π = Archimedes' constant, m_+ = the proton mass, m_{tri} = the triton mass, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$g_{\text{tri}} = 5.95792495126071 \dots$	prediction
$g_{\text{tri}} = 5.957924930(12)$	CODATA 2022, $\sigma = +1.77$
$g_{\text{tri}} = 5.957924931(12)$	CODATA 2018, $\sigma = -1.69$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -0.08$

electron-muon magnetic moment ratio

-62.35(22)

$$\frac{\mu_e}{\mu_\mu} = \left(\frac{m_\mu}{m_e} \right) \left(1 - \left(\frac{4\pi}{18} \right) \mathfrak{K}_r^3 \boxtimes \right)$$

Where m_μ = the muon mass, m_e = the electron mass, π = Archimedes' constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_e/\mu_\mu = 2.06766985041323 \dots \times 10^2$	prediction
$\mu_e/\mu_\mu = 2.067669881(46) \times 10^2$	CODATA 2022, $\sigma = +0.66$
$\mu_e/\mu_\mu = 2.067669883(46) \times 10^2$	CODATA 2018, $\sigma = -0.71$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -0.04$

Newtonian constant of gravitation

-60.(103)

$$G = \left(\frac{l_p^3}{t_p^2 m_p} \right) \left(1 - \left(\frac{4\pi}{18} \right) \mathfrak{K}_r^3 \boxtimes \right)$$

Where l_p = the Planck length, t_p = the Planck time, m_p = the Planck mass, π = Archimedes' constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$G = 6.67430315504533 \dots \times 10^{-11} \text{ m}^3/\text{s}^2 \cdot \text{kg}$	prediction
$G = 6.67430(15) \times 10^{-11} \text{ m}^3/\text{s}^2 \cdot \text{kg}$	CODATA 2022, $\sigma = +0.02$
$G = 6.67430(15) \times 10^{-11} \text{ m}^3/\text{s}^2 \cdot \text{kg}$	CODATA 2018, $\sigma = +0.02$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

Also called the *gravitational constant*.

$$\frac{G}{\hbar c} = \text{GeV}^2 \left(\frac{t_p^4}{l_p^4 m_p^2} \right) \left(1 - \left(\frac{4\pi}{18} \right) \mathfrak{K}_r^3 \boxtimes \right)$$

Where GeV = the gigaelectron volt, t_p = the Planck time, l_p = the Planck length, m_p = the Planck mass, π = Archimedes' constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$G/\hbar c = 6.70882525604658 \dots \times 10^{-39} \text{ c}^4/\text{GeV}^2$

prediction

$G/\hbar c = 6.70883(15) \times 10^{-39} \text{ c}^4/\text{GeV}^2$

CODATA 2022, $\sigma = -0.03$

$G/\hbar c = 6.70883(15) \times 10^{-39} \text{ c}^4/\text{GeV}^2$

CODATA 2018, $\sigma = -0.03$

$\Delta_{\text{precision}} = +0.00$

$\Delta_{\text{scaled}} = +0.00$

The Planck energy is equal to $E_p = \frac{l_p^2 m_p}{t_p^2}$.

neutron mass

−74.9526(39)

$$m_n = \left(\frac{2\pi}{2s}\right)^2 \frac{\text{GeV}}{\varphi^{-2}} \left(\frac{t_p^2}{l_p^2}\right) \left(1 - \left(\frac{2s}{4\pi}\right) \mathfrak{K}_r^3 \boxtimes \right)$$

Where π = Archimedes’ constant, s = the arc length of the unit lemniscate, GeV = the gigaelectron volt, φ = the golden ratio, t_p = the Planck time, l_p = the Planck length, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_n = 1.67492749887876 \dots \times 10^{-27} \text{ kg}$	prediction
$m_n = 1.67492750056(85) \times 10^{-27} \text{ kg}$	CODATA 2022, $\sigma = -1.98$
$m_n = 1.67492749804(95) \times 10^{-27} \text{ kg}$	CODATA 2018, $\sigma = +0.88$
	$\Delta_{\text{precision}} = +0.05$
	$\Delta_{\text{scaled}} = +2.65$

proton mass

−179.826407(84)

$$m_+ = \left(\frac{2\pi}{s}\right)^2 \frac{\text{GeV}}{\omega_2} \left(\frac{t_p^2}{l_p^2}\right) \left(1 - \frac{\sqrt{2}}{e^{\gamma/3}} \left(\frac{18}{2s}\right) \mathfrak{K}_r^3 \boxtimes \right)$$

Where π = Archimedes’ constant, s = the arc length of the unit lemniscate, GeV = the gigaelectron volt, ω_2 = the omega_2 constant, t_p = the Planck time, l_p = the Planck length, e = Euler’s number, γ = the Euler-Mascheroni constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_+ = 1.67262192525165 \dots \times 10^{-27} \text{ kg}$	prediction
$m_+ = 1.67262192595(52) \times 10^{-27} \text{ kg}$	CODATA 2022, $\sigma = -1.34$
$m_+ = 1.67262192369(51) \times 10^{-27} \text{ kg}$	CODATA 2018, $\sigma = +3.06$
	$\Delta_{\text{precision}} = -0.01$
	$\Delta_{\text{scaled}} = +4.35$

deuteron mass

130.771469(75)

$$m_{\text{de}} = \frac{18}{6 C_{R1}} \sqrt{\frac{2\pi}{14}} (A_{\text{mass}}) \left(1 + (\pi - x_{\infty}) \left(\frac{18}{2s} \right) \mathfrak{K}_r^3 \boxtimes \right)$$

Where C_{R1} = Ramanujan's 1st continued fraction constant, π = Archimedes' constant, A_{mass} = the atomic mass constant, x_{∞} = the 1st Foias constant, s = the arc length of the unit lemniscate, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_{\text{de}} = 3.343583777536356 \dots \times 10^{-27} \text{ kg}$	prediction
$m_{\text{de}} = 3.3435837768(10) \times 10^{-27} \text{ kg}$	CODATA 2022, $\sigma = -1.44$
$m_{\text{de}} = 3.3435837724(10) \times 10^{-27} \text{ kg}$	CODATA 2018, $\sigma = +2.96$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +4.40$

muon mass

−1869.61(23)

$$m_{\mu} = \frac{2\pi}{35} \sqrt{\frac{4\pi}{6s}} (A_{\text{mass}}) \left(1 - 18 K_{-6} \mathfrak{K}_r^3 \boxtimes \right)$$

Where π = Archimedes' constant, s = the arc length of the unit lemniscate, A_{mass} = the atomic mass constant, K_{-6} = the Khinchin mean of order −6, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_{\mu} = 1.88353161502363 \dots \times 10^{-28} \text{ kg}$	prediction
$m_{\mu} = 1.883531627(42) \times 10^{-28} \text{ kg}$	CODATA 2022, $\sigma = -0.29$
$m_{\mu} = 1.883531627(42) \times 10^{-28} \text{ kg}$	CODATA 2018, $\sigma = -0.29$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

deuteron-electron magnetic moment ratio

-1851.087(25)

$$\frac{\mu_{\text{de}}}{\mu_e} = -P_{up} \sqrt{\frac{2s}{6\pi} \left(\frac{m_e}{m_{\text{de}}} \right)} \left(1 - 18 \sqrt{L_1} \mathfrak{K}_r^3 \boxtimes \right)$$

Where P_{up} = the universal parabolic constant, s = the arc length of the unit lemniscate, π = Archimedes' constant, m_e = the electron mass, m_{de} = the deuteron mass, L_1 = the 1st lemniscate constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_{\text{de}}/\mu_e = -4.6643455\mathbf{8972163} \dots \times 10^{-4}$	prediction
$\mu_{\text{de}}/\mu_e = -4.664345550(12) \times 10^{-4}$	CODATA 2022, $\sigma = -3.31$
$\mu_{\text{de}}/\mu_e = -4.664345551(12) \times 10^{-4}$	CODATA 2018, $\sigma = -3.23$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -0.08$

electron-deuteron magnetic moment ratio

1851.431(26)

$$\frac{\mu_e}{\mu_{\text{de}}} = -\frac{1}{P_{up}} \sqrt{\frac{6\pi}{2s} \left(\frac{m_{\text{de}}}{m_e} \right)} \left(1 - 18 \sqrt{L_1} \mathfrak{K}_r^3 \boxtimes \right)^{-1}$$

Where P_{up} = the universal parabolic constant, π = Archimedes' constant, s = the arc length of the unit lemniscate, m_{de} = the deuteron mass, m_e = the electron mass, L_1 = the 1st lemniscate constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_e/\mu_{\text{de}} = -2.1439234\mathbf{7386009} \dots \times 10^3$	prediction
$\mu_e/\mu_{\text{de}} = -2.1439234921(56) \times 10^3$	CODATA 2022, $\sigma = +3.26$
$\mu_e/\mu_{\text{de}} = -2.1439234915(56) \times 10^3$	CODATA 2018, $\sigma = 3.15$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.11$

deuteron-proton magnetic moment ratio-876.881(26)

$$\frac{\mu_{de}}{\mu_+} = S \sqrt{\frac{2s}{6\pi} \left(\frac{m_+}{m_{de}}\right)} \left(1 - \frac{!5}{\varphi} \left(\frac{4\pi}{35}\right) \mathfrak{K}_r^3 \boxtimes \right)$$

Where S = Sierpiński’s constant, s = the arc length of the unit lemniscate, π = Archimedes’ constant, m_+ = the proton mass, m_{de} = the deuteron mass, $!n$ = the derangement function, φ = the golden ratio, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\mu_{de}/\mu_+ = 3.07012209390825 \dots \times 10^{-1}$$
$$\mu_{de}/\mu_+ = 3.0701220930(79) \times 10^{-1}$$
$$\mu_{de}/\mu_+ = 3.0701220939(79) \times 10^{-1}$$

prediction

CODATA 2022, $\sigma = +0.11$

CODATA 2018, $\sigma = +0.00$

$\Delta_{\text{precision}} = +0.00$

$\Delta_{\text{scaled}} = -0.11$

deuteron-neutron magnetic moment ratio-1637.0(25)

$$\frac{\mu_{de}}{\mu_n} = -C_{\text{CFP}} \sqrt{\frac{2s}{6\pi} \left(\frac{m_n}{m_{de}}\right)} \left(1 - 35 \operatorname{Im}(\omega_1) \left(\frac{4\pi}{32}\right) \mathfrak{K}_r^3 \boxtimes \right)$$

Where C_{CFP} = the fixed point of the hyperbolic cotangent, s = the arc length of the unit lemniscate, π = Archimedes’ constant, m_n = the neutron mass, m_{de} = the deuteron mass, ω_1 = the omega_1 constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\mu_{de}/\mu_n = -4.48206589662326 \dots \times 10^{-1}$$
$$\mu_{de}/\mu_n = -4.4820652(11) \times 10^{-1}$$
$$\mu_{de}/\mu_n = -4.4820653(11) \times 10^{-1}$$

prediction

CODATA 2022, $\sigma = -0.63$

CODATA 2018, $\sigma = -0.54$

$\Delta_{\text{precision}} = +0.00$

$\Delta_{\text{scaled}} = -0.09$

$$\mu_{\text{de}} = \kappa_1^2 \kappa_2^2 \sqrt{\frac{2s}{6\pi} \left(\frac{l_p^2 q_p m_p}{t_p m_{\text{de}}} \right)} \left(1 - \frac{8}{\omega_2} \left(\frac{32}{4\pi} \right) \kappa_r^3 \boxtimes \right)$$

Where κ_1 = the 1st hyperbolic partition constant, κ_2 = 2nd hyperbolic partition constant, s = the arc length of the unit lemniscate, π = Archimedes' constant, l_p = the Planck length, q_p = the Planck charge, m_p = the Planck mass, t_p = the Planck time, m_{de} = the deuteron mass, ω_2 = the omega_2 constant, κ_r = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_{\text{de}} = 4.33073505993240 \times 10^{-27} \text{ J/T}$	prediction
$\mu_{\text{de}} = 4.330735087(11) \times 10^{-27} \text{ J/T}$	CODATA 2022, $\sigma = -2.46$
$\mu_{\text{de}} = 4.330735094(11) \times 10^{-27} \text{ J/T}$	CODATA 2018, $\sigma = -3.10$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -0.63$

shielded helion magnetic moment to Bohr magneton ratio

347.5705(81)

$$\frac{\mu_{\text{he}}'}{\mu_B} = -\frac{1}{2} \sqrt{\frac{1}{4\pi} \frac{1}{4s} \left(\frac{m_\mu}{m_{\text{he}}} \right)} \left(1 + \frac{1}{\sqrt{7}} \left(\frac{2\pi}{3K} \right) \mathfrak{K}_r^4 \boxtimes \right)$$

Where π = Archimedes' constant, s = the arc length of the unit lemniscate, m_μ = the muon mass, m_{he} = the helion mass, K = Catalan's constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} \mu_{\text{he}}'/\mu_B &= -1.15867149\textcolor{teal}{694878} \dots \times 10^{-3} && \text{prediction} \\ \mu_{\text{he}}'/\mu_B &= -1.15867149457(94) \times 10^{-3} && \text{CODATA 2022, } \sigma = -2.53 \\ \mu_{\text{he}}'/\mu_B &= -1.158671471(14) \times 10^{-3} && \text{CODATA 2018, } \sigma = -1.85 \\ &&& \Delta_{\text{precision}} = +2.17 \\ &&& \Delta_{\text{scaled}} = +1.64 \end{aligned}$$

shielded helion magnetic moment to nuclear magneton ratio

347.5705(81)

$$\frac{\mu_{\text{he}}'}{\mu_N} = -\frac{1}{2} \sqrt{\frac{1}{4\pi} \frac{1}{4s} \left(\frac{m_+ m_\mu}{m_e m_{\text{he}}} \right)} \left(1 + \frac{1}{\sqrt{7}} \left(\frac{2\pi}{3K} \right) \mathfrak{K}_r^4 \boxtimes \right)$$

Where π = Archimedes' constant, s = the arc length of the unit lemniscate, m_+ = the proton mass, m_μ = the muon mass, m_e = the electron mass, m_{he} = the helion mass, K = Catalan's constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} \mu_{\text{he}}'/\mu_N &= -2.12749776\textcolor{teal}{673056} \dots && \text{prediction} \\ \mu_{\text{he}}'/\mu_N &= -2.1274977624(17) && \text{CODATA 2022, } \sigma = -2.55 \\ \mu_{\text{he}}'/\mu_N &= -2.127497719(25) && \text{CODATA 2018, } \sigma = -2.07 \\ &&& \Delta_{\text{precision}} = +1.17 \\ &&& \Delta_{\text{scaled}} = +1.72 \end{aligned}$$

$$g_{\text{he}} = -\sqrt{\frac{1}{4\pi} \frac{1}{4s} \left(\frac{m_+ m_\mu}{m_e m_{\text{he}}} \right) \left(1 + 18 \left(\frac{2K}{14} \right) \mathfrak{K}_r^4 \boxtimes \right)}$$

Where π = Archimedes' constant, s = the arc length of the unit lemniscate, m_+ = the proton mass, m_μ = the muon mass, m_e = the electron mass, m_{he} = the helion mass, K = Catalan's constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$g_{\text{he}} = -4.25525070422298 \dots$	prediction
$g_{\text{he}} = -4.2552506995(34)$	CODATA 2022, $\sigma = -1.39$
$g_{\text{he}} = -4.255250615(50)$	CODATA 2018, $\sigma = -1.78$
	$\Delta_{\text{precision}} = +0.17$
	$\Delta_{\text{scaled}} = +1.16$

helion magnetic moment to nuclear magneton ratio 947.2981(82)

$$\frac{\mu_{\text{he}}}{\mu_N} = -\frac{1}{2} \sqrt{\frac{1}{4\pi} \frac{1}{4s} \left(\frac{m_+ m_\mu}{m_e m_{\text{he}}} \right)} \left(1 + 18 \left(\frac{2K}{14} \right) \mathfrak{K}_r^4 \boxtimes \right)$$

Where π = Archimedes' constant, s = the arc length of the unit lemniscate, m_+ = the proton mass, m_μ = the muon mass, m_e = the electron mass, m_{he} = the helion mass, K = Catalan's constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_{\text{he}}/\mu_N = -2.12762535211149$	prediction
$\mu_{\text{he}}/\mu_N = -2.1276253498(17)$	CODATA 2018, $\sigma = -1.36$
$\mu_{\text{he}}/\mu_N = -2.127625307(25)$	CODATA 2018, $\sigma = -1.80$
	$\Delta_{\text{precision}} = +1.17$
	$\Delta_{\text{scaled}} = +1.16$

helion magnetic moment to Bohr magneton ratio 947.2981(82)

$$\frac{\mu_{\text{he}}}{\mu_B} = -\frac{1}{2} \sqrt{\frac{1}{4\pi} \frac{1}{4s} \left(\frac{m_\mu}{m_{\text{he}}} \right)} \left(1 + 18 \left(\frac{2K}{14} \right) \mathfrak{K}_r^4 \boxtimes \right)$$

Where π = Archimedes' constant, s = the arc length of the unit lemniscate, m_μ = the muon mass, m_{he} = the helion mass, K = Catalan's constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_{\text{he}}/\mu_B = -1.15874098211893 \dots \times 10^{-3}$	prediction
$\mu_{\text{he}}/\mu_B = -1.15874098083(94) \times 10^{-3}$	CODATA 2022, $\sigma = -1.37$
$\mu_{\text{he}}/\mu_B = -1.158740958(14) \times 10^{-3}$	CODATA 2018, $\sigma = -1.72$
	$\Delta_{\text{precision}} = +1.17$
	$\Delta_{\text{scaled}} = +1.57$

vacuum magnetic permeability

8.0057(15)

$$\mu_0 = 4\pi \left(\frac{l_p m_p}{q_p^2} \right) \left(1 + \frac{G_{Ga}}{28} \left(\frac{14}{4s} \right) \mathfrak{K}_r^4 \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, m_p = the Planck mass, q_p = the Planck charge, G_{Ga} = Gauss's constant, s = the arc length of the unit lemniscate, $\mathfrak{K}_r$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_0 = 1.25663706158249 \dots \times 10^{-6} \text{ N/A}^2$	prediction
$\mu_0 = 1.25663706127(20) \times 10^{-6} \text{ N/A}^2$	CODATA 2022, $\sigma = +1.56$
$\mu_0 = 1.25663706212(19) \times 10^{-6} \text{ N/A}^2$	CODATA 2018, $\sigma = -2.83$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = -4.47$

Also called the *permeability of free space*, or the *magnetic constant*.

electron mass

268.4338(18)

$$m_e = 2V_{\text{fe}} \left(\frac{m_p^4}{\text{kg}^3} \right) \left(1 + \left(\frac{14}{4s} \right) \mathfrak{K}_r^4 \boxtimes \right)$$

Where V_{fe} = the figure-eight knot hyperbolic volume, m_p = the Planck mass, kg = the kilogram, s = the arc length of the unit lemniscate, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_e = 9.10938371013637 \dots \times 10^{-31} \text{ kg}$	prediction
$m_e = 9.1093837139(28) \times 10^{-31} \text{ kg}$	CODATA 2022, $\sigma = -1.34$
$m_e = 9.1093837015(28) \times 10^{-31} \text{ kg}$	CODATA 2018, $\sigma = +3.08$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +4.42$

Also listed as the *atomic unit of mass*, and the *natural unit of mass*.

$$m_{\tau} = \sqrt{\frac{35 L_{LL}}{\mathfrak{x}_1^6}} \sqrt{\frac{4s}{2\pi}} \left(\frac{m_p^4}{\text{kg}^3} \right) \left(1 + \left(\frac{14}{4s} \right) \mathfrak{x}_r^4 \boxtimes \right)$$

Where L_{LL} = the Laplace limit, $\mathfrak{x}_1$ = the 1st hyperbolic partition constant, s = the arc length of the unit lemniscate, π = Archimedes' constant, m_p = the Planck mass, kg = the kilogram, $\mathfrak{x}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_{\tau} = 3.16754546668955 \dots \times 10^{-27} \text{ kg}$
 $m_{\tau} = 3.16754(21) \times 10^{-27} \text{ kg}$
 $m_{\tau} = 3.16754(21) \times 10^{-27} \text{ kg}$

prediction

CODATA 2022, $\sigma = +0.03$

CODATA 2018, $\sigma = +0.03$

$\Delta_{\text{precision}} = +0.00$

$\Delta_{\text{scaled}} = +0.00$

$$m_{\text{tri}} = \frac{\mathfrak{x}_1^{-5}}{18} \sqrt{\frac{4s}{2\pi}} \left(\frac{m_p^4}{\text{kg}^3} \right) \left(1 + \sqrt{\frac{G_{Ga}}{14}} \left(\frac{18}{2K} \right) \mathfrak{x}_r^4 \boxtimes \right)$$

Where $\mathfrak{x}_1$ = the 1st hyperbolic partition constant, s = the arc length of the unit lemniscate, π = Archimedes' constant, m_p = the Planck mass, kg = the kilogram, G_{Ga} = Gauss's constant, K = Catalan's constant, $\mathfrak{x}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_{\text{tri}} = 5.00735674868075 \dots \times 10^{-27} \text{ kg}$
 $m_{\text{tri}} = 5.0073567512(16) \times 10^{-27} \text{ kg}$
 $m_{\text{tri}} = 5.0073567446(15) \times 10^{-27} \text{ kg}$

prediction

CODATA 2022, $\sigma = -1.57$

CODATA 2018, $\sigma = +2.72$

$\Delta_{\text{precision}} = -0.03$

$\Delta_{\text{scaled}} = +4.40$

$$K_{cd} = \frac{\varkappa_1^{-7}}{f_1} \sqrt{\frac{4s}{2\pi}} \left(\frac{t_p^2}{l_p^2 A_{\text{mass}}} \right) \left(1 + \sqrt{\frac{G_{Ga}}{14}} \left(\frac{18}{2K} \right) \varkappa_r^4 \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, f_1 = the frequency corresponding to the peak sensitivity wavelength where energy is maximally converted to light, s = the arc length of the unit lemniscate, π = Archimedes' constant, t_p = the Planck time, l_p = the Planck length, A_{mass} = the atomic mass constant, G_{Ga} = Gauss's constant, K = Catalan's constant, $\varkappa_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$K_{cd} = 6.83002503039578 \dots \times 10^2 \text{ lm/W}$	prediction
$K_{cd} = 6.83 \times 10^2 \text{ lm/W}$	CODATA 2022, 3-digit match
$K_{cd} = 6.83 \times 10^2 \text{ lm/W}$	CODATA 2018, 3-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

$f_1 = 540 \times 10^{12}$ hertz, corresponding to the peak sensitivity wavelength of 555 nm.

helion mass

-23.50214(24)

$$m_{\text{he}} = \frac{36}{e^{\pi}} \sqrt{\frac{9s}{4\pi}} \left(A_{\text{mass}} \right) \left(1 - \sqrt{\frac{1}{14}} \left(\frac{x_{\infty}}{2s} \right) \mathfrak{K}_r^4 \boxtimes \right)$$

Where e = Euler’s number, π = Archimedes’ constant, s = the arc length of the unit lemniscate, A_{mass} = the atomic mass constant, x_{∞} = the 1st Foias constant, $\mathfrak{K}_r$ = the hyperbolic radius constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_{\text{he}} = 5.00641278404447 \dots \times 10^{-27} \text{ kg}$	prediction
$m_{\text{he}} = 5.0064127862(16) \times 10^{-27} \text{ kg}$	CODATA 2022, $\sigma = -1.35$
$m_{\text{he}} = 5.0064127796(15) \times 10^{-27} \text{ kg}$	CODATA 2018, $\sigma = +2.92$
	$\Delta_{\text{precision}} = -0.03$
	$\Delta_{\text{scaled}} = +4.40$

alpha particle mass

-469.43891(15)

$$m_{\alpha} = \frac{12}{\omega_2^4} \sqrt{\frac{4s}{2\pi}} \left(A_{\text{mass}} \right) \left(1 - \sqrt{\frac{32}{14}} \left(\frac{2V_{\text{fe}}}{x_{\infty}^2} \right) \mathfrak{K}_r^4 \boxtimes \right)$$

Where ω_2 = the omega_2 constant, s = the arc length of the unit lemniscate, π = Archimedes’ constant, A_{mass} = the atomic mass constant, V_{fe} = the figure eight knot hyperbolic volume, x_{∞} = the 1st Foias constant, $\mathfrak{K}_r$ = the hyperbolic radian constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_{\alpha} = 6.64465734214075 \dots \times 10^{-27} \text{ kg}$	prediction
$m_{\alpha} = 6.6446573450(21) \times 10^{-27} \text{ kg}$	CODATA 2022, $\sigma = -1.36$
$m_{\alpha} = 6.6446573357(20) \times 10^{-27} \text{ kg}$	CODATA 2018, $\sigma = +3.22$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = +4.65$

$$E_p = \frac{\text{joule}}{\text{GeV}} \left(\frac{l_p^2 m_p}{t_p^2} \right) \left(1 - \frac{1}{2} \text{Re} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where GeV = the gigaelectron volt, l_p = the Planck length, m_p = the Planck mass, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_p = 1.22088572754077 \dots \times 10^{19} \text{ GeV}$	prediction
$E_p = 1.220890(14) \times 10^{19} \text{ GeV}$	CODATA 2022, $\sigma = -0.31$
$E_p = 1.220890(14) \times 10^{19} \text{ GeV}$	CODATA 2018, $\sigma = -0.31$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

von Klitzing constant

4.3925(33)

$$R_K = \frac{2\pi}{\mathfrak{K}_1^2} \left(\frac{l_p^2 m_p}{t_p q_p^2} \right) \left(1 + \frac{1}{2} \text{Re} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where π = Archimedes' constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, m_p = the Planck mass, t_p = the Planck time, q_p = the Planck charge, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$R_K = 2.58128074493918 \dots \times 10^4 \Omega$	prediction
$R_K = 2.581280745 \times 10^4 \Omega$	CODATA 2022, 10.63-digit match
$R_K = 2.581280745 \times 10^4 \Omega$	CODATA 2018, 10.63-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

Also listed as the *conventional value of von Klitzing constant*.

characteristic impedance of vacuum

4.3936(15)

$$Z_0 = 4\pi \left(\frac{l_p^2 m_p}{t_p q_p^2} \right) \left(1 + \frac{1}{2} \operatorname{Re} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, m_p = the Planck mass, t_p = the Planck time, q_p = the Planck charge, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$Z_0 = 3.76730313712033 \dots \times 10^2 \Omega$	prediction
$Z_0 = 3.76730313412(59) \times 10^2 \Omega$	CODATA 2022, $\sigma = +5.09$
$Z_0 = 3.76730313668(57) \times 10^2 \Omega$	CODATA 2018, $\sigma = +0.77$
	$\Delta_{\text{precision}} = -0.05$
	$\Delta_{\text{scaled}} = -4.49$

Planck electric impedance

4.404(16)

$$Z_p = \left(\frac{l_p^2 m_p}{t_p q_p^2} \right) \left(1 + \frac{1}{2} \operatorname{Re} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where l_p = the Planck length, m_p = the Planck mass, t_p = the Planck time, q_p = the Planck charge, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$Z_p = 2.99792458199153 \dots \times 10^1 \Omega$	prediction
$Z_p = 2.99792458(45) \times 10^1 \Omega$	WolframAlpha 2022, $\sigma = +0.00$
$Z_p = 2.99792458(45) \times 10^1 \Omega$	WolframAlpha 2018, $\sigma = +0.00$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

hertz-inverse meter relationship

3.6106(15)

$$\text{Hz} : \frac{1}{\text{m}} = \left(\frac{t_p}{s l_p} \right) \left(1 + \frac{1}{2} \text{Im} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where t_p = the Planck time, s = the second, l_p = the Planck length, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

Hz : 1/m = 3.33564095322875 ... $\times 10^{-9}$ cycles/m	prediction
Hz : 1/m = 3.335640951 $\times 10^{-9}$ cycles/m	CODATA 2022, 9.18-digit match
Hz : 1/m = 3.335640951 $\times 10^{-9}$ cycles/m	CODATA 2018, 9.18-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

atomic unit of velocity

-3.6170(15)

$$A_{\text{vel}} = \mathfrak{K}_1^2 \left(\frac{l_p}{t_p} \right) \left(1 - \frac{1}{2} \text{Im} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$A_{\text{vel}} = 2.18769126392058 \dots \times 10^6$ m/s	prediction
$A_{\text{vel}} = 2.18769126216(34) \times 10^6$ m/s	CODATA 2022, $\sigma = +5.18$
$A_{\text{vel}} = 2.18769126364(33) \times 10^6$ m/s	CODATA 2018, $\sigma = +3.88$
	$\Delta_{\text{precision}} = -0.03$
	$\Delta_{\text{scaled}} = -4.48$

atomic unit of momentum

-3.6171(15)

$$A_{\text{mom}} = \kappa_1^2 \left(\frac{l_p m_e}{t_p} \right) \left(1 - \frac{1}{2} \text{Im} \left(i^{i^{i^{\ddots}}} \right) \kappa_3 \kappa_4 \boxtimes \right)$$

Where κ_1 = the 1st hyperbolic partition constant, l_p = the Planck length, m_e = the electron mass, t_p = the Planck time, i = the imaginary unit, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} A_{\text{mom}} &= 1.99285191\mathbf{623658} \dots \times 10^{-24} \text{ m} \cdot \text{kg/s} && \text{prediction} \\ A_{\text{mom}} &= 1.99285191545(31) \times 10^{-24} \text{ m} \cdot \text{kg/s} && \text{CODATA 2018, } \sigma = +2.54 \\ A_{\text{mom}} &= 1.99285191410(30) \times 10^{-24} \text{ m} \cdot \text{kg/s} && \text{CODATA 2018, } \sigma = +7.12 \\ &&& \Delta_{\text{precision}} = -0.03 \\ &&& \Delta_{\text{scaled}} = +4.50 \end{aligned}$$

natural unit of momentum

-3.6121(30)

$$\mathcal{N}_{\text{mom}} = \left(\frac{l_p m_e}{t_p} \right) \left(1 - \frac{1}{2} \text{Im} \left(i^{i^{i^{\ddots}}} \right) \kappa_3 \kappa_4 \boxtimes \right)$$

Where l_p = the Planck length, m_e = the electron mass, t_p = the Planck time, i = the imaginary unit, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} \mathcal{N}_{\text{mom}} &= 2.73092453\mathbf{230547} \dots \times 10^{-22} \text{ m} \cdot \text{kg/s} && \text{prediction} \\ \mathcal{N}_{\text{mom}} &= 2.73092453446(85) \times 10^{-22} \text{ m} \cdot \text{kg/s} && \text{CODATA 2022, } \sigma = -2.53 \\ \mathcal{N}_{\text{mom}} &= 2.73092453075(82) \times 10^{-22} \text{ m} \cdot \text{kg/s} && \text{CODATA 2018, } \sigma = +1.90 \\ &&& \Delta_{\text{precision}} = -0.01 \\ &&& \Delta_{\text{scaled}} = +4.52 \end{aligned}$$

electron volt-kilogram relationship

7.22329(90)

$$\text{eV} : \text{kg} = \text{eV} \left(\frac{t_p^2}{l_p^2} \right) \left(1 + \text{Im} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where eV = the electron-volt, t_p = the Planck time, l_p = the Planck length, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\text{eV} : \text{kg} = 1.78266192320573 \dots \times 10^{-36} \text{ kg}$	prediction
$\text{eV} : \text{kg} = 1.782661921 \times 10^{-36} \text{ kg}$	CODATA 2022, 9.91-digit match
$\text{eV} : \text{kg} = 1.782661921 \times 10^{-36} \text{ kg}$	CODATA 2018, 9.91-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

joule-kilogram relationship

7.22398(56)

$$\text{J} : \text{kg} = \text{joule} \left(\frac{t_p^2}{l_p^2} \right) \left(1 + \text{Im} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where t_p = the Planck time, l_p = the Planck length, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\text{J} : \text{kg} = 1.11265005688553 \dots \times 10^{-17} \text{ kg}$	prediction
$\text{J} : \text{kg} = 1.112650056 \times 10^{-17} \text{ kg}$	CODATA 2022, 10-digit match
$\text{J} : \text{kg} = 1.112650056 \times 10^{-17} \text{ kg}$	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

joule-atomic mass unit relationship

7.2242(30)

$$J : A_{\text{mass}} = \text{joule} \left(\frac{t_p^2}{l_p^2 A_{\text{mass}}} \right) \left(1 + \text{Im} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where t_p = the Planck time, l_p = the Planck length, A_{mass} = the atomic mass constant, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} J : A_{\text{mass}} &= 6.70053525499264 \dots \times 10^9 \text{ u} && \text{prediction} \\ J : A_{\text{mass}} &= 6.7005352471(21) \times 10^9 \text{ u} && \text{CODATA 2022, } \sigma = +3.77 \\ J : A_{\text{mass}} &= 6.7005352565(20) \times 10^9 \text{ u} && \text{CODATA 2018, } \sigma = -0.75 \\ &&& \Delta_{\text{precision}} = -0.04 \\ &&& \Delta_{\text{scaled}} = -4.70 \end{aligned}$$

atomic mass unit-joule relationship

-7.2242(30)

$$A_{\text{mass}} : J = \left(\frac{l_p^2 A_{\text{mass}}}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where l_p = the Planck length, A_{mass} = the atomic mass constant, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} A_{\text{mass}} : J &= 1.49241808593480 \dots \times 10^{-10} \text{ J} && \text{prediction} \\ A_{\text{mass}} : J &= 1.49241808768(46) \times 10^{-10} \text{ J} && \text{CODATA 2022, } \sigma = -3.79 \\ A_{\text{mass}} : J &= 1.49241808560(45) \times 10^{-10} \text{ J} && \text{CODATA 2018, } \sigma = +0.74 \\ &&& \Delta_{\text{precision}} = -0.00 \\ &&& \Delta_{\text{scaled}} = +4.62 \end{aligned}$$

Also listed as the *atomic mass constant energy equivalent*.

natural unit of momentum in MeV/c

-7.2227(29)

$$\mathcal{N}_{\text{mom}}^{\cdot} = \frac{\text{joule}}{\text{MeV}} \left(\frac{s l_p^2 m_e}{t_p^2 \text{m}} \right) \left(1 - \text{Im} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where MeV = the megaelectron-volt, s = the second, l_p = the Planck length, m_e = the electron mass, t_p = the Planck time, m = the meter, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} \mathcal{N}_{\text{mom}}^{\cdot} &= 5.10998950028076 \dots \times 10^{-1} \text{ MeV}/c && \text{prediction} \\ \mathcal{N}_{\text{mom}}^{\cdot} &= 5.1099895069(16) \times 10^{-1} \text{ MeV}/c && \text{CODATA 2022, } \sigma = -4.14 \\ \mathcal{N}_{\text{mom}}^{\cdot} &= 5.1099895000(15) \times 10^{-1} \text{ MeV}/c && \text{CODATA 2018, } \sigma = +0.19 \\ &&& \Delta_{\text{precision}} = -0.02 \\ &&& \Delta_{\text{scaled}} = +4.60 \end{aligned}$$

atomic mass constant energy equivalent in MeV

-7.2227(30)

$$E_{A_{\text{mass}}}^{\cdot} = \frac{1}{\text{MeV}} \left(\frac{l_p^2 A_{\text{mass}}}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where MeV = the megaelectron-volt, l_p = the Planck length, A_{mass} = the atomic mass constant, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} E_{A_{\text{mass}}}^{\cdot} &= 9.31494102497143 \dots \times 10^2 \text{ MeV} && \text{prediction} \\ E_{A_{\text{mass}}}^{\cdot} &= 9.3149410372(29) \times 10^2 \text{ MeV} && \text{CODATA 2022, } \sigma = -4.22 \\ E_{A_{\text{mass}}}^{\cdot} &= 9.3149410242(28) \times 10^2 \text{ MeV} && \text{CODATA 2018, } \sigma = -0.25 \\ &&& \Delta_{\text{precision}} = -0.01 \\ &&& \Delta_{\text{scaled}} = +4.64 \end{aligned}$$

kilogram-joule relationship

-7.22398(56)

$$\text{kg} : \text{J} = \left(\frac{l_p^2 \text{ kg}}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i'}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where l_p = the Planck length, kg = the kilogram, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\text{kg} : \text{J} = 8.98755178064359 \dots \times 10^{16} \text{ J}$	prediction
$\text{kg} : \text{J} = 8.987551787 \times 10^{16} \text{ J}$	CODATA 2022, 9.15-digit match
$\text{kg} : \text{J} = 8.987551787 \times 10^{16} \text{ J}$	CODATA 2018, 9.15-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

kilogram-electron volt relationship

-7.22329(90)

$$\text{kg} : \text{eV} = \frac{\text{joule}}{\text{eV}} \left(\frac{l_p^2 \text{ kg}}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i'}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where eV = the electron-volt, l_p = the Planck length, kg = the kilogram, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\text{Kg} : \text{eV} = 5.60958859883646 \dots \times 10^{35} \text{ eV}$	prediction
$\text{kg} : \text{eV} = 5.609588603 \times 10^{35} \text{ eV}$	CODATA 2022, 9.13-digit match
$\text{kg} : \text{eV} = 5.609588603 \times 10^{35} \text{ eV}$	CODATA 2018, 9.13-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

neutron-proton mass difference energy equivalent**-7.9(35)**

$$E_{\Delta} = (m_n - m_+) \left(\frac{l_p^2}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where m_n = the neutron mass, m_+ = the proton mass, l_p = the Planck length, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_{\Delta} = 2.07214623577065 \dots \times 10^{-13} \text{ J}$	prediction
$E_{\Delta} = 2.07214712(60) \times 10^{-13} \text{ J}$	CODATA 2022, $\sigma = -1.47$
$E_{\Delta} = 2.07214689(74) \times 10^{-13} \text{ J}$	CODATA 2018, $\sigma = -0.88$
	$\Delta_{\text{precision}} = +0.09$
	$\Delta_{\text{scaled}} = +0.31$

neutron-proton mass difference energy equivalent in MeV**-7.9(52)**

$$\dot{E}_{\Delta} = \frac{\text{joule}}{\text{MeV}} (m_n - m_+) \left(\frac{l_p^2}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where, MeV = the megaelectron-volt, m_n = the neutron mass, m_+ = the proton mass, l_p = the Planck length, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\dot{E}_{\Delta} = 1.29333195323950 \dots \text{ MeV}$	prediction
$\dot{E}_{\Delta} = 1.29333251(38) \text{ MeV}$	CODATA 2022, $\sigma = -1.47$
$\dot{E}_{\Delta} = 1.29333236(46) \text{ MeV}$	CODATA 2018, $\sigma = -0.88$
	$\Delta_{\text{precision}} = +0.11$
	$\Delta_{\text{scaled}} = +0.32$

natural unit of energy in MeV

-7.2227(29)

$$\dot{E}_e = \frac{\text{joule}}{\text{MeV}} \left(\frac{l_p^2 m_e}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i^{\ddots}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where MeV = the megaelectron-volt, l_p = the Planck length, m_e = the electron mass, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\dot{E}_e = 5.10998950028076 \dots \times 10^{-1} \text{ MeV}$	prediction
$\dot{E}_e = 5.1099895069(16) \times 10^{-1} \text{ MeV}$	CODATA 2022, $\sigma = -4.14$
$\dot{E}_e = 5.1099895000(15) \times 10^{-1} \text{ MeV}$	CODATA 2018, $\sigma = +0.19$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = +4.60$

Also listed as the *electron mass energy equivalent in MeV*.

muon mass energy equivalent in MeV

-7.20(21)

$$\dot{E}_\mu = \frac{\text{joule}}{\text{MeV}} \left(\frac{l_p^2 m_\mu}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i^{\ddots}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where MeV = the megaelectron-volt, l_p = the Planck length, m_μ = the muon mass, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\dot{E}_\mu = 1.05658374731846 \dots \times 10^2 \text{ MeV}$	prediction
$\dot{E}_\mu = 1.056583755(23) \times 10^2 \text{ MeV}$	CODATA 2022, $\sigma = -0.33$
$\dot{E}_\mu = 1.056583755(23) \times 10^2 \text{ MeV}$	CODATA 2018, $\sigma = -0.33$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

proton mass energy equivalent in MeV

-7.2229(31)

$$E_+ = \frac{\text{joule}}{\text{MeV}} \left(\frac{l_p^2 m_+}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i^{\ddots}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where MeV = the megaelectron-volt, l_p = the Planck length, m_+ = the proton mass, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_+ = 9.38272088205557 \dots \times 10^2 \text{ MeV}$	prediction
$E_+ = 9.3827208943(29) \times 10^2 \text{ MeV}$	CODATA 2022, $\sigma = -4.22$
$E_+ = 9.3827208816(29) \times 10^2 \text{ MeV}$	CODATA 2018, $\sigma = +0.16$
	$\Delta_{\text{precision}} = -0.01$
	$\Delta_{\text{scaled}} = +4.37$

neutron mass energy equivalent in MeV

-7.2239(58)

$$E_n = \frac{\text{joule}}{\text{MeV}} \left(\frac{l_p^2 m_n}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i^{\ddots}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where MeV = the megaelectron-volt, l_p = the Planck length, m_n = the neutron mass, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_n = 9.39565420158796 \dots \times 10^2 \text{ MeV}$	prediction
$E_n = 9.3956542194(48) \times 10^2 \text{ MeV}$	CODATA 2022, $\sigma = -3.71$
$E_n = 9.3956542052(54) \times 10^2 \text{ MeV}$	CODATA 2018, $\sigma = -0.67$
	$\Delta_{\text{precision}} = +0.08$
	$\Delta_{\text{scaled}} = +2.63$

tau energy equivalent

-23.5(722)

$$\dot{E}_{\tau} = \frac{\text{joule}}{\text{MeV}} \left(\frac{l_p^2 m_{\tau}}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i^{\ddots}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where MeV = the megaelectron-volt, l_p = the Planck length, m_{τ} = the tau mass, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\dot{E}_{\tau} = 1.77686\textcolor{green}{269362378} \dots \times 10^3 \text{ MeV}$	prediction
$\dot{E}_{\tau} = 1.77686(12) \times 10^3 \text{ MeV}$	CODATA 2022, $\sigma = +0.02$
$\dot{E}_{\tau} = 1.77686(12) \times 10^3 \text{ MeV}$	CODATA 2018, $\sigma = +0.02$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

deuteron mass energy equivalent in MeV

-7.2225(30)

$$\dot{E}_{\text{de}} = \frac{\text{joule}}{\text{MeV}} \left(\frac{l_p^2 m_{\text{de}}}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i^{\ddots}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where MeV = the megaelectron-volt, l_p = the Planck length, m_{de} = the deuteron mass, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\dot{E}_{\text{de}} = 1.87561294\textcolor{green}{255340} \dots \times 10^3 \text{ MeV}$	prediction
$\dot{E}_{\text{de}} = 1.87561294500(58) \times 10^3 \text{ MeV}$	CODATA 2022, $\sigma = -4.22$
$\dot{E}_{\text{de}} = 1.87561294257(57) \times 10^3 \text{ MeV}$	CODATA 2018, $\sigma = +0.03$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = +4.26$

helion mass energy equivalent in MeV

-7.2228(30)

$$E_{\text{he}} = \frac{\text{joule}}{\text{MeV}} \left(\frac{l_p^2 m_{\text{he}}}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i^{\ddots}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where MeV = the megaelectron-volt, l_p = the Planck length, m_{he} = the helion mass, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_{\text{he}} = 2.80839160744450 \dots \times 10^3 \text{ MeV}$	prediction
$E_{\text{he}} = 2.80839161112(88) \times 10^3 \text{ MeV}$	CODATA 2022, $\sigma = -4.18$
$E_{\text{he}} = 2.80839160743(85) \times 10^3 \text{ MeV}$	CODATA 2018, $\sigma = +0.02$
	$\Delta_{\text{precision}} = -0.01$
	$\Delta_{\text{scaled}} = +4.34$

triton mass energy equivalent in MeV

-7.2221(30)

$$E_{\text{tri}} = \frac{\text{joule}}{\text{MeV}} \left(\frac{l_p^2 m_{\text{tri}}}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i^{\ddots}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where MeV = the megaelectron-volt, l_p = the Planck length, m_{tri} = the triton mass, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_{\text{tri}} = 2.80892113277064 \dots \times 10^3 \text{ MeV}$	prediction
$E_{\text{tri}} = 2.80892113668(88) \times 10^3 \text{ MeV}$	CODATA 2018, $\sigma = -4.44$
$E_{\text{tri}} = 2.80892113298(85) \times 10^3 \text{ MeV}$	CODATA 2018, $\sigma = -0.25$
	$\Delta_{\text{precision}} = -0.03$
	$\Delta_{\text{scaled}} = +4.35$

alpha particle mass energy equivalent in MeV

-7.2235(30)

$$\dot{E}_{\alpha} = \frac{\text{joule}}{\text{MeV}} \left(\frac{l_p^2 m_{\alpha}}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i^{\ddots}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where MeV = the megaelectron-volt, l_p = the Planck length, m_{α} = the alpha particle mass, l_p = the Planck length, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\dot{E}_{\alpha} = 3.72737940696477 \dots \times 10^3 \text{ MeV}$	prediction
$\dot{E}_{\alpha} = 3.7273794118(12) \times 10^3 \text{ MeV}$	CODATA 2022, $\sigma = -4.03$
$\dot{E}_{\alpha} = 3.7273794066(11) \times 10^3 \text{ MeV}$	CODATA 2018, $\sigma = +0.33$
	$\Delta_{\text{precision}} = -0.03$
	$\Delta_{\text{scaled}} = +4.72$

electron mass energy equivalent

-7.2241(30)

$$E_e = \left(\frac{l_p^2 m_e}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where l_p = the Planck length, m_e = the electron mass, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_e = 8.18710577846019 \dots \times 10^{-14} \text{ J}$	prediction
$E_e = 8.1871057880(26) \times 10^{-14} \text{ J}$	CODATA 2018, $\sigma = -3.67$
$E_e = 8.1871057769(25) \times 10^{-14} \text{ J}$	CODATA 2018, $\sigma = +0.62$
	$\Delta_{\text{precision}} = -0.01$
	$\Delta_{\text{scaled}} = +4.44$

Also listed as the *natural unit of energy*.

muon mass energy equivalent

-7.19(24)

$$E_\mu = \left(\frac{l_p^2 m_\mu}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where l_p = the Planck length, m_μ = the muon mass, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_\mu = 1.69283379205041 \dots \times 10^{-11} \text{ J}$	prediction
$E_\mu = 1.692833804(38) \times 10^{-11} \text{ J}$	CODATA 2022, $\sigma = -0.31$
$E_\mu = 1.692833804(38) \times 10^{-11} \text{ J}$	CODATA 2018, $\sigma = -0.31$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

proton mass energy equivalent

-7.2242(31)

$$E_+ = \left(\frac{l_p^2 m_+}{t_p^2} \right) \left(1 - \operatorname{Im} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where l_p = the Planck length, m_+ = the proton mass, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_+ = 1.50327761626390 \dots \times 10^{-10} \text{ J}$	prediction
$E_+ = 1.50327761802(47) \times 10^{-10} \text{ J}$	CODATA 2022, $\sigma = -3.74$
$E_+ = 1.50327761598(46) \times 10^{-10} \text{ J}$	CODATA 2018, $\sigma = +0.62$
	$\Delta_{\text{precision}} = -0.04$
	$\Delta_{\text{scaled}} = +4.43$

neutron mass energy equivalent

-7.2253(57)

$$E_n = \left(\frac{l_p^2 m_n}{t_p^2} \right) \left(1 - \operatorname{Im} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where l_p = the Planck length, m_n = the neutron mass, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_n = 1.50534976249967 \dots \times 10^{-10} \text{ J}$	prediction
$E_n = 1.50534976514(76) \times 10^{-10} \text{ J}$	CODATA 2022, $\sigma = -3.47$
$E_n = 1.50534976287(86) \times 10^{-10} \text{ J}$	CODATA 2018, $\sigma = -0.43$
	$\Delta_{\text{precision}} = +0.07$
	$\Delta_{\text{scaled}} = +2.63$

tau mass energy equivalent

-40(668)

$$E_{\tau} = \left(\frac{l_p^2 m_{\tau}}{t_p^2} \right) \left(1 - \operatorname{Im} \left(i^{i^{i^{\ddots}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where l_p = the Planck length, m_{τ} = the tau mass, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_{\tau} = 2.84684788994152 \dots \times 10^{-10} \text{ J}$	prediction
$E_{\tau} = 2.84684(19) \times 10^{-10} \text{ J}$	CODATA 2022, $\sigma = +0.04$
$E_{\tau} = 2.84684(19) \times 10^{-10} \text{ J}$	CODATA 2018, $\sigma = +0.04$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

deuteron mass energy equivalent

-7.2239(30)

$$E_{\text{de}} = \left(\frac{l_p^2 m_{\text{de}}}{t_p^2} \right) \left(1 - \operatorname{Im} \left(i^{i^{i^{\ddots}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where l_p = the Planck length, m_{de} = the deuteron mass, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_{\text{de}} = 3.00506323139998 \dots \times 10^{-10} \text{ J}$	prediction
$E_{\text{de}} = 3.00506323491(94) \times 10^{-10} \text{ J}$	CODATA 2022, $\sigma = -3.73$
$E_{\text{de}} = 3.00506323102(91) \times 10^{-10} \text{ J}$	CODATA 2018, $\sigma = +0.42$
	$\Delta_{\text{precision}} = -0.04$
	$\Delta_{\text{scaled}} = +4.27$

helion mass energy equivalent

-7.2243(31)

$$E_{\text{he}} = \left(\frac{l_p^2 m_{\text{he}}}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where l_p = the Planck length, m_{he} = the helion mass, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_{\text{he}} = 4.49953941318757 \dots \times 10^{-10} \text{ J}$	prediction
$E_{\text{he}} = 4.4995394185(14) \times 10^{-10} \text{ J}$	CODATA 2022, $\sigma = -3.79$
$E_{\text{he}} = 4.4995394125(14) \times 10^{-10} \text{ J}$	CODATA 2018, $\sigma = +0.49$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +4.29$

triton mass energy equivalent

-7.2234(31)

$$E_{\text{tri}} = \left(\frac{l_p^2 m_{\text{tri}}}{t_p^2} \right) \left(1 - \text{Im} \left(i^{i^{i^{\cdot}}} \right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where l_p = the Planck length, m_{tri} = the triton mass, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_{\text{tri}} = 4.50038780629234 \dots \times 10^{-10} \text{ J}$	prediction
$E_{\text{tri}} = 4.5003878119(14) \times 10^{-10} \text{ J}$	CODATA 2022, $\sigma = -4.01$
$E_{\text{tri}} = 4.5003878060(14) \times 10^{-10} \text{ J}$	CODATA 2018, $\sigma = +0.21$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +4.21$

$$E_{\alpha} = \left(\frac{l_p^2 m_{\alpha}}{t_p^2}\right) \left(1 - Im\left(i^{i^{i^{\ddots}}}\right) \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where l_p = the Planck length, m_{α} = the alpha particle mass, t_p = the Planck time, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_{\alpha} = 5.97192019271236 \dots \times 10^{-10} \text{ J}$	prediction
$E_{\alpha} = 5.9719201997(19) \times 10^{-10} \text{ J}$	CODATA 2022, $\sigma = -3.68$
$E_{\alpha} = 5.9719201914(18) \times 10^{-10} \text{ J}$	CODATA 2018, $\sigma = +0.73$
	$\Delta_{\text{precision}} = -0.01$
	$\Delta_{\text{scaled}} = +4.61$

deuteron g factor

-181.339(25)

$$g_{\text{de}} = \frac{2\pi}{8K} \left(1 - 8 \left(\frac{2\pi}{W_{\text{We}}} \right) \mathfrak{K}_1 \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where π = Archimedes' constant, K = Catalan's constant, W_{We} = the Weierstrass constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$g_{\text{de}} = 8.57438235960775 \dots \times 10^{-1}$	prediction
$g_{\text{de}} = 8.574382335(22) \times 10^{-1}$	CODATA 2022, $\sigma = +1.12$
$g_{\text{de}} = 8.574382338(22) \times 10^{-1}$	CODATA 2018, $\sigma = +0.98$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -0.14$

deuteron magnetic moment to nuclear magneton ratio

-181.339(26)

$$\frac{\mu_{\text{de}}}{\mu_N} = \frac{2\pi}{8K} \left(1 - 8 \left(\frac{2\pi}{W_{\text{We}}} \right) \mathfrak{K}_1 \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where π = Archimedes' constant, K = Catalan's constant, W_{We} = the Weierstrass constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_{\text{de}}/\mu_N = 8.57438235960775 \times 10^{-1}$	prediction
$\mu_{\text{de}}/\mu_N = 8.574382335(22) \times 10^{-1}$	CODATA 2022, $\sigma = +1.12$
$\mu_{\text{de}}/\mu_N = 8.574382338(22) \times 10^{-1}$	CODATA 2018, $\sigma = +0.98$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -0.14$

deuteron magnetic moment to Bohr magneton ratio -1201.313(25)

$$\frac{\mu_{de}}{\mu_B} = \frac{2\pi}{8K} \left(\frac{m_e}{m_+} \right) \left(1 - 8 \left(\frac{2\pi}{W_{We}} \right) \mathfrak{K}_1 \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where π = Archimedes’ constant, K = Catalan’s constant, W_{We} = the Weierstrass constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_{de}/\mu_B = 4.66975458182757 \times 10^{-4}$	prediction
$\mu_{de}/\mu_B = 4.669754568(12) \times 10^{-4}$	CODATA 2022, $\sigma = +1.15$
$\mu_{de}/\mu_B = 4.669754570(12) \times 10^{-4}$	CODATA 2018, $\sigma = +0.99$
	$\Delta_{precision} = +0.00$
	$\Delta_{scaled} = -0.17$

muon magnetic moment

717.99(23)

$$\mu_{\mu} = -\frac{\mathfrak{K}_1^2 \mathfrak{K}_2^2}{P_{up}} \left(\frac{l_p^2 q_p m_p}{t_p m_{\mu}} \right) \left(1 + \frac{4\pi}{\mathcal{L}_{Li}} (\pi - 1) \mathfrak{K}_1^2 \mathfrak{K}_3^2 \mathfrak{K}_4^2 \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, P_{up} = the universal parabolic constant, l_p = the Planck length, q_p = the Planck charge, m_p = the Planck mass, t_p = the Planck time, m_{μ} = the muon mass, π = Archimedes' constant, $\mathcal{L}_{Li}$ = Liouville's constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_{\mu} = -4.490448303035195 \dots \times 10^{-26} \text{ J/T}$	prediction
$\mu_{\mu} = -4.49044830(10) \times 10^{-26} \text{ J/T}$	CODATA 2022, $\sigma = -0.30$
$\mu_{\mu} = -4.49044830(10) \times 10^{-26} \text{ J/T}$	CODATA 2018, $\sigma = -0.30$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

electron magnetic moment anomaly

-650.6071858(18)

$$a_e = \frac{2 \mathfrak{K}_1 \mathfrak{K}_2^2}{P_{up}} - 1 \left(1 - \frac{35}{\mathcal{L}_{Li}} \left(\frac{\sqrt{2}}{V_{fe}} \right) \mathfrak{K}_1^2 \mathfrak{K}_3^2 \mathfrak{K}_4^2 \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, P_{up} = the universal parabolic constant, $\mathcal{L}_{Li}$ = Liouville's constant, V_{fe} = the figure-eight knot hyperbolic volume, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$a_e = 1.15965218025405 \dots \times 10^{-3}$	prediction
$a_e = 1.15965218046(18) \times 10^{-3}$	CODATA 2022, $\sigma = -1.14$
$a_e = 1.15965218128(18) \times 10^{-3}$	CODATA 2018, $\sigma = -5.70$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -4.56$

muon magnetic moment anomaly

-713.2916(41)

$$a_{\mu} = \frac{2 \, \mathfrak{x}_1 \, \mathfrak{x}_2^2}{P_{up}} - 1 \left(1 - \frac{!5}{\sqrt{\mathcal{L}_{Li}}} \left(2K \right) \mathfrak{x}_1^2 \, \mathfrak{x}_3^2 \, \mathfrak{x}_4^2 \, \boxtimes \right)$$

Where $\mathfrak{x}_1$ = the 1st hyperbolic partition constant, $\mathfrak{x}_2$ = the 2nd hyperbolic partition constant, P_{up} = the universal parabolic constant, $!n$ = the derangement function, $\mathcal{L}_{Li}$ = Liouville’s constant, K = Catalan’s constant, $\mathfrak{x}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{x}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$a_{\mu} = 1.16592060110720 \dots \times 10^{-3}$	prediction
$a_{\mu} = 1.16592062(41) \times 10^{-3}$	CODATA 2022, $\sigma = -0.05$
$a_{\mu} = 1.16592089(63) \times 10^{-3}$	CODATA 2018, $\sigma = -0.46$
	$\Delta_{\text{precision}} = +0.18$
	$\Delta_{\text{scaled}} = -0.43$

muon g factor

712.5140(65)

$$g_{\mu} = -\frac{4 \, \mathfrak{x}_1 \, \mathfrak{x}_2^2}{P_{up}} \left(1 + \zeta(2)^{1/3} \left(\frac{35}{C_d} \right) \mathfrak{x}_1^3 \, \mathfrak{x}_3^3 \, \mathfrak{x}_4^3 \, \boxtimes \right)$$

Where $\mathfrak{x}_1$ = the 1st hyperbolic partition constant, $\mathfrak{x}_2$ = the 2nd hyperbolic partition constant, P_{up} = the universal parabolic constant, $\zeta(x)$ = the Riemann zeta function, C_d = the domino tiling constant, $\mathfrak{x}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{x}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$g_{\mu} = -2.00233184172840 \dots$	prediction
$g_{\mu} = -2.00233184123(82)$	CODATA 2022, $\sigma = -0.61$
$g_{\mu} = -2.0023318418(13)$	CODATA 2018, $\sigma = +0.06$
	$\Delta_{\text{precision}} = +1.80$
	$\Delta_{\text{scaled}} = +0.50$

muon magnetic moment to Bohr magneton ratio

712.48(22)

$$\frac{\mu_\mu}{\mu_B} = -\frac{2 \mathfrak{K}_1 \mathfrak{K}_2^2}{P_{up}} \left(\frac{m_e}{m_\mu} \right) \left(1 + \zeta(2)^{1/3} \left(\frac{35}{C_d} \right) \mathfrak{K}_1^3 \mathfrak{K}_3^3 \mathfrak{K}_4^3 \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, P_{up} = the universal parabolic constant, m_e = the electron mass, m_μ = the muon mass, $\zeta(x)$ = the Riemann zeta function, C_d = the domino tiling constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_\mu/\mu_B = -4.84197050791187 \dots \times 10^{-3}$	prediction
$\mu_\mu/\mu_B = -4.84197048(11) \times 10^{-3}$	CODATA 2022, $\sigma = -0.25$
$\mu_\mu/\mu_B = -4.84197047(11) \times 10^{-3}$	CODATA 2018, $\sigma = -0.34$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.09$

muon magnetic moment to nuclear magneton ratio

712.48(22)

$$\frac{\mu_\mu}{\mu_N} = -\frac{2 \mathfrak{K}_1 \mathfrak{K}_2^2}{P_{up}} \left(\frac{m_+}{m_\mu} \right) \left(1 + \zeta(2)^{1/3} \left(\frac{35}{C_d} \right) \mathfrak{K}_1^3 \mathfrak{K}_3^3 \mathfrak{K}_4^3 \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, P_{up} = the universal parabolic constant, m_+ = the proton mass, m_μ = the muon mass, $\zeta(x)$ = the Riemann zeta function, C_d = the domino tiling constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_\mu/\mu_N = -8.89059709269183 \dots$	prediction
$\mu_\mu/\mu_N = -8.89059704(20)$	CODATA 2022, $\sigma = -0.26$
$\mu_\mu/\mu_N = -8.89059703(20)$	CODATA 2018, $\sigma = -0.31$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.05$

molar gas constant

-236.35823(30)

$$R = \frac{6}{e^\gamma} \kappa_1^2 \left(\frac{l_p^2 C}{t_p^2 m q_p T_p} \right) \left(1 - \frac{8}{C_d} \kappa_1^4 \kappa_3^4 \kappa_4^4 \boxtimes \right)$$

Where e = Euler’s number, γ = the Euler-Mascheroni constant, κ_1 = the 1st hyperbolic partition constant, l_p = the Planck length, C = the coulomb, t_p = the Planck time, m = the meter, q_p = the Planck charge, T_p = The Planck temperature, C_d = the domino tiling constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$R = 8.31446261978165 \dots$ J/mol · K	prediction
$R = 8.314462618$ J/mol · K	CODATA 2022, 9.67-digit match
$R = 8.314462618$ J/mol · K	CODATA 2018, 9.67-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

molar volume of ideal gas (273.15 K, 100 kPa)

−236.357(02)

$$V_{m_0} = \frac{6}{e^\gamma} \kappa_1^2 \frac{T_0}{p_0} \left(\frac{l_p^2 C}{t_p^2 m q_p T_p} \right) \left(1 - \frac{8}{C_d} \kappa_1^4 \kappa_3^4 \kappa_4^4 \boxtimes \right)$$

Where e = Euler’s number, γ = the Euler-Mascheroni constant, κ_1 = the 1st hyperbolic partition constant, $T_0 = 273.150000000000$ K, $p_0 = 100.000000000000$ kPa, l_p = the Planck length, C = the coulomb, t_p = the Planck time, m = the meter, q_p = the Planck charge, T_p = the Planck temperature, C_d = the domino tiling constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$V_{m_0} = 2.27109546459336 \dots \times 10^{-2}$ m ³ /mol	prediction
$V_{m_0} = 2.271095464 \times 10^{-2}$ m ³ /mol	CODATA 2022, 10-digit match
$V_{m_0} = 2.271095464 \times 10^{-2}$ m ³ /mol	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

molar volume of ideal gas (273.15 K, 101.325 kPa) −236.359(22)

$$V_{m_1} = \frac{6}{e^\gamma} \kappa_1^2 \frac{T_0}{p_1} \left(\frac{l_p^2 C}{t_p^2 m q_p T_p} \right) \left(1 - \frac{8}{C_d} \kappa_1^4 \kappa_3^4 \kappa_4^4 \boxtimes \right)$$

Where e = Euler's number, γ = the Euler-Mascheroni constant, κ_1 = the 1st hyperbolic partition constant, $T_0 = 273.150000000000\text{ K}$, $p_1 = 101.325000000000\text{ kPa}$, l_p = the Planck length, C = the coulomb, t_p = the Planck time, m = the meter, q_p = the Planck charge, T_p = the Planck temperature, C_d = the domino tiling constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$V_{m_1} = 2.241396954\textcolor{teal}{94040} \dots \times 10^{-2} \text{ m}^3/\text{mol}$	prediction
$V_{m_1} = 2.241396954 \times 10^{-2} \text{ m}^3/\text{mol}$	CODATA 2022, 10-digit match
$V_{m_1} = 2.241396954 \times 10^{-2} \text{ m}^3/\text{mol}$	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

electron g factor 649.8958153(18)

$$g_e = -\frac{4 \kappa_1 \kappa_2^2}{P_{up}} \left(1 + C_{\text{CFP}}^2 \left(\frac{32}{D_d} \right) \kappa_1^6 \kappa_3^6 \kappa_4^6 \boxtimes \right)$$

Where κ_1 = the 1st hyperbolic partition constant, κ_2 = the 2nd hyperbolic partition constant, P_{up} = the universal parabolic constant, C_{CFP} = the fixed point of the hyperbolic cotangent, D_d = the dimer constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$g_e = -2.002319304360\textcolor{teal}{65} \dots$	prediction
$g_e = -2.00231930436092(36)$	CODATA 2022, $\sigma = +0.74$
$g_e = -2.00231930436256(35)$	CODATA 2018, $\sigma = -5.46$
	$\Delta_{\text{precision}} = -0.01$
	$\Delta_{\text{scaled}} = -4.69$

electron magnetic moment to Bohr magneton ratio 649.8958153(18)

$$\frac{\mu_e}{\mu_B} = -\frac{2 \mathfrak{K}_1 \mathfrak{K}_2^2}{P_{up}} \left(1 + C_{\text{CFP}}^2 \left(\frac{32}{D_d} \right) \mathfrak{K}_1^6 \mathfrak{K}_3^6 \mathfrak{K}_4^6 \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, P_{up} = the universal parabolic constant, C_{CFP} = the fixed point of the hyperbolic cotangent, D_d = the dimer constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_e/\mu_B = -1.00115965218033 \dots$	prediction
$\mu_e/\mu_B = -1.00115965218046(18)$	CODATA 2022, $\sigma = +0.74$
$\mu_e/\mu_B = -1.00115965218128(18)$	CODATA 2018, $\sigma = -5.28$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -4.56$

electron magnetic moment to nuclear magneton ratio 649.89584(17)

$$\frac{\mu_e}{\mu_N} = -\frac{2 \mathfrak{K}_1 \mathfrak{K}_2^2}{P_{up}} \left(\frac{m_+}{m_e} \right) \left(1 + C_{\text{CFP}}^2 \left(\frac{32}{D_d} \right) \mathfrak{K}_1^6 \mathfrak{K}_3^6 \mathfrak{K}_4^6 \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, P_{up} = the universal parabolic constant, m_+ = the proton mass, m_e = the electron mass, C_{CFP} = the fixed point of the hyperbolic cotangent, D_d = the dimer constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_e/\mu_N = -1.83828197186466 \dots \times 10^3$	prediction
$\mu_e/\mu_N = -1.838281971877(32) \times 10^3$	CODATA 2022, $\sigma = +0.39$
$\mu_e/\mu_N = -1.83828197188(11) \times 10^3$	CODATA 2018, $\sigma = -0.14$
	$\Delta_{\text{precision}} = +0.54$
	$\Delta_{\text{scaled}} = -0.03$

electron magnetic moment

655.4130(32)

$$\mu_e = -\frac{\mathfrak{x}_1^2 \mathfrak{x}_2^2}{P_{up}} \left(\frac{l_p^2 q_p m_p}{t_p m_e} \right) \left(1 + \sqrt{7} \left(\frac{8s}{4\pi} \right) \mathfrak{x}_1^8 \mathfrak{x}_3^8 \mathfrak{x}_4^8 \boxtimes \right)$$

Where $\mathfrak{x}_1$ = the 1st hyperbolic partition constant, $\mathfrak{x}_2$ = the 2nd hyperbolic partition constant, P_{up} = the universal parabolic constant, l_p = the Planck length, q_p = the Planck charge, m_p = the Planck mass, t_p = the Planck time, m_e = the electron mass, π = Archimedes' constant, $\mathfrak{x}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{x}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_e = -9.28476469207109 \dots \times 10^{-24} \text{ J/T}$	prediction
$\mu_e = -9.2847646917(29) \times 10^{-24} \text{ J/T}$	CODATA 2022, $\sigma = -0.13$
$\mu_e = -9.2847647043(28) \times 10^{-24} \text{ J/T}$	CODATA 2018, $\sigma = -4.37$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = -4.50$

electron gyromagnetic ratio

650.2658(31)

$$\gamma_e = \frac{2 \mathfrak{x}_1^2 \mathfrak{x}_2^2}{P_{up}} \left(\frac{q_p}{m_e} \right) \left(1 + 7 \left(\frac{3s}{4\pi} \right) \mathfrak{x}_1^8 \mathfrak{x}_3^8 \mathfrak{x}_4^8 \boxtimes \right)$$

Where $\mathfrak{x}_1$ = the 1st hyperbolic partition constant, $\mathfrak{x}_2$ = the 2nd hyperbolic partition constant, P_{up} = the universal parabolic constant, q_p = the Planck charge, m_e = the electron mass, s = the arc length of the unit lemniscate, π = Archimedes' constant, $\mathfrak{x}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{x}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\gamma_e = 1.76085962907620 \dots \times 10^{11} \text{ 1/s} \cdot \text{T}$	prediction
$\gamma_e = 1.76085962784(55) \times 10^{11} \text{ 1/s} \cdot \text{T}$	CODATA 2022, $\sigma = +2.25$
$\gamma_e = 1.76085963023(53) \times 10^{11} \text{ 1/s} \cdot \text{T}$	CODATA 2018, $\sigma = -2.17$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = 4.51$

$$\dot{\gamma}_e = \frac{\mathfrak{x}_1^2 \mathfrak{x}_2^2}{\pi P_{up}} \frac{1}{\text{MHz}} \left(\frac{q_p}{s m_e} \right) \left(1 + \textcolor{blue}{7} \left(\frac{3s}{4\pi} \right) \mathfrak{x}_1^8 \mathfrak{x}_3^8 \mathfrak{x}_4^8 \boxtimes \right)$$

Where $\mathfrak{x}_1$ = the 1st hyperbolic partition constant, $\mathfrak{x}_2$ = the 2nd hyperbolic partition constant, π = Archimedes' constant, P_{up} = the universal parabolic constant, MHz = the mega hertz, q_p = the Planck charge, s = the second, m_e = the electron mass, s = the arc length of the unit lemniscate, π = Archimedes' constant, $\mathfrak{x}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{x}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\dot{\gamma}_e = 2.8024951\textcolor{green}{4058438}... \times 10^4 \text{ MHz/T}$

prediction

 $\dot{\gamma}_e = 2.80249513861(87) \times 10^4 \text{ MHz/T}$ CODATA 2022, $\sigma = +2.27$ $\dot{\gamma}_e = 2.80249514242(85) \times 10^4 \text{ MHz/T}$ CODATA 2018, $\sigma = -2.16$

$\Delta_{\text{precision}} = -0.02$ $\Delta_{\text{scaled}} = -4.48$

shielded proton gyromagnetic ratio in MHz/T

-580.805(41)

$$\dot{\gamma}_+' = \frac{\mathfrak{x}_1^2 \mathfrak{x}_2^2}{\pi S} \frac{1}{\text{MHz}} \left(\frac{q_p}{s m_+} \right) \left(1 - \sqrt{35} \left(\frac{7}{S} \right) \mathfrak{x}_2 \mathfrak{x}_3 \mathfrak{x}_4 \boxtimes \right)$$

Where $\mathfrak{x}_1$ = the 1st hyperbolic partition constant, $\mathfrak{x}_2$ = the 2nd hyperbolic partition constant, π = Archimedes' constant, S = Sierpiński's constant, MHz = the megahertz, q_p = the Planck charge, s = the second, m_+ = the proton mass, s = the arc length of the unit lemniscate, $\mathfrak{x}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{x}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} \dot{\gamma}_+' &= 4.2576385\textcolor{brown}{2810155} \dots \times 10^1 \text{ MHz/T} && \text{prediction} \\ \dot{\gamma}_+' &= 4.257638543(17) \times 10^1 \text{ MHz/T} && \text{CODATA 2022, } \sigma = -0.88 \\ \dot{\gamma}_+' &= 4.257638474(46) \times 10^1 \text{ MHz/T} && \text{CODATA 2018, } \sigma = +1.18 \\ &&& \Delta_{\text{precision}} = +0.44 \\ &&& \Delta_{\text{scaled}} = +1.50 \end{aligned}$$

shielded proton gyromagnetic ratio

-580.799(41)

$$\gamma_+' = \frac{2 \mathfrak{x}_1^2 \mathfrak{x}_2^2}{S} \left(\frac{q_p}{m_+} \right) \left(1 - \sqrt{35} \left(\frac{7}{S} \right) \mathfrak{x}_2 \mathfrak{x}_3 \mathfrak{x}_4 \boxtimes \right)$$

Where $\mathfrak{x}_1$ = the 1st hyperbolic partition constant, $\mathfrak{x}_2$ = the 2nd hyperbolic partition constant, S = Sierpiński's constant, q_p = the Planck charge, m_+ = the proton mass, s = the arc length of the unit lemniscate, $\mathfrak{x}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{x}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} \gamma_+' &= 2.6751531\textcolor{brown}{8430494} \dots \times 10^8 \text{ 1/s} \cdot \text{T} && \text{prediction} \\ \gamma_+' &= 2.675153194(11) \times 10^8 \text{ 1/s} \cdot \text{T} && \text{CODATA 2022, } \sigma = -0.88 \\ \gamma_+' &= 2.675153151(29) \times 10^8 \text{ 1/s} \cdot \text{T} && \text{CODATA 2018, } \sigma = -0.33 \\ &&& \Delta_{\text{precision}} = +0.53 \\ &&& \Delta_{\text{scaled}} = +1.48 \end{aligned}$$

$$\mu_{+}' = \frac{\mathfrak{x}_1^2 \mathfrak{x}_2^2}{S} \left(\frac{l_p^2 q_p m_p}{t_p m_+} \right) \left(1 + \frac{\sqrt{3 \cdot 35}}{M} \left(\frac{7}{s} \right) \mathfrak{x}_2 \mathfrak{x}_3 \mathfrak{x}_4 \boxtimes \right)$$

Where $\mathfrak{x}_1$ = the 1st hyperbolic partition constant, $\mathfrak{x}_2$ = the 2nd hyperbolic partition constant, S = Sierpiński's constant, l_p = the Planck length, q_p = the Planck charge, m_p = the Planck mass, t_p = the Planck time, m_+ = the proton mass, M = the Madelung constant, s = the arc length of the unit lemniscate, $\mathfrak{x}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{x}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_{+}' = 1.41057057939798 \dots \times 10^{-26} \text{ J/T}$
 $\mu_{+}' = 1.4105705830(58) \times 10^{-26} \text{ J/T}$
 $\mu_{+}' = 1.410570560(15) \times 10^{-26} \text{ J/T}$

prediction

CODATA 2022, $\sigma = -0.62$

CODATA 2018, $\sigma = +1.29$

$\Delta_{\text{precision}} = +0.43$

$\Delta_{\text{scaled}} = +1.53$

proton magnetic moment to Bohr magneton ratio -324.4677(31)

$$\frac{\mu_+}{\mu_B} = \frac{2 \, \mathfrak{K}_1 \, \mathfrak{K}_2^2}{S} \left(\frac{m_e}{m_+} \right) \left(1 - 6 \left(\frac{2\pi}{8K} \right)^2 \mathfrak{K}_2 \, \mathfrak{K}_3 \, \mathfrak{K}_4 \, \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, S = Sierpiński's constant, m_e = the electron mass, m_+ = the proton mass, π = Archimedes' constant, K = Catalan's constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_+/\mu_B = 1.52103220255243 \dots \times 10^{-3}$	prediction
$\mu_+/\mu_B = 1.52103220230(45) \times 10^{-3}$	CODATA 2022, $\sigma = +0.56$
$\mu_+/\mu_B = 1.52103220230(46) \times 10^{-3}$	CODATA 2018, $\sigma = +0.55$
	$\Delta_{\text{precision}} = +0.01$
	$\Delta_{\text{scaled}} = +0.00$

proton magnetic moment to nuclear magneton ratio -324.4678(29)

$$\frac{\mu_+}{\mu_N} = \frac{2 \, \mathfrak{K}_1 \, \mathfrak{K}_2^2}{S} \left(1 - 6 \left(\frac{2\pi}{8K} \right)^2 \mathfrak{K}_2 \, \mathfrak{K}_3 \, \mathfrak{K}_4 \, \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, S = Sierpiński's constant, π = Archimedes' constant, K = Catalan's constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_+/\mu_N = 2.79284734506470 \dots$	prediction
$\mu_+/\mu_N = 2.79284734463(82)$	CODATA 2022, $\sigma = +0.53$
$\mu_+/\mu_N = 2.79284734463(82)$	CODATA 2018, $\sigma = +0.53$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

proton g factor

-324.4677(18)

$$g_+ = \frac{4 \, \mathfrak{x}_1 \, \mathfrak{x}_2^2}{S} \left(1 - 6 \left(\frac{2\pi}{8K} \right)^2 \mathfrak{x}_2 \, \mathfrak{x}_3 \, \mathfrak{x}_4 \, \boxtimes \right)$$

Where $\mathfrak{x}_1$ = the 1st hyperbolic partition constant, $\mathfrak{x}_2$ = the 2nd hyperbolic partition constant, S = Sierpiński's constant, π = Archimedes' constant, K = Catalan's constant, $\mathfrak{x}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{x}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$g_+ = 5.58569469012941 \dots$	prediction
$g_+ = 5.5856946893(16)$	CODATA 2022, $\sigma = +0.52$
$g_+ = 5.5856946893(16)$	CODATA 2018, $\sigma = +0.52$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

neutron to shielded proton magnetic moment ratio

1017.2(23)

$$\frac{\mu_n}{\mu_+'} = -\frac{S}{C_{\text{CFP}}} \left(\frac{m_+}{m_n} \right) \left(1 + \sqrt{5} \left(\frac{4\pi}{V_{\text{fe}}} \right) \mathfrak{K}_2 \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where S = Sierpiński's constant, C_{CFP} = the fixed point of the hyperbolic cotangent, m_+ = the proton mass, m_n = the neutron mass, π = Archimedes' constant, V_{fe} = the figure eight knot hyperbolic volume, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_n/\mu_+' = -6.84997004903262 \dots \times 10^{-1}$	prediction
$\mu_n/\mu_+' = -6.8499694(16) \times 10^{-1}$	CODATA 2022, $\sigma = -0.41$
$\mu_n/\mu_+' = -6.8499694(16) \times 10^{-1}$	CODATA 2018, $\sigma = -0.41$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

electron-proton magnetic moment ratio

974.3959(29)

$$\frac{\mu_e}{\mu_+} = -\frac{S}{P_{up}} \left(\frac{m_+}{m_e} \right) \left(1 + \sqrt{7} \left(\frac{4x_{\infty}}{2K} \right) \mathfrak{K}_2 \mathfrak{K}_3 \mathfrak{K}_4 \boxtimes \right)$$

Where S = Sierpiński's constant, P_{up} = the universal parabolic constant, m_+ = the proton mass, m_e = the electron mass, x_{∞} = the 1st Foias constant, K = Catalan's constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_e/\mu_+ = -6.58210687490440 \dots \times 10^2$	prediction
$\mu_e/\mu_+ = -6.5821068789(19) \times 10^2$	CODATA 2022, $\sigma = +2.10$
$\mu_e/\mu_+ = -6.5821068789(20) \times 10^2$	CODATA 2018, $\sigma = -2.00$
	$\Delta_{\text{precision}} = +0.02$
	$\Delta_{\text{scaled}} = +0.00$

electron to shielded proton magnetic moment ratio

1231.146(41)

$$\frac{\mu_e}{\mu_+}' = -\frac{S}{P_{up}} \left(\frac{m_+}{m_e} \right) \left(1 + \sqrt{5} \left(\frac{2K}{18} \right) \mathfrak{K}_2^2 \mathfrak{K}_3^2 \mathfrak{K}_4^2 \boxtimes \right)$$

Where S = Sierpiński's constant, P_{up} = the universal parabolic constant, m_+ = the proton mass, m_e = the electron mass, K = Catalan's constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_e/\mu_+' = -6.58227587029051 \dots \times 10^2$	prediction
$\mu_e/\mu_+' = -6.582275856(27) \times 10^2$	CODATA 2022, $\sigma = -0.53$
$\mu_e/\mu_+' = -6.582275971(72) \times 10^2$	CODATA 2018, $\sigma = -1.40$
	$\Delta_{\text{precision}} = +0.42$
	$\Delta_{\text{scaled}} = -1.60$

muon-proton magnetic moment ratio

1036.95(23)

$$\frac{\mu_\mu}{\mu_+} = -\frac{S}{P_{up}} \left(\frac{m_+}{m_\mu} \right) \left(1 + \frac{\sqrt{35}}{2\pi} \left(\frac{4K}{18} \right) \mathfrak{K}_2^2 \mathfrak{K}_3^2 \mathfrak{K}_4^2 \boxtimes \right)$$

Where S = Sierpiński's constant, P_{up} = the universal parabolic constant, m_+ = the proton mass, m_μ = the muon mass, π = Archimedes' constant, K = Catalan's constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_\mu/\mu_+ = -3.18334511221316 \dots$	prediction
$\mu_\mu/\mu_+ = -3.183345146(71)$	CODATA 2022, $\sigma = +0.48$
$\mu_\mu/\mu_+ = -3.183345142(71)$	CODATA 2018, $\sigma = -0.42$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.06$

shielded proton magnetic moment to nuclear magneton ratio

-581.173(40)

$$\frac{\mu_+'}{\mu_N} = \frac{2 \mathfrak{x}_1 \mathfrak{x}_2^2}{S} \left(1 - \frac{1}{e^{6\gamma}} \left(\frac{2\pi}{2K} \right) \mathfrak{x}_2^2 \mathfrak{x}_3^2 \mathfrak{x}_4^2 \boxtimes \right)$$

Where $\mathfrak{x}_1$ = the 1st hyperbolic partition constant, $\mathfrak{x}_2$ = the 2nd hyperbolic partition constant, S = Sierpiński's constant, e = Euler's number, γ = the Euler-Mascheroni constant, π = Archimedes' constant, K = Catalan's constant, $\mathfrak{x}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{x}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_+'/\mu_N = 2.79277562234849 \dots$	prediction
$\mu_+'/\mu_N = 2.792775648(11)$	CODATA 2022, $\sigma = -2.33$
$\mu_+'/\mu_N = 2.792775599(30)$	CODATA 2018, $\sigma = +0.78$
	$\Delta_{\text{precision}} = +0.44$
	$\Delta_{\text{scaled}} = +1.63$

shielded proton magnetic moment to Bohr magneton ratio

-581.174(41)

$$\frac{\mu_+'}{\mu_B} = \frac{2 \mathfrak{x}_1 \mathfrak{x}_2^2}{S} \left(\frac{m_e}{m_+} \right) \left(1 - \frac{1}{e^{6\gamma}} \left(\frac{2\pi}{2K} \right) \mathfrak{x}_2^2 \mathfrak{x}_3^2 \mathfrak{x}_4^2 \boxtimes \right)$$

Where $\mathfrak{x}_1$ = the 1st hyperbolic partition constant, $\mathfrak{x}_2$ = the 2nd hyperbolic partition constant, S = Sierpiński's constant, m_e = the electron mass, m_+ = the proton mass, e = Euler's number, γ = the Euler-Mascheroni constant, π = Archimedes' constant, K = Catalan's constant, $\mathfrak{x}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{x}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_+'/\mu_B = 1.52099314114035 \dots \times 10^{-3}$	prediction
$\mu_+'/\mu_B = 1.5209931551(62) \times 10^{-3}$	CODATA 2022, $\sigma = -2.25$
$\mu_+'/\mu_B = 1.520993128(17) \times 10^{-3}$	CODATA 2018, $\sigma = +0.78$
	$\Delta_{\text{precision}} = +1.44$
	$\Delta_{\text{scaled}} = +1.59$

proton gyromagnetic ratio
 -324.098045(83)

$$\gamma_+ = \frac{2 \, \varkappa_1^2 \, \varkappa_2^2}{S} \left(\frac{q_p}{m_+} \right) \left(1 - \frac{1}{\textcolor{red}{5}} \left(\frac{2\pi}{4s} \right) \varkappa_2^2 \, \varkappa_3^2 \, \varkappa_4^2 \, \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, S = Sierpiński's constant, q_p = the Planck charge, m_+ = the proton mass, π = Archimedes' constant, s = the arc length of the unit lemniscate, $\varkappa_3$ = the 3rd hyperbolic partition constant, $\varkappa_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\gamma_+ = 2.67522187\textcolor{green}{170411} \dots \times 10^8 \text{ 1/s} \cdot \text{T}$	prediction
$\gamma_+ = 2.6752218708(11) \times 10^8 \text{ 1/s} \cdot \text{T}$	CODATA 2018, $\sigma = +0.82$
$\gamma_+ = 2.6752218744(11) \times 10^8 \text{ 1/s} \cdot \text{T}$	CODATA 2018, $\sigma = -2.45$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = +3.27$

proton gyromagnetic ratio in MHz/T
 -324.097998(83)

$$\dot{\gamma}_+ = \frac{\varkappa_1^2 \, \varkappa_2^2}{\pi \, S} \frac{1}{\text{MHz}} \left(\frac{s \, q_p}{m_+} \right) \left(1 - \frac{1}{\textcolor{red}{5}} \left(\frac{2\pi}{4s} \right) \varkappa_2^2 \, \varkappa_3^2 \, \varkappa_4^2 \, \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, π = Archimedes' constant, S = Sierpiński's constant, MHz = the megahertz, s = the second, q_p = the Planck charge, m_+ = the proton mass, π = Archimedes' constant, s = the arc length of the unit lemniscate, $\varkappa_3$ = the 3rd hyperbolic partition constant, $\varkappa_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\dot{\gamma}_+ = 4.25774784\textcolor{green}{749262} \dots \times 10^1 \text{ MHz/T}$	prediction
$\dot{\gamma}_+ = 4.2577478461(18) \times 10^1 \text{ MHz/T}$	CODATA 2022, $\sigma = +0.77$
$\dot{\gamma}_+ = 4.2577478518(18) \times 10^1 \text{ MHz/T}$	CODATA 2018, $\sigma = -2.39$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -3.17$

$$\mu_+ = \frac{\mathfrak{x}_1^2 \mathfrak{x}_2^2}{S} \left(\frac{l_p^2 q_p m_p}{t_p m_+} \right) \left(1 - \frac{\textcolor{blue}{7}}{\textcolor{red}{5}} \left(\frac{Im(\omega_1)}{6s} \right) \mathfrak{x}_2^2 \mathfrak{x}_3^2 \mathfrak{x}_4^2 \boxtimes \right)$$

Where $\mathfrak{x}_1$ = the 1st hyperbolic partition constant, $\mathfrak{x}_2$ = the 2nd hyperbolic partition constant, S = Sierpiński's constant, l_p = the Planck length, q_p = the Planck charge, m_p = the Planck mass, t_p = the Planck time, m_+ = the proton mass, ω_1 = the omega_1 constant, s = the arc length of the unit lemniscate, $\mathfrak{x}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{x}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_+ = 1.410606797\textcolor{green}{16311} \dots \times 10^{-26} \text{ J/T}$	prediction
$\mu_+ = 1.41060679545(60) \times 10^{-26} \text{ J/T}$	CODATA 2018, $\sigma = +2.86$
$\mu_+ = 1.41060679736(60) \times 10^{-26} \text{ J/T}$	CODATA 2018, $\sigma = -0.32$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -3.18$

Faraday constant

-283.17057(51)

$$F = \frac{6}{e^\gamma} \mathfrak{x}_1^3 \left(\frac{1}{m_p} \right) \left(1 - \left(\frac{2K}{35} \right) \mathfrak{x}_2^2 \mathfrak{x}_3^2 \mathfrak{x}_4^2 \boxtimes \right)$$

Where e = Euler’s number, γ = the Euler-Mascheroni constant, $\mathfrak{x}_1$ = the 1st hyperbolic partition constant, m_p = the Planck mass, K = Catalan’s constant, $\mathfrak{x}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{x}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{x}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$F = 9.64853322042857 \dots \times 10^4 \text{ C/mol}$
 $F = 9.648533212 \times 10^4 \text{ C/mol}$
 $F = 9.648533212 \times 10^4 \text{ C/mol}$

prediction

CODATA 2022, 9.06-digit match

CODATA 2018, 9.06-digit match

$\Delta_{\text{precision}} = +0.00$

$\Delta_{\text{scaled}} = +0.00$

molar Planck constant

-278.3994(13)

$$M_p = \frac{6}{e^\gamma} \mathfrak{x}_1^2 2\pi \left(\frac{l_p^2}{s^2 t_p q_p} \right) \left(1 - \frac{5}{\sqrt{32}} \left(\frac{2K}{6s} \right) \mathfrak{x}_2^2 \mathfrak{x}_3^2 \mathfrak{x}_4^2 \boxtimes \right)$$

Where e = Euler’s number, γ = the Euler-Mascheroni constant, π = Archimedes’ constant, $\mathfrak{x}_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, s = the second, t_p = the Planck time, q_p = the Planck charge, K = Catalan’s constant, s = the arc length of the unit lemniscate, $\mathfrak{x}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{x}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{x}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$M_p = 3.99031271062295 \dots \times 10^{-10} \text{ J/Hz mol}$
 $M_p = 3.990312712 \times 10^{-10} \text{ J/Hz mol}$
 $M_p = 3.990312712 \times 10^{-10} \text{ J/Hz mol}$

prediction

CODATA 2022, 9.46-digit match

CODATA 2018, 9.46-digit match

$\Delta_{\text{precision}} = +0.00$

$\Delta_{\text{scaled}} = +0.00$

$$V_m(Si) = \frac{6}{e^\gamma} \left(\frac{2\pi}{32\,s} \right)^3 \frac{1}{\mathfrak{K}_1^{10}} \left(\frac{l_p^3\,m_p^2}{q_p\,m_e^3} \right) \left(1 + \frac{32}{18^2} \left(\frac{4\pi}{D_d} \right) \mathfrak{K}_2^2\,\mathfrak{K}_3^2\,\mathfrak{K}_4^2\,\boxtimes \right)$$

Where e = Euler’s number, γ = the Euler-Mascheroni constant, π = Archimedes’ constant, s = the arc length of the unit lemniscate, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, m_p = the Planck mass, q_p = the Planck charge, m_e = the electron mass, D_d = the dimer constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$V_m(Si) = 1.20588317215772 \dots \times 10^{-5} \text{ m}^3/\text{mol}$

prediction

 $V_m(Si) = 1.205883199(60) \times 10^{-5} \text{ m}^3/\text{mol}$

CODATA 2018, $\sigma = -0.45$

 $V_m(Si) = 1.205883199(60) \times 10^{-5} \text{ m}^3/\text{mol}$

CODATA 2018, $\sigma = -0.45$

$\Delta_{\text{precision}} = +0.00$
 $\Delta_{\text{scaled}} = +0.00$

speed of light in vacuum

$$c = \left(\frac{l_p}{t_p} \right) \left(1 + \sqrt{G_{Gi}} (\mathfrak{K}_1 - \mathfrak{K}_2) \boxtimes \right)$$

Where l_p = the Planck length, t_p = the Planck time, G_{Gi} = Gieseeking's constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$c = 2.99792458097898 \dots \times 10^8 \text{ m/s}$	prediction
$c = 2.99792458 \times 10^8 \text{ m/s}$	CODATA 2022, 9-digit match
$c = 2.99792458 \times 10^8 \text{ m/s}$	CODATA 2018, 9-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

Also listed as the *natural unit of velocity*.

inverse meter-hertz relationship

-3.595(16)

$$\frac{1}{\text{m}} : \text{Hz} = \left(\frac{l_p}{t_p \text{ m}} \right) \left(1 + \sqrt{G_{Gi}} (\mathfrak{K}_1 - \mathfrak{K}_2) \boxtimes \right)$$

Where l_p = the Planck length, t_p = the Planck time, m = the meter, G_{Gi} = Gieseeking's constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$1/\text{m} : \text{Hz} = 2.99792458097898 \dots \times 10^8 \text{ Hz}$	prediction
$1/\text{m} : \text{Hz} = 2.99792458 \times 10^8 \text{ Hz}$	CODATA 2022, 9-digit match
$1/\text{m} : \text{Hz} = 2.99792458 \times 10^8 \text{ Hz}$	CODATA 2018, 9-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

classical electron radius

8.7468(46)

$$r_e = \kappa_1^2 \left(\frac{l_p m_p}{m_e} \right) \left(1 - \sqrt{6 C_{R1}} (\kappa_1 - \kappa_2) \boxtimes \right)$$

Where κ_1 = the 1st hyperbolic partition constant, l_p = the Planck length, m_p = the Planck mass, m_e = the electron mass, C_{R1} = Ramanujan's 1st continued fraction constant, κ_2 = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$r_e = 2.81794032699479 \dots \times 10^{-15} \text{ m}$	prediction
$r_e = 2.8179403205(13) \times 10^{-15} \text{ m}$	CODATA 2022, $\sigma = +5.00$
$r_e = 2.8179403262(13) \times 10^{-15} \text{ m}$	CODATA 2018, $\sigma = +0.61$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -4.38$

atomic unit of length

8.7701(15)

$$a_0 = \frac{1}{\kappa_1^2} \left(\frac{l_p m_p}{m_e} \right) \left(1 - \sqrt{6 C_{R1}} (\kappa_1 - \kappa_2) \boxtimes \right)$$

Where κ_1 = the 1st hyperbolic partition constant, l_p = the Planck length, m_p = the Planck mass, m_e = the electron mass, C_{R1} = Ramanujan's 1st continued fraction constant, κ_2 = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$a_0 = 5.29177210521101 \dots \times 10^{-11} \text{ m}$	prediction
$a_0 = 5.29177210544(82) \times 10^{-11} \text{ m}$	CODATA 2022, $\sigma = -0.28$
$a_0 = 5.29177210903(80) \times 10^{-11} \text{ m}$	CODATA 2018, $\sigma = -4.77$
	$\Delta_{\text{precision}} = -0.03$
	$\Delta_{\text{scaled}} = -4.48$

Also listed as the *Bohr radius*.

Thomson cross section

17.5080(90)

$$\sigma_e = \sqrt{\frac{32}{18}} 2\pi \kappa_1^4 \left(\frac{l_p m_p}{m_e} \right)^2 \left(1 - 2 \sqrt{6 C_{R1}} (\kappa_1 - \kappa_2) \boxtimes \right)$$

Where π = Archimedes' constant, κ_1 = the 1st hyperbolic partition constant, l_p = the Planck length, m_p = the Planck mass, m_e = the electron mass, C_{R1} = Ramanujan's 1st continued fraction constant, κ_2 = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\sigma_e = 6.6524587\textcolor{green}{3589771} \dots \times 10^{-29} \text{ m}^2$	prediction
$\sigma_e = 6.6524587051(62) \times 10^{-29} \text{ m}^2$	CODATA 2022, $\sigma = +4.97$
$\sigma_e = 6.6524587321(60) \times 10^{-29} \text{ m}^2$	CODATA 2018, $\sigma = +0.63$
	$\Delta_{\text{precision}} = -0.03$
	$\Delta_{\text{scaled}} = -4.50$

Fermi coupling constant

−243.6(51)

$$\frac{G_F}{(\hbar c)^3} = \sqrt{\frac{32}{18}} \left(2\pi \frac{\text{joule}}{\text{GeV}} \right)^2 \left(\frac{t_p^4}{l_p^4 m_p \text{ kg}} \right) \left(1 + 28 \sqrt{6 C_{R1}} (\kappa_1 - \kappa_2) \boxtimes \right)$$

Where π = Archimedes' constant, GeV = the gigaelectron volt, t_p = the Planck time, l_p = the Planck length, m_p = the Planck mass, kg = the kilogram, C_{R1} = Ramanujan's 1st continued fraction constant, κ_2 = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$G_F/(\hbar c)^3 = 1.1663785\textcolor{green}{2183353} \dots \times 10^{-5} \text{ GeV}^{-2}$	prediction
$G_F/(\hbar c)^3 = 1.1663787(06) \times 10^{-5} \text{ GeV}^{-2}$	CODATA 2022, $\sigma = -0.30$
$G_F/(\hbar c)^3 = 1.1663787(06) \times 10^{-5} \text{ GeV}^{-2}$	CODATA 2018, $\sigma = -0.30$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

atomic mass constant

-1.3664(31)

$$A_{\text{mass}} = \sqrt{\frac{18}{32} \frac{P_{up}}{\mathfrak{K}_1^6}} (m_e) \left(1 + \frac{1}{L} (\mathfrak{K}_1 - \mathfrak{K}_2) \boxtimes \right)$$

Where P_{up} = the universal parabolic constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, m_e = the electron mass, L = the lemniscate constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$A_{\text{mass}} = 1.66053906821322 \dots \times 10^{-27} \text{ kg}$	prediction
$A_{\text{mass}} = 1.66053906892(52) \times 10^{-27} \text{ kg}$	CODATA 2022, $\sigma = -1.36$
$A_{\text{mass}} = 1.66053906660(50) \times 10^{-27} \text{ kg}$	CODATA 2018, $\sigma = +3.23$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +4.64$

Also listed as the *atomic mass unit-kilogram relationship*, and the *unified atomic mass unit*.

kilogram-atomic mass unit relationship

1.3665(30)

$$\text{kg} : A_{\text{mass}} = \sqrt{\frac{32}{18} \frac{\mathfrak{K}_1^6}{P_{up}}} \left(\frac{\text{kg}}{m_e} \right) \left(1 - \frac{1}{L} (\mathfrak{K}_1 - \mathfrak{K}_2) \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, kg = the kilogram, m_e = the electron mass, L = the lemniscate constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\text{Kg} : A_{\text{mass}} = 6.02214075623046 \dots \times 10^{26} \text{ u}$	prediction
$\text{kg} : A_{\text{mass}} = 6.0221407537(19) \times 10^{26} \text{ u}$	CODATA 2022, $\sigma = +1.33$
$\text{kg} : A_{\text{mass}} = 6.0221407621(18) \times 10^{26} \text{ u}$	CODATA 2018, $\sigma = -3.26$
	$\Delta_{\text{precision}} = -0.03$
	$\Delta_{\text{scaled}} = -4.66$

electron relative atomic mass

1.36644(29)

$$A_r(e) = \sqrt{\frac{32}{18} \frac{\varkappa_1^6}{P_{up}}} \left(1 - \frac{1}{L} (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, L = the lemniscate constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$A_r(e) = 5.48579909049541 \dots \times 10^{-4}$	prediction
$A_r(e) = 5.485799090441(97) \times 10^{-4}$	CODATA 2022, $\sigma = +0.56$
$A_r(e) = 5.48579909065(16) \times 10^{-4}$	CODATA 2018, $\sigma = -0.97$
	$\Delta_{\text{precision}} = -0.78$
	$\Delta_{\text{scaled}} = 1.31$

proton relative atomic mass

1.36639(53)

$$A_r(+) = \sqrt{\frac{32}{18} \frac{\varkappa_1^6}{P_{up}}} \left(\frac{m_+}{m_e} \right) \left(1 - \frac{1}{L} (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, m_+ = the proton mass, m_e = the electron mass, L = the lemniscate constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$A_r(+) = 1.00727646658226 \dots$	prediction
$A_r(+) = 1.0072764665789(83)$	CODATA 2022, $\sigma = +0.41$
$A_r(+) = 1.007276466621(53)$	CODATA 2018, $\sigma = -0.73$
	$\Delta_{\text{precision}} = -0.20$
	$\Delta_{\text{scaled}} = -0.81$

neutron relative atomic mass

1.3653(48)

$$A_r(n) = \sqrt{\frac{32}{18} \frac{\varkappa_1^6}{P_{up}}} \left(\frac{m_n}{m_e} \right) \left(1 - \frac{1}{L} (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, m_n = the neutron mass, m_e = the electron mass, L = the lemniscate constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$A_r(n) = 1.00866491547289 \dots$$

prediction

$$A_r(n) = 1.00866491606(40)$$

CODATA 2022, $\sigma = -1.47$

$$A_r(n) = 1.00866491595(49)$$

CODATA 2018, $\sigma = -0.97$

$$\Delta_{\text{precision}} = +0.09$$

$$\Delta_{\text{scaled}} = +0.22$$

deuteron relative atomic mass

1.36674(20)

$$A_r(\text{de}) = \sqrt{\frac{32}{18} \frac{\varkappa_1^6}{P_{up}}} \left(\frac{m_{\text{de}}}{m_e} \right) \left(1 - \frac{1}{L} (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, m_{de} = the deuteron mass, m_e = the electron mass, L = the lemniscate constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$A_r(\text{de}) = 2.01355321254847 \dots$$

prediction

$$A_r(\text{de}) = 2.013553212544(15)$$

CODATA 2022, $\sigma = +0.32$

$$A_r(\text{de}) = 2.013553212745(40)$$

CODATA 2018, $\sigma = -4.91$

$$\Delta_{\text{precision}} = +0.43$$

$$\Delta_{\text{scaled}} = -5.03$$

helion relative atomic mass

1.36637(32)

$$A_r(\text{he}) = \sqrt{\frac{32}{18} \frac{\varkappa_1^6}{P_{up}}} \left(\frac{m_{\text{he}}}{m_e} \right) \left(1 - \frac{1}{L} (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, m_{he} = the helion mass, m_e = the electron mass, L = the lemniscate constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$A_r(\text{he}) = 3.01493224693\textcolor{brown}{074} \dots$$

prediction

$$A_r(\text{he}) = 3.014932246932(74)$$

CODATA 2022, $\sigma = -0.02$

$$A_r(\text{he}) = 3.014932247175(97)$$

CODATA 2018, $\sigma = -2.52$

$$\Delta_{\text{precision}} = +0.12$$

$$\Delta_{\text{scaled}} = -2.51$$

triton relative atomic mass

1.36615(40)

$$A_r(\text{tri}) = \sqrt{\frac{32}{18} \frac{\varkappa_1^6}{P_{up}}} \left(\frac{m_{\text{tri}}}{m_e} \right) \left(1 - \frac{1}{L} (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, m_{tri} = the triton mass, m_e = the electron mass, L = the lemniscate constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$A_r(\text{tri}) = 3.015500715\textcolor{brown}{72160} \dots$$

prediction

$$A_r(\text{tri}) = 3.01550071597(10)$$

CODATA 2022, $\sigma = -2.48$

$$A_r(\text{tri}) = 3.01550071621(12)$$

CODATA 2018, $\sigma = -4.07$

$$\Delta_{\text{precision}} = +0.08$$

$$\Delta_{\text{scaled}} = -2.00$$

alpha particle relative atomic mass

1.36577(15)

$$A_r(\alpha) = \sqrt{\frac{32}{18} \frac{\varkappa_1^6}{P_{up}}} \left(\frac{m_\alpha}{m_e} \right) \left(1 - \frac{1}{L} (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, m_α = the alpha particle mass, m_e = the electron mass, L = the lemniscate constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$A_r(\alpha) = 4.00150617912918 \dots$	prediction
$A_r(\alpha) = 4.001506179129(62)$	CODATA 2022, $\sigma = +0.00$
$A_r(\alpha) = 4.001506179127(63)$	CODATA 2018, $\sigma = +0.03$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.03$

electron mass in u

1.36644(29)

$$\frac{m_e}{A_{\text{mass}}} = \sqrt{\frac{32}{18} \frac{\varkappa_1^6}{P_{up}}} \left(1 - \frac{1}{L} (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where A_{mass} = atomic mass constant, $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, L = the lemniscate constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_e/A_{\text{mass}} = 5.48579909049541 \dots \times 10^{-4} \text{ u}$	prediction
$m_e/A_{\text{mass}} = 5.485799090441(97) \times 10^{-4} \text{ u}$	CODATA 2022, $\sigma = +0.56$
$m_e/A_{\text{mass}} = 5.48579909065(16) \times 10^{-4} \text{ u}$	CODATA 2018, $\sigma = -0.96$
	$\Delta_{\text{precision}} = +0.22$
	$\Delta_{\text{scaled}} = -1.25$

neutron-proton mass difference in u

0.63(3.52)

$$\frac{m_{\Delta}}{A_{\text{mass}}} = \sqrt{\frac{32}{18} \frac{\varkappa_1^6}{P_{up}}} \left(\frac{m_n - m_+}{m_e} \right) \left(1 - \frac{1}{L} (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, m_n = the neutron mass, m_+ = the proton mass, m_e = the electron mass, L = the lemniscate constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_{\Delta}/A_{\text{mass}} = 1.38844\textcolor{green}{889062885} \dots \times 10^{-3} \text{ u}$	prediction
$m_{\Delta}/A_{\text{mass}} = 1.38844948(40) \times 10^{-3} \text{ u}$	CODATA 2022, $\sigma = -1.40$
$m_{\Delta}/A_{\text{mass}} = 1.38844933(49) \times 10^{-3} \text{ u}$	CODATA 2018, $\sigma = -0.90$
	$\Delta_{\text{precision}} = +0.09$
	$\Delta_{\text{scaled}} = +0.31$

muon mass in u

1.40(22)

$$\frac{m_{\mu}}{A_{\text{mass}}} = \sqrt{\frac{32}{18} \frac{\varkappa_1^6}{P_{up}}} \left(\frac{m_{\mu}}{m_e} \right) \left(1 - \frac{1}{L} (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, m_{μ} = the muon mass, m_e = the electron mass, L = the lemniscate constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_{\mu}/A_{\text{mass}} = 1.1342892\textcolor{green}{5044824} \dots \times 10^{-1} \text{ u}$	prediction
$m_{\mu}/A_{\text{mass}} = 1.134289257(25) \times 10^{-1} \text{ u}$	CODATA 2022, $\sigma = -0.26$
$m_{\mu}/A_{\text{mass}} = 1.134289259(25) \times 10^{-1} \text{ u}$	CODATA 2018, $\sigma = -0.34$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -0.08$

proton mass in u

1.36639(53)

$$\frac{m_+}{A_{\text{mass}}} = \sqrt{\frac{32}{18} \frac{\varkappa_1^6}{P_{up}}} \left(\frac{m_+}{m_e} \right) \left(1 - \frac{1}{L} (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, m_+ = the proton mass, m_e = the electron mass, L = the lemniscate constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_+/A_{\text{mass}} = 1.0072764665$	8226 ... u	prediction
$m_+/A_{\text{mass}} = 1.0072764665789(83)$	u	CODATA 2022, $\sigma = +0.41$
$m_+/A_{\text{mass}} = 1.007276466621(53)$	u	CODATA 2018, $\sigma = -0.73$
		$\Delta_{\text{precision}} = +0.81$
		$\Delta_{\text{scaled}} = -2.06$

neutron mass in u

1.3653(48)

$$\frac{m_n}{A_{\text{mass}}} = \sqrt{\frac{32}{18} \frac{\varkappa_1^6}{P_{up}}} \left(\frac{m_n}{m_e} \right) \left(1 - \frac{1}{L} (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, m_n = the neutron mass, m_e = the electron mass, L = the lemniscate constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_n/A_{\text{mass}} = 1.00866491$	547289 ... u	prediction
$m_n/A_{\text{mass}} = 1.00866491606(40)$	u	CODATA 2022, $\sigma = -1.47$
$m_n/A_{\text{mass}} = 1.00866491595(49)$	u	CODATA 2018, $\sigma = -0.97$
		$\Delta_{\text{precision}} = +0.09$
		$\Delta_{\text{scaled}} = +0.22$

tau mass in u

-3.0(678)

$$\frac{m_{\tau}}{A_{\text{mass}}} = \sqrt{\frac{32}{18} \frac{\varkappa_1^6}{P_{up}}} \left(\frac{m_{\tau}}{m_e} \right) \left(1 - \frac{1}{L} (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, m_{τ} = the tau mass, m_e = the electron mass, L = the lemniscate constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_{\tau}/A_{\text{mass}} = 1.90754046521642 \dots \text{ u}$	prediction
$m_{\tau}/A_{\text{mass}} = 1.90754(13) \text{ u}$	CODATA 2022, $\sigma = +0.00$
$m_{\tau}/A_{\text{mass}} = 1.90754(13) \text{ u}$	CODATA 2018, $\sigma = +0.00$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

deuteron mass in u

1.36674(20)

$$\frac{m_{\text{de}}}{A_{\text{mass}}} = \sqrt{\frac{32}{18} \frac{\varkappa_1^6}{P_{up}}} \left(\frac{m_{\text{de}}}{m_e} \right) \left(1 - \frac{1}{L} (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, m_{de} = the deuteron mass, m_e = the electron mass, L = the lemniscate constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_{\text{de}}/A_{\text{mass}} = 2.01355321254878 \dots \text{ u}$	prediction
$m_{\text{de}}/A_{\text{mass}} = 2.013553212544(15) \text{ u}$	CODATA 2022, $\sigma = +0.32$
$m_{\text{de}}/A_{\text{mass}} = 2.013553212745(40) \text{ u}$	CODATA 2018, $\sigma = -4.91$
	$\Delta_{\text{precision}} = +0.43$
	$\Delta_{\text{scaled}} = -5.03$

helion mass in u

1.36637(32)

$$\frac{m_{\text{he}}}{A_{\text{mass}}} = \sqrt{\frac{32}{18} \frac{\varkappa_1^6}{P_{up}}} \left(\frac{m_{\text{he}}}{m_e} \right) \left(1 - \frac{1}{L} (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, m_{he} = the helion mass, m_e = the electron mass, L = the lemniscate constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_{\text{he}}/A_{\text{mass}} = 3.01493224693\textcolor{green}{074} \dots \text{ u}$	prediction
$m_{\text{he}}/A_{\text{mass}} = 3.014932246932(74) \text{ u}$	CODATA 2022, $\sigma = -0.02$
$m_{\text{he}}/A_{\text{mass}} = 3.014932247175(97) \text{ u}$	CODATA 2018, $\sigma = -2.52$
	$\Delta_{\text{precision}} = +0.12$
	$\Delta_{\text{scaled}} = -2.51$

triton mass in u

1.36715(40)

$$\frac{m_{\text{tri}}}{A_{\text{mass}}} = \sqrt{\frac{32}{18} \frac{\varkappa_1^6}{P_{up}}} \left(\frac{m_{\text{tri}}}{m_e} \right) \left(1 - \frac{1}{L} (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, m_{tri} = the triton mass, m_e = the electron mass, L = the lemniscate constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_{\text{tri}}/A_{\text{mass}} = 3.015500715\textcolor{green}{72160} \dots \text{ u}$	prediction
$m_{\text{tri}}/A_{\text{mass}} = 3.01550071597(10) \text{ u}$	CODATA 2022, $\sigma = -2.48$
$m_{\text{tri}}/A_{\text{mass}} = 3.01550071621(12) \text{ u}$	CODATA 2018, $\sigma = -4.15$
	$\Delta_{\text{precision}} = +0.08$
	$\Delta_{\text{scaled}} = -2.00$

$$\frac{m_{\alpha}}{A_{\text{mass}}} = \sqrt{\frac{32}{18} \frac{\varkappa_1^6}{P_{up}}} \left(\frac{m_{\alpha}}{m_e} \right) \left(1 - \frac{1}{L} (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, m_{α} = the alpha particle mass, m_e = the electron mass, L = the lemniscate constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_{\alpha}/A_{\text{mass}} = 4.001506179129\textcolor{brown}{18} \dots \text{ u}$	prediction
$m_{\alpha}/A_{\text{mass}} = 4.001506179129(62) \text{ u}$	CODATA 2022, $\sigma = +0.00$
$m_{\alpha}/A_{\text{mass}} = 4.001506179127(63) \text{ u}$	CODATA 2018, $\sigma = +0.03$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.03$

electron volt-atomic mass unit relationship

8.9646(30)

$$\text{eV} : A_{\text{mass}} = \text{joule}^2 \sqrt{\frac{32}{18} \frac{\varkappa_1^8}{P_{up}}} \left(\frac{t_p^2 q_p}{l_p^2 C m_e} \right) \left(1 - \alpha_F (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, t_p = the Planck time, q_p = the Planck charge, l_p = the Planck length, C = the coulomb, m_e = the electron mass, α_F = the alpha Feigenbaum constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} \text{eV} : A_{\text{mass}} &= 1.07354410\mathbf{142667} \dots \times 10^{-9} \text{ u} && \text{prediction} \\ \text{eV} : A_{\text{mass}} &= 1.07354410083(33) \times 10^{-9} \text{ u} && \text{CODATA 2022, } \sigma = +1.75 \\ \text{eV} : A_{\text{mass}} &= 1.07354410233(32) \times 10^{-9} \text{ u} && \text{CODATA 2018, } \sigma = -2.82 \\ &&& \Delta_{\text{precision}} = -0.03 \\ &&& \Delta_{\text{scaled}} = -4.68 \end{aligned}$$

atomic mass unit-electron volt relationship

-8.9646(30)

$$A_{\text{mass}} : \text{eV} = \sqrt{\frac{18}{32} \frac{P_{up}}{\varkappa_1^8}} \left(\frac{l_p^2 C m_e}{t_p^2 q_p} \right) \left(1 + \alpha_F (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where P_{up} = the universal parabolic constant, $\varkappa_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, C = the coulomb, m_e = the electron mass, t_p = the Planck time, q_p = the Planck charge, α_F = the alpha Feigenbaum constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} A_{\text{mass}} : \text{eV} &= 9.31494103\mathbf{195445} \dots \times 10^8 \text{ eV} && \text{prediction} \\ A_{\text{mass}} : \text{eV} &= 9.3149410372(29) \times 10^8 \text{ eV} && \text{CODATA 2022, } \sigma = -1.81 \\ A_{\text{mass}} : \text{eV} &= 9.3149410242(28) \times 10^8 \text{ eV} && \text{CODATA 2018, } \sigma = +2.77 \\ &&& \Delta_{\text{precision}} = -0.04 \\ &&& \Delta_{\text{scaled}} = +4.64 \end{aligned}$$

joule-inverse meter relationship

-1.5349(10)

$$J : \frac{1}{m} = \frac{\text{joule}}{2\pi} \left(\frac{t_p^2}{l_p^3 m_p} \right) \left(1 + \frac{2}{\delta_F} (\mathfrak{K}_1 - \mathfrak{K}_2) \boxtimes \right)$$

Where π = Archimedes' constant, t_p = the Planck time, l_p = the Planck length, m_p = the Planck mass, δ_F = the delta Feigenbaum constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} J : 1/m &= 5.03411656778384 \dots \times 10^{24} \text{ cycles/m} && \text{prediction} \\ J : 1/m &= 5.034116567 \times 10^{24} \text{ cycles/m} && \text{CODATA 2022, 10-digit match} \\ J : 1/m &= 5.034116567 \times 10^{24} \text{ cycles/m} && \text{CODATA 2018, 10-digit match} \\ &&& \Delta_{\text{precision}} = +0.00 \\ &&& \Delta_{\text{scaled}} = +0.00 \end{aligned}$$

electron volt-inverse meter relationship

-1.53673(62)

$$\text{eV} : \frac{1}{m} = \frac{\text{eV}}{2\pi} \left(\frac{t_p^2}{l_p^3 m_p} \right) \left(1 + \frac{2}{\delta_F} (\mathfrak{K}_1 - \mathfrak{K}_2) \boxtimes \right)$$

Where eV = the electron-volt, π = Archimedes' constant, t_p = the Planck time, l_p = the Planck length, m_p = the Planck mass, δ_F = the delta Feigenbaum constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} \text{eV} : 1/m &= 8.06554393884387 \dots \times 10^5 \text{ cycles/m} && \text{prediction} \\ \text{eV} : 1/m &= 8.065543937 \times 10^5 \text{ cycles/m} && \text{CODATA 2022, 9.64-digit match} \\ \text{eV} : 1/m &= 8.065543937 \times 10^5 \text{ cycles/m} && \text{CODATA 2018, 9.64-digit match} \\ &&& \Delta_{\text{precision}} = +0.00 \\ &&& \Delta_{\text{scaled}} = +0.00 \end{aligned}$$

inverse meter-electron volt relationship

1.5376(41)

$$\frac{1}{\text{m}} : \text{eV} = 2\pi \frac{\text{joule}}{\text{eV}} \left(\frac{l_p^3 m_p}{t_p^2 \text{m}} \right) \left(1 - \frac{2}{\delta_F} (\mathfrak{K}_1 - \mathfrak{K}_2) \boxtimes \right)$$

Where π = Archimedes' constant, eV = the electron-volt, l_p = the Planck length, m_p = the Planck mass, t_p = the Planck time, m = the meter, δ_F = the delta Feigenbaum constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$1/\text{m} : \text{eV} = 1.239841984\textcolor{brown}{10221} \dots \times 10^{-6} \text{ eV}$	prediction
$1/\text{m} : \text{eV} = 1.239841984 \times 10^{-6} \text{ eV}$	CODATA 2022, 10-digit match
$1/\text{m} : \text{eV} = 1.239841984 \times 10^{-6} \text{ eV}$	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

inverse meter-joule relationship

1.5349(10)

$$\frac{1}{\text{m}} : \text{J} = 2\pi \left(\frac{l_p^3 m_p}{t_p^2 \text{m}} \right) \left(1 - \frac{2}{\delta_F} (\mathfrak{K}_1 - \mathfrak{K}_2) \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, m_p = the Planck mass, t_p = the Planck time, m = the meter, δ_F = the delta Feigenbaum constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$1/\text{m} : \text{J} = 1.986445857\textcolor{brown}{05373} \dots \times 10^{-25} \text{ J}$	prediction
$1/\text{m} : \text{J} = 1.986445857 \times 10^{-25} \text{ J}$	CODATA 2022, 10-digit match
$1/\text{m} : \text{J} = 1.986445857 \times 10^{-25} \text{ J}$	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

reduced Planck constant times c in MeV fm
1.5345(13)

$$\hbar c = \frac{\text{joule}}{\text{MeV fm}} \left(\frac{l_p^3 \text{ m } m_p}{t_p^2} \right) \left(1 - \frac{2}{\delta_F} (\varkappa_1 - \varkappa_2) \boxtimes \right)$$

Where MeV = the megaelectron-volt, fm = the femtometer, l_p = the Planck length, m = the meter, m_p = the Planck mass, t_p = the Planck time, δ_F = the delta Feigenbaum constant, $\varkappa_1$ = the 1st hyperbolic partition constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\hbar c = 1.97326980422731 \dots \times 10^2 \text{ MeV fm}$
prediction

$\hbar c = 1.973269804 \times 10^2 \text{ MeV fm}$
CODATA 2022, 10-digit match

$\hbar c = 1.973269804 \times 10^2 \text{ MeV fm}$
CODATA 2018, 10-digit match

$\Delta_{\text{precision}} = +0.00$
 $\Delta_{\text{scaled}} = +0.00$

atomic unit of electric potential

-6.251583(11)

$$A_{\text{e pot}} = \sqrt{\frac{32}{18} \frac{\varkappa_1^{12}}{P_{up}}} \left(\frac{l_p^2 A_{\text{mass}}}{t_p^2 q_p} \right) \left(1 - \frac{35}{18} \left(\frac{4}{\delta_F} \right) (\varkappa_1 + \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, l_p = the Planck length, A_{mass} = the atomic mass constant, t_p = the Planck time, q_p = the Planck charge, δ_F = the delta Feigenbaum constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$A_{\text{e pot}} = 2.72113862459\mathbf{609} \dots \times 10^1 \text{ V} \quad \text{prediction}$$

$$A_{\text{e pot}} = 2.7211386245981(30) \times 10^1 \text{ V} \quad \text{CODATA 2022, } \sigma = -0.67$$

$$A_{\text{e pot}} = 2.7211386245988(53) \times 10^1 \text{ V} \quad \text{CODATA 2018, } \sigma = -0.51$$

$$\Delta_{\text{precision}} = +0.25$$

$$\Delta_{\text{scaled}} = -0.13$$

Rydberg constant times hc in eV

-6.251583(11)

$$R_{\infty} hc = \frac{1}{2} \sqrt{\frac{32}{18} \frac{\varkappa_1^{12}}{P_{up}}} \left(\frac{l_p^2 C A_{\text{mass}}}{t_p^2 q_p} \right) \left(1 - \frac{35}{18} \left(\frac{4}{\delta_F} \right) (\varkappa_1 + \varkappa_2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, l_p = the Planck length, C = the coulomb, A_{mass} = the atomic mass constant, t_p = the Planck time, q_p = the Planck charge, δ_F = the delta Feigenbaum constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$R_{\infty} hc = 1.36056931229\mathbf{804} \dots \times 10^1 \text{ eV} \quad \text{prediction}$$

$$R_{\infty} hc = 1.3605693122990(15) \times 10^1 \text{ eV} \quad \text{CODATA 2022, } \sigma = -0.64$$

$$R_{\infty} hc = 1.3605693122994(26) \times 10^1 \text{ eV} \quad \text{CODATA 2018, } \sigma = -0.52$$

$$\Delta_{\text{precision}} = +0.24$$

$$\Delta_{\text{scaled}} = -0.15$$

hartree-electron volt relationship

−6.251583(11)

$$E_h : \text{eV} = \sqrt{\frac{32}{18} \frac{\mathfrak{K}_1^{12}}{P_{up}}} \left(\frac{l_p^2 C A_{\text{mass}}}{t_p^2 q_p} \right) \left(1 - \frac{35}{18} \left(\frac{4}{\delta_F} \right) (\mathfrak{K}_1 + \mathfrak{K}_2) \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, l_p = the Planck length, C = the coulomb, A_{mass} = the atomic mass constant, t_p = the Planck time, q_p = the Planck charge, δ_F = the delta Feigenbaum constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_h : \text{eV} = 2.72113862459\textcolor{teal}{609} \dots \times 10^1 \text{ eV}$	prediction
$E_h : \text{eV} = 2.7211386245981(30) \times 10^1 \text{ eV}$	CODATA 2022, $\sigma = -0.67$
$E_h : \text{eV} = 2.7211386245988(53) \times 10^1 \text{ eV}$	CODATA 2018, $\sigma = -0.51$
	$\Delta_{\text{precision}} = +0.25$
	$\Delta_{\text{scaled}} = -0.13$

Also listed as the *Hartree energy in eV*.

electron volt-hartree relationship

6.251583(11)

$$\text{eV} : E_h = \sqrt{\frac{18}{32} \frac{P_{up}}{\mathfrak{K}_1^{12}}} \left(\frac{t_p^2 q_p}{l_p^2 C A_{\text{mass}}} \right) \left(1 + \frac{35}{18} \left(\frac{4}{\delta_F} \right) (\mathfrak{K}_1 + \mathfrak{K}_2) \boxtimes \right)$$

Where P_{up} = the universal parabolic constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, t_p = the Planck time, q_p = the Planck charge, l_p = the Planck length, C = the coulomb, A_{mass} = the atomic mass constant, δ_F = the delta Feigenbaum constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\text{eV} : E_h = 3.67493221756\textcolor{teal}{773} \dots \times 10^{-2} E_h$	prediction
$\text{eV} : E_h = 3.6749322175665(40) \times 10^{-2} E_h$	CODATA 2022, $\sigma = +0.31$
$\text{eV} : E_h = 3.6749322175655(71) \times 10^{-2} E_h$	CODATA 2018, $\sigma = +0.31$
	$\Delta_{\text{precision}} = +0.25$
	$\Delta_{\text{scaled}} = +0.14$

conventional value of farad-90

-0.177949(50)

$$F_{90} = \text{farad} \left(1 + \frac{C_d}{6} (\mathfrak{K}_1 + \mathfrak{K}_3 + \mathfrak{K}_4) \boxtimes \right)$$

Where C_d = the domino tiling constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$F_{90} = 0.999999982178026 \dots$	F	prediction
$F_{90} = 0.99999998220$	F	CODATA 2022, 9.66-digit match
$F_{90} = 0.99999998220$	F	CODATA 2018, 9.66-digit match
		$\Delta_{\text{precision}} = +0.00$
		$\Delta_{\text{scaled}} = +0.00$

conventional value of henry-90

0.177949(50)

$$H_{90} = \text{henry} \left(1 - \frac{C_d}{6} (\mathfrak{K}_1 + \mathfrak{K}_3 + \mathfrak{K}_4) \boxtimes \right)$$

Where C_d = the domino tiling constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$H_{90} = 1.00000001782197 \dots$	H	prediction
$H_{90} = 1.00000001779$	H	CODATA 2022, 10.50-digit match
$H_{90} = 1.00000001779$	H	CODATA 2018, 10.50-digit match
		$\Delta_{\text{precision}} = +0.00$
		$\Delta_{\text{scaled}} = +0.00$

conventional value of ohm-90

0.177949(50)

$$\Omega_{90} = \text{ohm} \left(1 - \frac{C_d}{6} (\mathfrak{K}_1 + \mathfrak{K}_3 + \mathfrak{K}_4) \boxtimes \right)$$

Where C_d = the domino tiling constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\Omega_{90} = 1.00000001782197 \dots \Omega$	prediction
$\Omega_{90} = 1.00000001779 \Omega$	CODATA 2022, 10.50-digit match
$\Omega_{90} = 1.00000001779 \Omega$	CODATA 2018, 10.50-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

neutron-proton mass difference

−7.3(29)

$$m_{\Delta} = (m_n - m_+) \left(1 - 35 \frac{C_d}{6} (\mathfrak{K}_1 + \mathfrak{K}_3 + \mathfrak{K}_4) \boxtimes \right)$$

Where m_n = the neutron mass, m_+ = the proton mass, C_d = the domino tiling constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constants, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$m_{\Delta} = 2.30557506525214 \dots \times 10^{-30} \text{ kg}$	prediction
$m_{\Delta} = 2.30557461(67) \times 10^{-30} \text{ kg}$	CODATA 2022, $\sigma = +0.68$
$m_{\Delta} = 2.30557435(82) \times 10^{-30} \text{ kg}$	CODATA 2018, $\sigma = -2.86$
	$\Delta_{\text{precision}} = +0.09$
	$\Delta_{\text{scaled}} = +0.32$

helion shielding shift

-599.67077(23)

$$\sigma_{\text{he}} = 1 - \left(1 - \sqrt{\frac{7 s^4}{\mathcal{P}^8 (2\pi)^3} (\mathfrak{K}_1 - \mathfrak{K}_3 - \mathfrak{K}_4) \boxtimes} \right)$$

Where $\mathcal{P}$ = the prime constant, s = the arc length of the unit lemniscate, π = Archimedes' constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\sigma_{\text{he}} = 5.99670299153336 \dots \times 10^{-5}$	prediction
$\sigma_{\text{he}} = 5.9967029(23) \times 10^{-5}$	CODATA 2022, $\sigma = +0.04$
$\sigma_{\text{he}} = 5.996743(10) \times 10^{-5}$	CODATA 2018, $\sigma = -4.00$
	$\Delta_{\text{precision}} = +0.64$
	$\Delta_{\text{scaled}} = -4.10$

shielding difference of t and p in HT

-0.239450(20)

$$\sigma_{\text{tp}} = 1 - \left(1 - \frac{1}{7} \sqrt{\frac{s^4}{2^4 (2\pi)^3} (\mathfrak{K}_1 - \mathfrak{K}_3 - \mathfrak{K}_4) \boxtimes} \right)$$

Where s = the arc length of the unit lemniscate, π = Archimedes' constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\sigma_{\text{tp}} = 2.39369251464032 \dots \times 10^{-8}$	prediction
$\sigma_{\text{tp}} = 2.39450(20) \times 10^{-8}$	CODATA 2022, $\sigma = -4.04$
$\sigma_{\text{tp}} = 2.4140(20) \times 10^{-8}$	CODATA 2018, $\sigma = -10.15$
	$\Delta_{\text{precision}} = +1.00$
	$\Delta_{\text{scaled}} = -9.75$

shielding difference of d and p in HD

−0.198770(10)

$$\sigma_{dp} = 1 - \left(1 - \frac{1}{3} \sqrt{\frac{\textcolor{red}{5} s^4}{2^4 (2\pi)^{\textcolor{red}{5}}}} (\kappa_1 - \kappa_3 - \kappa_4) \boxtimes \right)$$

Where s = the arc length of the unit lemniscate, π = Archimedes' constant, κ_1 = the 1st hyperbolic partition constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\sigma_{dp} = 1.987\textcolor{green}{69745110889} \dots \times 10^{-8}$	prediction
$\sigma_{dp} = 1.98770(10) \times 10^{-8}$	CODATA 2022, $\sigma = -0.03$
$\sigma_{dp} = 2.0200(20) \times 10^{-8}$	CODATA 2018, $\sigma = -16.15$
	$\Delta_{\text{precision}} = +1.30$
	$\Delta_{\text{scaled}} = -16.15$

proton magnetic shielding correction

−256.715(41)

$$\sigma_+' = 1 - \left(1 - 35 \sqrt{\frac{s^8}{2^4 (2\pi)^{\textcolor{red}{5}}}} (\kappa_1 - \kappa_3 - \kappa_4) \boxtimes \right)$$

Where s = the arc length of the unit lemniscate, π = Archimedes' constant, κ_1 = the 1st hyperbolic partition constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\sigma_+' = 2.56\textcolor{green}{684124759787} \dots \times 10^{-5}$	prediction
$\sigma_+' = 2.56715(41) \times 10^{-5}$	CODATA 2022, $\sigma = -0.75$
$\sigma_+' = 2.5689(11) \times 10^{-5}$	CODATA 2018, $\sigma = -1.87$
	$\Delta_{\text{precision}} = +0.43$
	$\Delta_{\text{scaled}} = -1.64$

atomic unit of electric polarizability

25.5252(31)

$$A_{\text{ep}} = \frac{1}{\varkappa_1^6} \left(\frac{t_p^2 q_p^2 m_p^2}{m_e^3} \right) \left(1 - 18 \sqrt{\frac{s^9}{2^{11} (2\pi)^5}} (\varkappa_1 + \varkappa_3 + \varkappa_4) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, t_p = the Planck time, q_p = the Planck charge, m_p = the Planck mass, m_e = the electron mass, L = the lemniscate constant, π = Archimedes' constant, $\varkappa_3$ = the 3rd hyperbolic partition constant, $\varkappa_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} A_{\text{ep}} &= 1.64877727300424 \dots \times 10^{-41} \text{ m}^2 \cdot \text{C}^2 / \text{J} && \text{prediction} \\ A_{\text{ep}} &= 1.64877727212(51) \times 10^{-41} \text{ m}^2 \cdot \text{C}^2 / \text{J} && \text{CODATA 2018, } \sigma = +1.73 \\ A_{\text{ep}} &= 1.64877727436(50) \times 10^{-41} \text{ m}^2 \cdot \text{C}^2 / \text{J} && \text{CODATA 2018, } \sigma = -2.71 \\ &&& \Delta_{\text{precision}} = -0.01 \\ &&& \Delta_{\text{scaled}} = -4.48 \end{aligned}$$

atomic unit of electric field

-16.3913(16)

$$A_{\text{ef}} = \varkappa_1^5 \left(\frac{l_p m_e^2}{t_p^2 q_p m_p} \right) \left(1 + 18 \sqrt{\frac{s^8}{24 (2\pi)^7}} (\varkappa_1 + \varkappa_3 + \varkappa_4) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, m_e = the electron mass, t_p = the Planck time, q_p = the Planck charge, m_p = the Planck mass, s = the arc length of the unit lemniscate, π = Archimedes' constant, $\varkappa_3$ = the 3rd hyperbolic partition constant, $\varkappa_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} A_{\text{ef}} &= 5.14220674774742 \dots \times 10^{11} \text{ V/m} && \text{prediction} \\ A_{\text{ef}} &= 5.14220675112(80) \times 10^{11} \text{ V/m} && \text{CODATA 2022, } \sigma = -4.22 \\ A_{\text{ef}} &= 5.14220674763(78) \times 10^{11} \text{ V/m} && \text{CODATA 2018, } \sigma = +0.15 \\ &&& \Delta_{\text{precision}} = -0.01 \\ &&& \Delta_{\text{scaled}} = +4.47 \end{aligned}$$

atomic unit of force

-16.0172(16)

$$A_{\text{force}} = \kappa_1^6 \left(\frac{l_p m_e^2}{t_p^2 m_p} \right) \left(1 + 18 \sqrt{\frac{s^8}{4 (2\pi)^8}} (\kappa_1 + \kappa_3 + \kappa_4) \boxtimes \right)$$

Where κ_1 = the 1st hyperbolic partition constant, l_p = the Planck length, m_e = the electron mass, t_p = the Planck time, m_p = the Planck mass, s = the arc length of the unit lemniscate, π = Archimedes' constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$A_{\text{force}} = 8.23872349803771 \dots \times 10^{-8} \text{ N}$	prediction
$A_{\text{force}} = 8.2387235038(13) \times 10^{-8} \text{ N}$	CODATA 2022, $\sigma = -4.43$
$A_{\text{force}} = 8.2387234983(12) \times 10^{-8} \text{ N}$	CODATA 2018, $\sigma = -0.22$
	$\Delta_{\text{precision}} = -0.04$
	$\Delta_{\text{scaled}} = +4.58$

atomic unit of charge density

-25.9362(45)

$$A_{\text{cd}} = \kappa_1^7 \left(\frac{q_p m_e^3}{l_p^3 m_p^3} \right) \left(1 + 18 \sqrt{\frac{s^9}{8 (2\pi)^8}} (\kappa_1 + \kappa_3 + \kappa_4) \boxtimes \right)$$

Where κ_1 = the 1st hyperbolic partition constant, q_p = the Planck charge, m_e = the electron mass, l_p = the Planck length, m_p = the Planck mass, s = the arc length of the unit lemniscate, π = Archimedes' constant, κ_1 = the 1st hyperbolic partition constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$A_{\text{cd}} = 1.08120238561797 \dots \times 10^{12} \text{ C/m}^3$	prediction
$A_{\text{cd}} = 1.08120238677(51) \times 10^{12} \text{ C/m}^3$	CODATA 2022, $\sigma = -2.26$
$A_{\text{cd}} = 1.08120238457(49) \times 10^{12} \text{ C/m}^3$	CODATA 2018, $\sigma = +2.13$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = +4.49$

neutron magnetic moment

441.5(25)

$$\mu_n = -\frac{\varkappa_1^2 \varkappa_2^2}{C_{\text{CFP}}} \left(\frac{l_p^2 q_p m_p}{t_p m_n} \right) \left(1 + 18 (2\pi) (\varkappa_1 - \varkappa_3 - \varkappa_4) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, C_{CFP} = the real fixed point of the hyperbolic cotangent, l_p = the Planck length, q_p = the Planck charge, m_p = the Planck mass, t_p = the Planck time, m_n = the neutron mass, π = Archimedes' constant, $\varkappa_3$ = the 3rd hyperbolic partition constant, $\varkappa_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_n = -9.66235818798183 \dots \times 10^{-27} \text{ J/T}$	prediction
$\mu_n = -9.6623653(23) \times 10^{-27} \text{ J/T}$	CODATA 2022, $\sigma = +3.09$
$\mu_n = -9.6623651(23) \times 10^{-27} \text{ J/T}$	CODATA 2018, $\sigma = +3.01$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.09$

neutron gyromagnetic ratio in MHz/T

436.1(24)

$$\dot{\gamma}_n = \frac{\varkappa_1^2 \varkappa_2^2}{\pi C_{\text{CFP}}} \frac{1}{\text{MHz}} \left(\frac{q_p}{s m_n} \right) \left(1 + 18 (2\pi) (\varkappa_1 - \varkappa_3 - \varkappa_4) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, π = Archimedes' constant, C_{CFP} = the fixed point of the hyperbolic cotangent, MHz = the megahertz, q_p = the Planck charge, s = the second, m_n = the neutron mass, $\varkappa_3$ = the 3rd hyperbolic partition constant, $\varkappa_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\dot{\gamma}_n = 2.91646871897931 \dots \times 10^1 \text{ MHz/T}$	prediction
$\dot{\gamma}_n = 2.91646935(69) \times 10^1 \text{ MHz/T}$	CODATA 2022, $\sigma = -0.91$
$\dot{\gamma}_n = 2.91646931(69) \times 10^1 \text{ MHz/T}$	CODATA 2018, $\sigma = -0.86$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.06$

neutron gyromagnetic ratio

436.1(24)

$$\gamma_n = \frac{2 \kappa_1^2 \kappa_2^2}{C_{\text{CFP}}} \left(\frac{q_p}{m_n} \right) \left(1 + 18 (2\pi) (\kappa_1 - \kappa_3 - \kappa_4) \boxtimes \right)$$

Where κ_1 = the 1st hyperbolic partition constant, κ_2 = the 2nd hyperbolic partition constant, C_{CFP} = the fixed point of the hyperbolic cotangent, q_p = the Planck charge, m_n = the neutron mass, π = Archimedes' constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\gamma_n = 1.83247134039396 \dots \times 10^8 \text{ 1/s} \cdot \text{T}$	prediction
$\gamma_n = 1.83247174(43) \times 10^8 \text{ 1/s} \cdot \text{T}$	CODATA 2022, $\sigma = -0.93$
$\gamma_n = 1.83247171(43) \times 10^8 \text{ 1/s} \cdot \text{T}$	CODATA 2018, $\sigma = -0.86$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.07$

neutron magnetic moment to Bohr magneton ratio

435.7(24)

$$\frac{\mu_n}{\mu_B} = -\frac{2 \kappa_1 \kappa_2^2}{C_{\text{CFP}}} \left(\frac{m_e}{m_n} \right) \left(1 + 18 (2\pi) (\kappa_1 - \kappa_3 - \kappa_4) \boxtimes \right)$$

Where κ_1 = the 1st hyperbolic partition constant, κ_2 = the 2nd hyperbolic partition constant, C_{CFP} = the fixed point of the hyperbolic cotangent, m_e = the electron mass, m_n = the neutron mass, π = Archimedes' constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_n/\mu_B = -1.04187546165272 \dots \times 10^{-3}$	prediction
$\mu_n/\mu_B = -1.04187565(25) \times 10^{-3}$	CODATA 2022, $\sigma = +0.75$
$\mu_n/\mu_B = -1.04187563(25) \times 10^{-3}$	CODATA 2018, $\sigma = +0.67$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.08$

neutron magnetic moment to nuclear magneton ratio

435.7(23)

$$\frac{\mu_n}{\mu_N} = -\frac{2 \mathfrak{K}_1 \mathfrak{K}_2^2}{C_{\text{CFP}}} \left(\frac{m_+}{m_n} \right) \left(1 + 18 (2\pi) (\mathfrak{K}_1 - \mathfrak{K}_3 - \mathfrak{K}_4) \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, C_{CFP} = the fixed point of the hyperbolic cotangent, m_+ = the proton mass, m_n = the neutron mass, π = Archimedes' constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\mu_n/\mu_N = -1.91304241427759 \dots$$

prediction

$$\mu_n/\mu_N = -1.91304276(45)$$

CODATA 2022, $\sigma = +0.77$

$$\mu_n/\mu_N = -1.91304273(45)$$

CODATA 2018, $\sigma = +0.70$

$$\Delta_{\text{precision}} = +0.00$$

$$\Delta_{\text{scaled}} = +0.07$$

neutron g factor

435.7(23)

$$g_n = -\frac{4 \mathfrak{K}_1 \mathfrak{K}_2^2}{C_{\text{CFP}}} \left(\frac{m_+}{m_n} \right) \left(1 + 18 (2\pi) (\mathfrak{K}_1 - \mathfrak{K}_3 - \mathfrak{K}_4) \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, C_{CFP} = the fixed point of the hyperbolic cotangent, m_+ = the proton mass, m_n = the neutron mass, π = Archimedes' constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$g_n = -3.82608482855518 \dots$$

prediction

$$g_n = -3.82608552(90)$$

CODATA 2022, $\sigma = +0.77$

$$g_n = -3.82608545(90)$$

CODATA 2018, $\sigma = +0.69$

$$\Delta_{\text{precision}} = +0.00$$

$$\Delta_{\text{scaled}} = +0.08$$

hertz-kilogram relationship

12.37064(68)

$$\text{Hz} : \text{kg} = 2\pi \left(\frac{t_p m_p}{s} \right) \left(1 + \frac{2}{C_{\text{CFP}}} (\varkappa_2 - \varkappa_3 - \varkappa_4) \boxtimes \right)$$

Where π = Archimedes' constant, t_p = the Planck time, m_p = the Planck mass, s = the second, C_{CFP} = the fixed point of the hyperbolic cotangent, $\varkappa_2$ = the 2nd hyperbolic partition constant, $\varkappa_3$ = the 3rd hyperbolic partition constant, $\varkappa_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\text{Hz} : \text{kg} = 7.372497323\textcolor{brown}{69515} \dots \times 10^{-51} \text{ kg}$	prediction
$\text{Hz} : \text{kg} = 7.372497323 \times 10^{-51} \text{ kg}$	CODATA 2022, 10-digit match
$\text{Hz} : \text{kg} = 7.372497323 \times 10^{-51} \text{ kg}$	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

hertz-atomic mass unit relationship

12.3711(29)

$$\text{Hz} : A_{\text{mass}} = 2\pi \left(\frac{t_p m_p}{s A_{\text{mass}}} \right) \left(1 + \frac{2}{C_{\text{CFP}}} (\varkappa_2 - \varkappa_3 - \varkappa_4) \boxtimes \right)$$

Where π = Archimedes' constant, t_p = the Planck time, m_p = the Planck mass, s = the second, A_{mass} = the atomic mass constant, C_{CFP} = the fixed point of the hyperbolic cotangent, $\varkappa_2$ = the 2nd hyperbolic partition constant, $\varkappa_3$ = the 3rd hyperbolic partition constant, $\varkappa_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\text{Hz} : A_{\text{mass}} = 4.4398216\textcolor{brown}{6082254} \dots \times 10^{-24} \text{ u}$	prediction
$\text{Hz} : A_{\text{mass}} = 4.4398216590(14) \times 10^{-24} \text{ u}$	CODATA 2022, $\sigma = +1.30$
$\text{Hz} : A_{\text{mass}} = 4.4398216652(13) \times 10^{-24} \text{ u}$	CODATA 2018, $\sigma = -3.37$
	$\Delta_{\text{precision}} = -0.03$
	$\Delta_{\text{scaled}} = -4.77$

natural unit of time

12.3711(30)

$$\mathcal{N}_{\text{time}} = \left(\frac{t_p m_p}{m_e} \right) \left(1 + \frac{2}{C_{\text{CFP}}} (\mathfrak{K}_2 - \mathfrak{K}_3 - \mathfrak{K}_4) \boxtimes \right)$$

Where t_p = the Planck time, m_p = the Planck mass, m_e = the electron mass, C_{CFP} = the fixed point of the hyperbolic cotangent, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mathcal{N}_{\text{time}} = 1.288088666$ 95581 ... $\times 10^{-21}$ s	prediction
$\mathcal{N}_{\text{time}} = 1.28808866644(40) \times 10^{-21}$ s	CODATA 2022, $\sigma = +1.29$
$\mathcal{N}_{\text{time}} = 1.28808866819(39) \times 10^{-21}$ s	CODATA 2018, $\sigma = -3.16$
	$\Delta_{\text{precision}} = -0.01$
	$\Delta_{\text{scaled}} = -4.49$

atomic mass unit-hertz relationship

-12.37064(68)

$$A_{\text{mass}} : \text{Hz} = \frac{1}{2\pi} \left(\frac{A_{\text{mass}}}{t_p m_p} \right) \left(1 - \frac{2}{C_{\text{CFP}}} (\mathfrak{K}_2 - \mathfrak{K}_3 - \mathfrak{K}_4) \boxtimes \right)$$

Where π = Archimedes' constant, A_{mass} = the atomic mass constant, t_p = the Planck time, m_p = the Planck mass, C_{CFP} = the fixed point of the hyperbolic cotangent, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$A_{\text{mass}} : \text{Hz} = 2.25234272$ 093084 ... $\times 10^{23}$ Hz	prediction
$A_{\text{mass}} : \text{Hz} = 2.25234272185(70) \times 10^{23}$ Hz	CODATA 2022, $\sigma = -1.31$
$A_{\text{mass}} : \text{Hz} = 2.25234271871(68) \times 10^{23}$ Hz	CODATA 2018, $\sigma = +3.27$
	$\Delta_{\text{precision}} = -0.01$
	$\Delta_{\text{scaled}} = +4.61$

kilogram-hertz relationship

-12.37064(68)

$$\text{kg} : \text{Hz} = \frac{1}{2\pi} \left(\frac{\text{kg}}{t_p m_p} \right) \left(1 - \frac{2}{C_{\text{CFP}}} (\varkappa_2 - \varkappa_3 - \varkappa_4) \boxtimes \right)$$

Where π = Archimedes’ constant, kg = the kilogram, t_p = the Planck time, m_p = the Planck mass, C_{CFP} = the fixed point of the hyperbolic cotangent, $\varkappa_2$ = the 2nd hyperbolic partition constant, $\varkappa_3$ = the 3rd hyperbolic partition constant, $\varkappa_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

Kg : Hz = 1.35639248967168 ... × 10 ⁵⁰ Hz	prediction
kg : Hz = 1.356392489 × 10 ⁵⁰ Hz	CODATA 2022, 10-digit match
kg : Hz = 1.356392489 × 10 ⁵⁰ Hz	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

Boltzmann constant

47.18(73)

$$k_B = \left(\frac{l_p^2 m_p}{t_p^2 T_p} \right) \left(1 + \frac{s}{\sqrt{2}} \left(\frac{2\pi}{4K} \right) (\mathfrak{K}_2 - \mathfrak{K}_3 - \mathfrak{K}_4) \boxtimes \right)$$

Where l_p = the Planck length, m_p = the Planck mass, t_p = the Planck time, T_p = the Planck temperature, s = the arc length of the unit lemniscate, π = Archimedes' constant, K = Catalan's constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$k_B = 1.38064900009246 \dots \times 10^{-23}$ J/K	prediction
$k_B = 1.380649 \times 10^{-23}$ J/K	CODATA 2022, 7-digit match
$k_B = 1.380649 \times 10^{-23}$ J/K	CODATA 2018, 7-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

Boltzmann constant in eV/K

47.18940(98)

$$\ddot{k}_B = \frac{\text{joule}}{\text{eV}} \left(\frac{l_p^2 m_p}{t_p^2 T_p} \right) \left(1 + \frac{s}{\sqrt{2}} \left(\frac{2\pi}{4K} \right) (\mathfrak{K}_2 - \mathfrak{K}_3 - \mathfrak{K}_4) \boxtimes \right)$$

Where eV = the electron-volt, l_p = the Planck length, m_p = the Planck mass, t_p = the Planck time, T_p = the Planck temperature, s = the arc length of the unit lemniscate, π = Archimedes' constant, K = Catalan's constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\ddot{k}_B = 8.61733326153815 \dots \times 10^{-5}$ eV/K	prediction
$\ddot{k}_B = 8.617333262 \times 10^{-5}$ eV/K	CODATA 2022, 9.50-digit match
$\ddot{k}_B = 8.617333262 \times 10^{-5}$ eV/K	CODATA 2018, 9.50-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

kelvin-joule relationship

47.18(73)

$$K : J = \left(\frac{l_p^2 m_p K}{t_p^2 T_p} \right) \left(1 + \frac{s}{\sqrt{2}} \left(\frac{2\pi}{4K} \right) (\mathfrak{K}_2 - \mathfrak{K}_3 - \mathfrak{K}_4) \boxtimes \right)$$

Where l_p = the Planck length, m_p = the Planck mass, K = the kelvin, t_p = the Planck time, T_p = the Planck temperature, s = the arc length of the unit lemniscate, π = Archimedes' constant, K = Catalan's constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} K : J &= 1.380649\textcolor{green}{00009246} \dots \times 10^{-23} \text{ J} && \text{prediction} \\ K : J &= 1.380649 \times 10^{-23} \text{ J} && \text{CODATA 2022, 7-digit match} \\ K : J &= 1.380649 \times 10^{-23} \text{ J} && \text{CODATA 2018, 7-digit match} \\ &&& \Delta_{\text{precision}} = +0.00 \\ &&& \Delta_{\text{scaled}} = +0.00 \end{aligned}$$

kelvin-electron volt relationship

47.18942(58)

$$K : \text{eV} = \frac{\text{joule}}{\text{eV}} \left(\frac{l_p^2 m_p K}{t_p^2 T_p} \right) \left(1 + \frac{s}{\sqrt{2}} \left(\frac{2\pi}{4K} \right) (\mathfrak{K}_2 - \mathfrak{K}_3 - \mathfrak{K}_4) \boxtimes \right)$$

Where eV = the electron-volt, l_p = the Planck length, m_p = the Planck mass, K = the kelvin, t_p = the Planck time, T_p = the Planck temperature, s = the arc length of the unit lemniscate, π = Archimedes' constant, K = Catalan's constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} K : \text{eV} &= 8.61733326\textcolor{green}{153815} \dots \times 10^{-5} \text{ eV} && \text{prediction} \\ K : \text{eV} &= 8.617333262 \times 10^{-5} \text{ eV} && \text{CODATA 2022, 9.50-digit match} \\ K : \text{eV} &= 8.617333262 \times 10^{-5} \text{ eV} && \text{CODATA 2018, 9.50-digit match} \\ &&& \Delta_{\text{precision}} = +0.00 \\ &&& \Delta_{\text{scaled}} = +0.00 \end{aligned}$$

electron volt-kelvin relationship

-47.18942(58)

$$\text{eV} : \text{K} = \text{eV} \left(\frac{t_p^2 T_p}{l_p^2 m_p} \right) \left(1 - \frac{s}{\sqrt{2}} \left(\frac{2\pi}{4K} \right) (\mathfrak{K}_2 - \mathfrak{K}_3 - \mathfrak{K}_4) \boxtimes \right)$$

Where eV = the electron-volt, t_p = the Planck time, T_p = the Planck temperature, l_p = the Planck length, m_p = the Planck mass, s = the arc length of the unit lemniscate, π = Archimedes' constant, K = Catalan's constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\text{eV} : \text{K} = 1.160451812\mathbf{21091} \dots \times 10^4 \text{ K} \quad \text{prediction}$$

$$\text{eV} : \text{K} = 1.160451812 \times 10^4 \text{ K} \quad \text{CODATA 2022, 10-digit match}$$

$$\text{eV} : \text{K} = 1.160451812 \times 10^4 \text{ K} \quad \text{CODATA 2018, 10-digit match}$$

$$\Delta_{\text{precision}} = +0.00$$

$$\Delta_{\text{scaled}} = +0.00$$

joule-kelvin relationship

-47.18675(69)

$$\text{J} : \text{K} = \text{joule} \left(\frac{t_p^2 T_p}{l_p^2 m_p} \right) \left(1 - \frac{s}{\sqrt{2}} \left(\frac{2\pi}{4K} \right) (\mathfrak{K}_2 - \mathfrak{K}_3 - \mathfrak{K}_4) \boxtimes \right)$$

Where t_p = the Planck time, T_p = the Planck temperature, l_p = the Planck length, m_p = the Planck mass, s = the arc length of the unit lemniscate, π = Archimedes' constant, K = Catalan's constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\text{J} : \text{K} = 7.24297051\mathbf{539357} \dots \times 10^{22} \text{ K} \quad \text{prediction}$$

$$\text{J} : \text{K} = 7.242970516 \times 10^{22} \text{ K} \quad \text{CODATA 2022, 9.92-digit match}$$

$$\text{J} : \text{K} = 7.242970516 \times 10^{22} \text{ K} \quad \text{CODATA 2018, 9.92-digit match}$$

$$\Delta_{\text{precision}} = +0.00$$

$$\Delta_{\text{scaled}} = +0.00$$

Loschmidt constant (273.15 K, 100 kPa)

-47.1889(19)

$$n_0 = \frac{p_0}{T_0} \left(\frac{t_p^2 T_p}{l_p^2 m_p} \right) \left(1 - \frac{s}{\sqrt{2}} \left(\frac{2\pi}{4K} \right) (\varkappa_2 - \varkappa_3 - \varkappa_4) \boxtimes \right)$$

Where $p_0 = 100.000000000000$ kPa, $T_0 = 273.150000000000$ K, t_p = the Planck time, T_p = the Planck temperature, l_p = the Planck length, m_p = the Planck mass, s = the arc length of the unit lemniscate, π = Archimedes' constant, K = Catalan's constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, $\varkappa_3$ = the 3rd hyperbolic partition constant, $\varkappa_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$n_0 = 2.65164580464711 \dots \times 10^{25} \text{ 1/m}^3$	prediction
$n_0 = 2.651645804 \times 10^{25} \text{ 1/m}^3$	CODATA 2022, 10-digit match
$n_0 = 2.651645804 \times 10^{25} \text{ 1/m}^3$	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

Loschmidt constant (273.15 K, 101.325 kPa)

-47.1885(19)

$$n_1 = \frac{p_1}{T_0} \left(\frac{t_p^2 T_p}{l_p^2 m_p} \right) \left(1 - \frac{s}{\sqrt{2}} \left(\frac{2\pi}{4K} \right) (\varkappa_2 - \varkappa_3 - \varkappa_4) \boxtimes \right)$$

Where $p_1 = 101.325000000000$ kPa, $T_0 = 273.150000000000$ K, t_p = the Planck time, T_p = the Planck temperature, l_p = the Planck length, m_p = the Planck mass, s = the arc length of the unit lemniscate, π = Archimedes' constant, K = Catalan's constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, $\varkappa_3$ = the 3rd hyperbolic partition constant, $\varkappa_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$n_1 = 2.686780111155868 \dots \times 10^{25} \text{ 1/m}^3$	prediction
$n_1 = 2.686780111 \times 10^{25} \text{ 1/m}^3$	CODATA 2022, 10-digit match
$n_1 = 2.686780111 \times 10^{25} \text{ 1/m}^3$	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

Bohr magneton in K/T

-41.6669(30)

$$\frac{\mu_B}{k_B} = \frac{\varkappa_1}{2} \left(\frac{t_p q_p T_p}{m_e} \right) \left(1 - \sqrt{\frac{32}{G_{\text{Gi}}}} (\varkappa_2 - \varkappa_3 - \varkappa_4) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, t_p = the Planck time, q_p = the Planck charge, T_p = the Planck temperature, m_e = the electron mass, G_{Gi} = Gieseking's constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, $\varkappa_3$ = the 3rd hyperbolic partition constant, $\varkappa_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_B/k_B = 6.71713814\textcolor{green}{961807} \dots \times 10^{-1} \text{ K/T}$	prediction
$\mu_B/k_B = 6.7171381472(21) \times 10^{-1} \text{ K/T}$	CODATA 2022, $\sigma = +1.15$
$\mu_B/k_B = 6.7171381563(20) \times 10^{-1} \text{ K/T}$	CODATA 2018, $\sigma = -3.34$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = -4.55$

nuclear magneton in K/T

-41.6669(30)

$$\frac{\mu_N}{k_B} = \frac{\varkappa_1}{2} \left(\frac{t_p q_p T_p}{m_+} \right) \left(1 - \sqrt{\frac{32}{G_{\text{Gi}}}} (\varkappa_2 - \varkappa_3 - \varkappa_4) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, t_p = the Planck time, q_p = the Planck charge, T_p = the Planck temperature, m_+ = the proton mass, G_{Gi} = Gieseking's constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, $\varkappa_3$ = the 3rd hyperbolic partition constant, $\varkappa_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_N/k_B = 3.65826777\textcolor{green}{199876} \dots \times 10^{-4} \text{ K/T}$	prediction
$\mu_N/k_B = 3.6582677706(11) \times 10^{-4} \text{ K/T}$	CODATA 2022, $\sigma = +1.27$
$\mu_N/k_B = 3.6582677756(11) \times 10^{-4} \text{ K/T}$	CODATA 2018, $\sigma = -3.27$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -4.55$

Rydberg constant times hc in J

−7.243621(11)

$$R_{\infty}^{\cdot\cdot}hc = \frac{\varkappa_1^4}{2} \left(\frac{l_p^2 m_e}{t_p^2} \right) \left(1 + \sqrt{35} (8 D_d) (\varkappa_2 + \varkappa_3 + \varkappa_4) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, m_e = the electron mass, t_p = the Planck time, D_d = the dimer constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, $\varkappa_3$ = the 3rd hyperbolic partition constant, $\varkappa_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} R_{\infty}^{\cdot\cdot}hc &= 2.1798723611\textcolor{green}{1219} \dots \times 10^{-18} \text{ J} && \text{prediction} \\ R_{\infty}^{\cdot\cdot}hc &= 2.1798723611030(24) \times 10^{-18} \text{ J} && \text{CODATA 2022, } \sigma = +3.83 \\ R_{\infty}^{\cdot\cdot}hc &= 2.1798723611035(42) \times 10^{-18} \text{ J} && \text{CODATA 2018, } \sigma = +2.07 \\ &&& \Delta_{\text{precision}} = +0.24 \\ &&& \Delta_{\text{scaled}} = -0.12 \end{aligned}$$

Hartree energy

−7.243622(11)

$$E_h = \varkappa_1^4 \left(\frac{l_p^2 m_e}{t_p^2} \right) \left(1 + \sqrt{35} (8 D_d) (\varkappa_2 + \varkappa_3 + \varkappa_4) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, m_e = the electron mass, t_p = the Planck time, D_d = the dimer constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, $\varkappa_3$ = the 3rd hyperbolic partition constant, $\varkappa_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} E_h &= 4.3597447222\textcolor{green}{2439} \dots \times 10^{-18} \text{ J} && \text{prediction} \\ E_h &= 4.3597447222060(48) \times 10^{-18} \text{ J} && \text{CODATA 2022, } \sigma = +3.83 \\ E_h &= 4.3597447222071(85) \times 10^{-18} \text{ J} && \text{CODATA 2018, } \sigma = +2.03 \\ &&& \Delta_{\text{precision}} = +0.25 \\ &&& \Delta_{\text{scaled}} = -0.13 \end{aligned}$$

Also listed as the *atomic unit of energy*, and the *hartree-joule relationship*.

$$J \colon E_h = \frac{\text{joule}^2}{\mathfrak{K}_1^4} \left(\frac{t_p^2}{l_p^2 m_e} \right) \left(1 + \sqrt{35} \left(8 D_d \right) \left(\mathfrak{K}_2 + \mathfrak{K}_3 + \mathfrak{K}_4 \right) \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, t_p = the Planck time, l_p = the Planck length, m_e = the electron mass, D_d = the dimer constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

J ∶ E_h = 2.29371227838603 ... × 10¹⁷ E_h

J ∶ E_h = 2.2937122783969(25) × 10¹⁷ E_h

J ∶ E_h = 2.2937122783963(45) × 10¹⁷ E_h

prediction

CODATA 2022, σ = −4.35

CODATA 2018, σ = −2.28

Δ_{precision} = +0.26

Δ_{scaled} = +0.13

Josephson constant

-4.7726(05)

$$K_J = \frac{\varkappa_1}{\pi} \left(\frac{t_p q_p}{l_p^2 m_p} \right) \left(1 - \frac{7}{8} \left(\frac{8}{4\pi} \right)^2 (\varkappa_1^2 + \varkappa_2^2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, π = Archimedes' constant, t_p = the Planck time, q_p = the Planck charge, l_p = the Planck length, m_p = the Planck mass, π = Archimedes' constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$K_J = 4.83597848424072 \dots \times 10^{14} \text{ Hz/V} \quad \text{prediction}$$

$$K_J = 4.835978484 \times 10^{14} \text{ Hz/V} \quad \text{CODATA 2022, 10-digit match}$$

$$K_J = 4.835978484 \times 10^{14} \text{ Hz/V} \quad \text{CODATA 2018, 10-digit match}$$

$$\Delta_{\text{precision}} = +0.00$$

$$\Delta_{\text{scaled}} = +0.00$$

Also listed as the *conventional value of Josephson constant*.

elementary charge over h-bar

-4.7771(14)

$$\frac{e}{\hbar} = \varkappa_1 \left(\frac{t_p q_p}{l_p^2 m_p} \right) \left(1 - \frac{7}{8} \left(\frac{8}{4\pi} \right)^2 (\varkappa_1^2 + \varkappa_2^2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, t_p = the Planck time, q_p = the Planck charge, l_p = the Planck length, m_p = the Planck mass, π = Archimedes' constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$e/\hbar = 1.51926744790089 \dots \times 10^{15} \text{ A/J} \quad \text{prediction}$$

$$e/\hbar = 1.519267447 \times 10^{15} \text{ A/J} \quad \text{CODATA 2022, 10-digit match}$$

$$e/\hbar = 1.519267447 \times 10^{15} \text{ A/J} \quad \text{CODATA 2018, 10-digit match}$$

$$\Delta_{\text{precision}} = +0.00$$

$$\Delta_{\text{scaled}} = +0.00$$

magnetic flux quantum

4.7717(12)

$$\Phi_0 = \frac{\pi}{\varkappa_1} \left(\frac{l_p^2 m_p}{t_p q_p} \right) \left(1 + \frac{7}{8} \left(\frac{8}{4\pi} \right)^2 (\varkappa_1^2 + \varkappa_2^2) \boxtimes \right)$$

Where π = Archimedes' constant, $\varkappa_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, m_p = the Planck mass, t_p = the Planck time, q_p = the Planck charge, $\varkappa_2$ = the 2nd hyperbolic partition constant, π = Archimedes' constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\Phi_0 = 2.06783384843115 \dots \times 10^{-15} \text{ Wb}$	prediction
$\Phi_0 = 2.067833848 \times 10^{-15} \text{ Wb}$	CODATA 2022, 10-digit match
$\Phi_0 = 2.067833848 \times 10^{-15} \text{ Wb}$	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

reduced Planck constant in eV s

4.77270(38)

$$\frac{\hbar}{e} = \frac{1}{\varkappa_1} \left(\frac{l_p^2 C m_p}{t_p q_p} \right) \left(1 + \frac{7}{8} \left(\frac{8}{4\pi} \right)^2 (\varkappa_1^2 + \varkappa_2^2) \boxtimes \right)$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, C = the coulomb, m_p = the Planck mass, t_p = the Planck time, q_p = the Planck charge, π = Archimedes' constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\hbar/e = 6.58211956941109 \dots \times 10^{-16} \text{ eV} \cdot \text{s}$	prediction
$\hbar/e = 6.582119569 \times 10^{-16} \text{ eV} \cdot \text{s}$	CODATA 2022, 10-digit match
$\hbar/e = 6.582119569 \times 10^{-16} \text{ eV} \cdot \text{s}$	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

Planck constant in eV/Hz4.77117(60)

$$\frac{h}{e} = \frac{2\pi}{\varkappa_1} \left(\frac{l_p^2 C m_p}{t_p q_p} \right) \left(1 + \frac{7}{8} \left(\frac{8}{4\pi} \right)^2 (\varkappa_1^2 + \varkappa_2^2) \boxtimes \right)$$

Where π = Archimedes' constant, $\varkappa_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, C = the coulomb, m_p = the Planck mass, t_p = the Planck time, q_p = the Planck charge, $\varkappa_2$ = the 2nd hyperbolic partition constant, π = Archimedes' constant, $\varkappa_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$h/e = 4.13566769686229 \dots \times 10^{-15} \text{ eV/Hz}$

$h/e = 4.135667696 \times 10^{-15} \text{ eV/Hz}$

$h/e = 4.135667696 \times 10^{-15} \text{ eV/Hz}$

prediction

CODATA 2022, 10-digit match

CODATA 2018, 10-digit match

$\Delta_{\text{precision}} = +0.00$

$\Delta_{\text{scaled}} = +0.00$

first radiation constant

-2.0763(14)

$$c_1 = (2\pi)^2 \left(\frac{l_p^4 m_p}{t_p^3} \right) \left(1 - \left(\frac{4\pi}{32} \right)^2 (\mathfrak{K}_1^2 + \mathfrak{K}_2^2) \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, m_p = the Planck mass, t_p = the Planck time, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} c_1 &= 3.741771852\textcolor{brown}{85687} \dots \times 10^{-16} \text{ W} \cdot \text{m}^2 && \text{prediction} \\ c_1 &= 3.741771852 \times 10^{-16} \text{ W} \cdot \text{m}^2 && \text{CODATA 2022, 10-digit match} \\ c_1 &= 3.741771852 \times 10^{-16} \text{ W} \cdot \text{m}^2 && \text{CODATA 2018, 10-digit match} \\ &&& \Delta_{\text{precision}} = +0.00 \\ &&& \Delta_{\text{scaled}} = +0.00 \end{aligned}$$

first radiation constant for spectral radiance

-2.0763(42)

$$c_{1L} = 4\pi \left(\frac{l_p^4 m_p}{t_p^3} \right) \left(1 - \left(\frac{4\pi}{32} \right)^2 (\mathfrak{K}_1^2 + \mathfrak{K}_2^2) \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, m_p = the Planck mass, t_p = the Planck time, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} c_{1L} &= 1.191042972\textcolor{brown}{60858} \dots \times 10^{-16} \text{ W} \cdot \text{m}^2/\text{sr} && \text{prediction} \\ c_{1L} &= 1.191042972 \times 10^{-16} \text{ W} \cdot \text{m}^2/\text{sr} && \text{CODATA 2022, 10-digit match} \\ c_{1L} &= 1.191042972 \times 10^{-16} \text{ W} \cdot \text{m}^2/\text{sr} && \text{CODATA 2018, 10-digit match} \\ &&& \Delta_{\text{precision}} = +0.00 \\ &&& \Delta_{\text{scaled}} = +0.00 \end{aligned}$$

shielded helion gyromagnetic ratio-3.9241(88)

$$\gamma_{\text{he}}' = \frac{C_{\text{CFP}}^2}{\sqrt{7}} \left(\frac{q_p}{m_{\text{he}}} \right) \left(1 - C_d \left(\mathfrak{K}_1^2 + \mathfrak{K}_2^2 \right) \boxtimes \right)$$

Where C_{CFP} = the fixed point of the hyperbolic cotangent, q_p = the Planck charge, m_{he} = the helion mass, C_d = the domino tiling constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\gamma_{\text{he}}' = 2.03789460878416 \dots \times 10^8 \text{ 1/s} \cdot \text{T}$$
$$\gamma_{\text{he}}' = 2.0378946078(18) \times 10^8 \text{ 1/s} \cdot \text{T}$$
$$\gamma_{\text{he}}' = 2.037894569(24) \times 10^8 \text{ 1/s} \cdot \text{T}$$

prediction

CODATA 2022, $\sigma = +0.55$

CODATA 2018, $\sigma = +1.66$

$\Delta_{\text{precision}} = +1.13$

$\Delta_{\text{scaled}} = +1.58$

shielded helion gyromagnetic ratio in MHz/T-3.9241(88)

$$\dot{\gamma}_{\text{he}}' = \frac{C_{\text{CFP}}^2}{2\pi \sqrt{7} \text{ MHz}} \left(\frac{q_p}{s m_{\text{he}}} \right) \left(1 - C_d \left(\mathfrak{K}_1^2 + \mathfrak{K}_2^2 \right) \boxtimes \right)$$

Where C_{CFP} = the fixed point of the hyperbolic cotangent, π = Archimedes' constant, MHz = the megahertz, q_p = the Planck charge, s = the second, m_{he} = the helion mass, C_d = the domino tiling constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\dot{\gamma}_{\text{he}}' = 3.24341000488323 \dots \times 10^1 \text{ MHz/T}$$
$$\dot{\gamma}_{\text{he}}' = 3.2434100033(28) \times 10^1 \text{ MHz/T}$$
$$\dot{\gamma}_{\text{he}}' = 3.243409942(38) \times 10^1 \text{ MHz/T}$$

prediction

CODATA 2022, $\sigma = +0.57$

CODATA 2018, $\sigma = +1.65$

$\Delta_{\text{precision}} = +0.13$

$\Delta_{\text{scaled}} = +1.61$

helion magnetic moment

600.9249(87)

$$\mu_{\text{he}} = -\frac{C_{\text{CFP}}^2}{2\sqrt{7}} \left(\frac{l_p^2 q_p m_p}{t_p m_{\text{he}}} \right) \left(1 + 5 \left(\frac{16}{D_d} \right) (\mathfrak{K}_1^2 + \mathfrak{K}_2^2) \boxtimes \right)$$

Where C_{CFP} = the fixed point of the hyperbolic cotangent, l_p = the Planck length, q_p = the Planck charge, m_p = the Planck mass, t_p = the Planck time, m_{he} = the helion mass, D_d = the dimer constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} \mu_{\text{he}} &= -1.07461755420270 \dots \times 10^{-26} \text{ J/T} && \text{prediction} \\ \mu_{\text{he}} &= -1.07461755198(93) \times 10^{-26} \text{ J/T} && \text{CODATA 2022, } \sigma = -2.39 \\ \mu_{\text{he}} &= -1.074617532(13) \times 10^{-26} \text{ J/T} && \text{CODATA 2018, } \sigma = -1.71 \\ &&& \Delta_{\text{precision}} = +1.15 \\ &&& \Delta_{\text{scaled}} = +1.38 \end{aligned}$$

shielded helion magnetic moment

1.2231(87)

$$\mu_{\text{he}}' = -\frac{C_{\text{CFP}}^2}{2\sqrt{7}} \left(\frac{l_p^2 q_p m_p}{t_p m_{\text{he}}} \right) \left(1 + \frac{1}{7} \left(\frac{8}{4\pi} \right) (\mathfrak{K}_1^2 + \mathfrak{K}_2^2) \boxtimes \right)$$

Where C_{CFP} = the fixed point of the hyperbolic cotangent, l_p = the Planck length, q_p = the Planck charge, m_p = the Planck mass, t_p = the Planck time, m_{he} = the helion mass, π = Archimedes' constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} \mu_{\text{he}}' &= -1.07455311098442 \dots \times 10^{-26} \text{ J/T} && \text{prediction} \\ \mu_{\text{he}}' &= -1.07455311035(93) \times 10^{-26} \text{ J/T} && \text{CODATA 2022, } \sigma = +0.68 \\ \mu_{\text{he}}' &= -1.074553090(13) \times 10^{-26} \text{ J/T} && \text{CODATA 2018, } \sigma = +1.61 \\ &&& \Delta_{\text{precision}} = +1.15 \\ &&& \Delta_{\text{scaled}} = +1.54 \end{aligned}$$

shielded helion to proton magnetic moment ratio

320.1798(87)

$$\frac{\mu_{\text{he}}'}{\mu_+} = -\frac{C_{\text{CFP}}^2}{2\sqrt{7}} \frac{S}{\mathfrak{K}_1^2 \mathfrak{K}_2^2} \left(\frac{m_+}{m_{\text{he}}} \right) \left(1 - \sqrt{12} L^2 (\mathfrak{K}_1^2 - \mathfrak{K}_2^2) \boxtimes \right)$$

Where C_{CFP} = the fixed point of the hyperbolic cotangent, S = Sierpiński's constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, m_+ = the proton mass, m_{he} = the helion mass, L = the lemniscate constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} \mu_{\text{he}}'/\mu_+ &= -7.617665777\mathbf{306786} \dots \times 10^{-1} && \text{prediction} \\ \mu_{\text{he}}'/\mu_+ &= -7.6176657721(66) \times 10^{-1} && \text{CODATA 2022, } \sigma = -0.15 \\ \mu_{\text{he}}'/\mu_+ &= -7.617665618(89) \times 10^{-1} && \text{CODATA 2018, } \sigma = +1.74 \\ &&& \Delta_{\text{precision}} = +1.13 \\ &&& \Delta_{\text{scaled}} = +1.73 \end{aligned}$$

shielded helion to shielded proton magnetic moment ratio

576.910(41)

$$\frac{\mu_{\text{he}}'}{\mu_+'} = -\frac{C_{\text{CFP}}^2}{2\sqrt{7}} \frac{S}{\mathfrak{K}_1^2 \mathfrak{K}_2^2} \left(\frac{m_+}{m_{\text{he}}} \right) \left(1 + \frac{\pi}{\sqrt{12}} L^4 (\mathfrak{K}_1^2 + \mathfrak{K}_2^2) \boxtimes \right)$$

Where C_{CFP} = the fixed point of the hyperbolic cotangent, S = Sierpiński's constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, m_+ = the proton mass, m_{he} = the helion mass, π = Archimedes' constant, L = the lemniscate constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} \mu_{\text{he}}'/\mu_+' &= -7.6178613\mathbf{4617088} \dots \times 10^{-1} && \text{prediction} \\ \mu_{\text{he}}'/\mu_+' &= -7.617861334(31) \times 10^{-1} && \text{CODATA 2022, } \sigma = -0.39 \\ \mu_{\text{he}}'/\mu_+' &= -7.617861313(33) \times 10^{-1} && \text{CODATA 2018, } \sigma = -1.00 \\ &&& \Delta_{\text{precision}} = +0.03 \\ &&& \Delta_{\text{scaled}} = +0.64 \end{aligned}$$

electron to shielded helion magnetic moment ratio 654.1948(81)

$$\frac{\mu_e}{\mu_{\text{he}}'} = \frac{2\sqrt{7}}{C_{\text{CFP}}^2} \frac{\mathfrak{K}_1^2 \mathfrak{K}_2^2}{P_{up}} \left(\frac{m_{\text{he}}}{m_e} \right) \left(1 + \frac{2e^\gamma}{\sqrt{12}} L^4 \left(\mathfrak{K}_1^2 + \mathfrak{K}_2^2 \right) \boxtimes \right)$$

Where C_{CFP} = the fixed point of the hyperbolic cotangent, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, P_{up} = the universal parabolic constant, m_{he} = the helion mass, m_e = the electron mass, e = Euler's number, γ = the Euler-Mascheroni constant, L = the lemniscate constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\mu_e/\mu_{\text{he}}' = 8.64058236541607 \dots \times 10^2$	prediction
$\mu_e/\mu_{\text{he}}' = 8.6405823986(70) \times 10^2$	CODATA 2022, $\sigma = -4.74$
$\mu_e/\mu_{\text{he}}' = 8.64058257(10) \times 10^2$	CODATA 2018, $\sigma = -2.05$
	$\Delta_{\text{precision}} = +1.15$
	$\Delta_{\text{scaled}} = -2.80$

atomic mass unit-hartree relationship

-1.3473(31)

$$A_{\text{mass}} : E_h = \sqrt{\frac{18}{32} \frac{P_{up}}{\mathfrak{K}_1^{14}}} \left(1 + \frac{\zeta(3)}{12} (\mathfrak{K}_1^2 - \mathfrak{K}_2^2) \boxtimes \right)$$

Where P_{up} = the universal parabolic constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\zeta(x)$ = the Riemann zeta function, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$A_{\text{mass}} : E_h = 3.42317769077001 \dots \times 10^7 E_h$	prediction
$A_{\text{mass}} : E_h = 3.4231776922(11) \times 10^7 E_h$	CODATA 2022, $\sigma = -1.30$
$A_{\text{mass}} : E_h = 3.4231776874(10) \times 10^7 E_h$	CODATA 2018, $\sigma = +3.37$
	$\Delta_{\text{precision}} = -0.04$
	$\Delta_{\text{scaled}} = +4.80$

hartree-atomic mass unit relationship

1.3473(31)

$$E_h : A_{\text{mass}} = \sqrt{\frac{32}{18} \frac{\mathfrak{K}_1^{14}}{P_{up}}} \left(1 - \frac{\zeta(3)}{12} (\mathfrak{K}_1^2 - \mathfrak{K}_2^2) \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, $\zeta(x)$ = the Riemann zeta function, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_h : A_{\text{mass}} = 2.92126231920798 \dots \times 10^{-8} \text{ u}$	prediction
$E_h : A_{\text{mass}} = 2.92126231797(91) \times 10^{-8} \text{ u}$	CODATA 2022, $\sigma = +1.36$
$E_h : A_{\text{mass}} = 2.92126232205(88) \times 10^{-8} \text{ u}$	CODATA 2018, $\sigma = -3.23$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = -4.64$

kilogram-hartree relationship

-1.346674(10)

$$\text{kg} : E_h = \sqrt{\frac{18}{32} \frac{P_{up}}{\mathfrak{K}_1^{14}}} \left(\frac{\text{kg}}{A_{\text{mass}}} \right) \left(1 + \frac{\zeta(3)}{12} (\mathfrak{K}_1^2 - \mathfrak{K}_2^2) \boxtimes \right)$$

Where P_{up} = the universal parabolic constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, kg = the kilogram, A_{mass} = the atomic mass constant, $\zeta(x)$ = the Riemann zeta function, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\text{kg} : E_h = 2.06148578874053 \dots \times 10^{34} E_h$	prediction
$\text{kg} : E_h = 2.0614857887415(22) \times 10^{34} E_h$	CODATA 2022, $\sigma = -0.44$
$\text{kg} : E_h = 2.0614857887409(40) \times 10^{34} E_h$	CODATA 2018, $\sigma = -0.09$
	$\Delta_{\text{precision}} = +0.26$
	$\Delta_{\text{scaled}} = +0.15$

hartree-kilogram relationship

1.346675(11)

$$E_h : \text{kg} = \sqrt{\frac{32}{18} \frac{\mathfrak{K}_1^{14}}{P_{up}}} (A_{\text{mass}}) \left(1 - \frac{\zeta(3)}{12} (\mathfrak{K}_1^2 - \mathfrak{K}_2^2) \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, P_{up} = the universal parabolic constant, A_{mass} = the atomic mass constant, $\zeta(x)$ = the Riemann zeta function, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_h : \text{kg} = 4.85087020954402 \dots \times 10^{-35} \text{ kg}$	prediction
$E_h : \text{kg} = 4.8508702095419(53) \times 10^{-35} \text{ kg}$	CODATA 2022, $\sigma = +0.40$
$E_h : \text{kg} = 4.8508702095432(94) \times 10^{-35} \text{ kg}$	CODATA 2018, $\sigma = +0.09$
	$\Delta_{\text{precision}} = +0.25$
	$\Delta_{\text{scaled}} = -0.14$

kelvin-kilogram relationship

54.41184(77)

$$K : \text{kg} = \left(\frac{K m_p}{T_p} \right) \left(1 - L_2 \sqrt{6V_{\text{fe}}} \left(\mathfrak{K}_3^2 + \mathfrak{K}_4^2 \right) \boxtimes \right)$$

Where K = the kelvin, m_p = the Planck mass, T_p = the Planck temperature, L_2 = the 2nd lemniscate constant, V_{fe} = the figure-eight knot hyperbolic volume, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$K : \text{kg} = 1.53617918688372 \dots \times 10^{-40} \text{ kg}$	prediction
$K : \text{kg} = 1.536179187 \times 10^{-40} \text{ kg}$	CODATA 2022, 9.93-digit match
$K : \text{kg} = 1.536179187 \times 10^{-40} \text{ kg}$	CODATA 2018, 9.93-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

kilogram-kelvin relationship

-54.41184(77)

$$\text{kg} : K = \left(\frac{T_p \text{ kg}}{m_p} \right) \left(1 + L_2 \sqrt{6V_{\text{fe}}} \left(\mathfrak{K}_3^2 + \mathfrak{K}_4^2 \right) \boxtimes \right)$$

Where T_p = the Planck temperature, m_p = the Planck mass, L_2 = the 2nd lemniscate constant, V_{fe} = the figure-eight knot hyperbolic volume, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\text{kg} : K = 6.50965726204757 \dots \times 10^{39} \text{ K}$	prediction
$\text{kg} : K = 6.509657260 \times 10^{39} \text{ K}$	CODATA 2022, 9.50-digit match
$\text{kg} : K = 6.509657260 \times 10^{39} \text{ K}$	CODATA 2018, 9.50-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

kelvin-atomic mass unit relationship

54.4119(31)

$$K : A_{\text{mass}} = \left(\frac{K m_p}{T_p A_{\text{mass}}} \right) \left(1 - L_2 \sqrt{6V_{\text{fe}}} \left(\mathfrak{K}_3^2 + \mathfrak{K}_4^2 \right) \boxtimes \right)$$

Where A_{mass} = the atomic mass constant, m_p = the Planck mass, T_p = the Planck temperature, L_2 = the 2nd lemniscate constant, V_{fe} = the figure-eight knot hyperbolic volume, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$K : A_{\text{mass}} = 9.25108729020561 \dots \times 10^{-14} \text{ u}$	prediction
$K : A_{\text{mass}} = 9.2510872884(29) \times 10^{-14} \text{ u}$	CODATA 2022, $\sigma = +0.62$
$K : A_{\text{mass}} = 9.2510873014(28) \times 10^{-14} \text{ u}$	CODATA 2018, $\sigma = -4.00$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = -4.64$

atomic mass unit-kelvin relationship

-54.4119(31)

$$A_{\text{mass}} : K = \left(\frac{T_p A_{\text{mass}}}{m_p} \right) \left(1 + L_2 \sqrt{6V_{\text{fe}}} \left(\mathfrak{K}_3^2 + \mathfrak{K}_4^2 \right) \boxtimes \right)$$

Where T_p = the Planck temperature, A_{mass} = the atomic mass constant, m_p = the Planck mass, L_2 = the 2nd lemniscate constant, V_{fe} = the figure-eight knot hyperbolic volume, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$A_{\text{mass}} : K = 1.08095402043079 \dots \times 10^{13} \text{ K}$	prediction
$A_{\text{mass}} : K = 1.08095402067(34) \times 10^{13} \text{ K}$	CODATA 2022, $\sigma = -0.70$
$A_{\text{mass}} : K = 1.08095401916(33) \times 10^{13} \text{ K}$	CODATA 2018, $\sigma = +3.58$
	$\Delta_{\text{precision}} = -0.01$
	$\Delta_{\text{scaled}} = +4.58$

kilogram-inverse meter relationship

8.7598(22)

$$\text{kg} : \frac{1}{\text{m}} = \frac{1}{2\pi} \left(\frac{\text{kg}}{l_p m_p} \right) \left(1 - \frac{P_{up}}{8} i (\kappa_3^2 - \kappa_4^2) \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, m_p = the Planck mass, P_{up} = the universal parabolic constant, i = the imaginary unit, κ_3 = the 3rd hyperbolic partition constant κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

Kg : 1/m = 4.52443833598084 ... × 10 ⁴¹ cycles/m	prediction
kg : 1/m = 4.524438335 × 10 ⁴¹ cycles/m	CODATA 2022, 10-digit match
kg : 1/m = 4.524438335 × 10 ⁴¹ cycles/m	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

atomic mass unit-inverse meter relationship

-8.7591(30)

$$A_{\text{mass}} : \frac{1}{\text{m}} = \frac{1}{2\pi} \left(\frac{A_{\text{mass}}}{l_p m_p} \right) \left(1 - \frac{P_{up}}{8} i (\kappa_3^2 - \kappa_4^2) \boxtimes \right)$$

Where π = Archimedes' constant, A_{mass} = the atomic mass constant, l_p = the Planck length, m_p = the Planck mass, P_{up} = the universal parabolic constant, i = the imaginary unit, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

A _{mass} : 1/m = 7.51300661861781 ... × 10 ¹⁴ cycles/m	prediction
A _{mass} : 1/m = 7.5130066209(23) × 10 ¹⁴ cycles/m	CODATA 2022, $\sigma = +0.99$
A _{mass} : 1/m = 7.5130066104(23) × 10 ¹⁴ cycles/m	CODATA 2018, $\sigma = +3.57$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +4.57$

inverse meter-kilogram relationship

8.7598(22)

$$\frac{1}{\text{m}} : \text{kg} = 2\pi \left(\frac{l_p m_p}{\text{m}} \right) \left(1 + \frac{P_{up} i}{8} (\mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, m_p = the Planck mass, m = the meter, P_{up} = the universal parabolic constant, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} 1/\text{m} : \text{kg} &= 2.210219094\mathbf{04020} \dots \times 10^{-42} \text{ kg} && \text{prediction} \\ 1/\text{m} : \text{kg} &= 2.210219094 \times 10^{-42} \text{ kg} && \text{CODATA 2022, 10-digit match} \\ 1/\text{m} : \text{kg} &= 2.210219094 \times 10^{-42} \text{ kg} && \text{CODATA 2018, 10-digit match} \\ &&& \Delta_{\text{precision}} = +0.00 \\ &&& \Delta_{\text{scaled}} = +0.00 \end{aligned}$$

inverse meter-atomic mass unit relationship

8.7591(30)

$$\frac{1}{\text{m}} : A_{\text{mass}} = 2\pi \left(\frac{l_p m_p}{\text{m} A_{\text{mass}}} \right) \left(1 + \frac{P_{up} i}{8} (\mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, m_p = the Planck mass, m = the meter, A_{mass} = the atomic mass constant, P_{up} = the universal parabolic constant, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} 1/\text{m} : A_{\text{mass}} &= 1.331025048\mathbf{64185} \dots \times 10^{-15} \text{ u} && \text{prediction} \\ 1/\text{m} : A_{\text{mass}} &= 1.33102504824(41) \times 10^{-15} \text{ u} && \text{CODATA 2022, } \sigma = +0.98 \\ 1/\text{m} : A_{\text{mass}} &= 1.33102505010(40) \times 10^{-15} \text{ u} && \text{CODATA 2018, } \sigma = -3.65 \\ &&& \Delta_{\text{precision}} = -0.01 \\ &&& \Delta_{\text{scaled}} = -4.65 \end{aligned}$$

Compton wavelength

8.7590(30)

$$\lambda_c = 2\pi \left(\frac{l_p m_p}{m_e} \right) \left(1 + \frac{P_{up} i}{8} (\mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, m_p = the Planck mass, m_e = the electron mass, P_{up} = the universal parabolic constant, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} \lambda_c &= 2.42631023609293 \dots \times 10^{-12} \text{ m/cycle} && \text{prediction} \\ \lambda_c &= 2.42631023538(76) \times 10^{-12} \text{ m/cycle} && \text{CODATA 2022, } \sigma = +0.94 \\ \lambda_c &= 2.42631023867(73) \times 10^{-12} \text{ m/cycle} && \text{CODATA 2018, } \sigma = -3.53 \\ &&& \Delta_{\text{precision}} = -0.02 \\ &&& \Delta_{\text{scaled}} = -4.63 \end{aligned}$$

muon Compton wavelength

8.72(22)

$$\lambda_\mu = 2\pi \left(\frac{l_p m_p}{m_\mu} \right) \left(1 + \frac{P_{up} i}{8} (\mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, m_p = the Planck mass, m_μ = the muon mass, P_{up} = the universal parabolic constant, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} \lambda_\mu &= 1.17344411764093 \dots \times 10^{-14} \text{ m/cycle} && \text{prediction} \\ \lambda_\mu &= 1.173444110(26) \times 10^{-14} \text{ m/cycle} && \text{CODATA 2022, } \sigma = +0.29 \\ \lambda_\mu &= 1.173444110(26) \times 10^{-14} \text{ m/cycle} && \text{CODATA 2018, } \sigma = +0.29 \\ &&& \Delta_{\text{precision}} = +0.00 \\ &&& \Delta_{\text{scaled}} = +0.00 \end{aligned}$$

proton Compton wavelength

8.7591(30)

$$\lambda_+ = 2\pi \left(\frac{l_p m_p}{m_+} \right) \left(1 + \frac{P_{up}}{8} i (\mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, m_p = the Planck mass, m_+ = the proton mass, P_{up} = the universal parabolic constant, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\lambda_+ = 1.32140985399774 \dots \times 10^{-15}$ m/cycle	prediction
$\lambda_+ = 1.32140985360(41) \times 10^{-15}$ m/cycle	CODATA 2022, $\sigma = +0.97$
$\lambda_+ = 1.32140985539(40) \times 10^{-15}$ m/cycle	CODATA 2018, $\sigma = -3.48$
	$\Delta_{\text{precision}} = -0.01$
	$\Delta_{\text{scaled}} = -4.48$

neutron Compton wavelength

8.7601(56)

$$\lambda_n = 2\pi \left(\frac{l_p m_p}{m_n} \right) \left(1 + \frac{P_{up}}{8} i (\mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, m_p = the Planck mass, m_n = the neutron mass, P_{up} = the universal parabolic constant, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\lambda_n = 1.31959090499128 \dots \times 10^{-15}$ m/cycle	prediction
$\lambda_n = 1.31959090382(67) \times 10^{-15}$ m/cycle	CODATA 2022, $\sigma = +1.75$
$\lambda_n = 1.31959090581(75) \times 10^{-15}$ m/cycle	CODATA 2018, $\sigma = -1.09$
	$\Delta_{\text{precision}} = +0.05$
	$\Delta_{\text{scaled}} = -2.65$

$$\lambda_{\tau} = 2\pi \left(\frac{l_p m_p}{m_{\tau}} \right) \left(1 + \frac{P_{up} i}{8} (\mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, m_p = the Planck mass, m_{τ} = the tau mass, P_{up} = the universal parabolic constant, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\lambda_{\tau} = 6.97770282157982 \dots \times 10^{-16} \text{ m/cycle}$	prediction
$\lambda_{\tau} = 6.97771(47) \times 10^{-16} \text{ m/cycle}$	CODATA 2022, $\sigma = -0.02$
$\lambda_{\tau} = 6.97771(47) \times 10^{-16} \text{ m/cycle}$	CODATA 2018, $\sigma = -0.02$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

$$\lambda_{-} = \left(\frac{l_p m_p}{m_e} \right) \left(1 + \frac{P_{up} i}{8} (\mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where l_p = the Planck length, m_p = the Planck mass, m_e = the electron mass, P_{up} = the universal parabolic constant, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\lambda_{-} = 3.86159267548653 \dots \times 10^{-13} \text{ m/cycle}$	prediction
$\lambda_{-} = 3.8615926744(12) \times 10^{-13} \text{ m/cycle}$	CODATA 2022, $\sigma = +0.91$
$\lambda_{-} = 3.8615926796(12) \times 10^{-13} \text{ m/cycle}$	CODATA 2018, $\sigma = -4.55$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -4.33$

Also listed as the *natural unit of length*.

reduced muon Compton wavelength

8.72(23)

$$\lambda_{\mu-} = \left(\frac{l_p m_p}{m_\mu} \right) \left(1 + \frac{P_{up} i}{8} (\mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where l_p = the Planck length, m_p = the Planck mass, m_μ = the muon mass, P_{up} = the universal parabolic constant, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\lambda_{\mu-} = 1.86759431764662 \dots \times 10^{-15} \text{ m/cycle} \quad \text{prediction}$$

$$\lambda_{\mu-} = 1.867594306(42) \times 10^{-15} \text{ m/cycle} \quad \text{CODATA 2022, } \sigma = +0.28$$

$$\lambda_{\mu-} = 1.867594306(42) \times 10^{-15} \text{ m/cycle} \quad \text{CODATA 2018, } \sigma = +0.28$$

$$\Delta_{\text{precision}} = +0.00$$

$$\Delta_{\text{scaled}} = +0.00$$

reduced proton Compton wavelength

8.7591(31)

$$\lambda_{+-} = \left(\frac{l_p m_p}{m_+} \right) \left(1 + \frac{P_{up} i}{8} (\mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where l_p = the Planck length, m_p = the Planck mass, m_+ = the proton mass, P_{up} = the universal parabolic constant, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\lambda_{+-} = 2.10308910114080 \dots \times 10^{-16} \text{ m/cycle} \quad \text{prediction}$$

$$\lambda_{+-} = 2.10308910051(66) \times 10^{-16} \text{ m/cycle} \quad \text{CODATA 2022, } \sigma = +0.96$$

$$\lambda_{+-} = 2.10308910336(64) \times 10^{-16} \text{ m/cycle} \quad \text{CODATA 2018, } \sigma = -3.47$$

$$\Delta_{\text{precision}} = -0.01$$

$$\Delta_{\text{scaled}} = -4.45$$

reduced neutron Compton wavelength

8.7603(58)

$$\lambda_{n_-} = \left(\frac{l_p m_p}{m_n} \right) \left(1 + \frac{P_{up} i}{8} (\mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where l_p = the Planck length, m_p = the Planck mass, m_n = the neutron mass, P_{up} = the universal parabolic constant, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\lambda_{n_-} = 2.10019415388470 \dots \times 10^{-16}$ m/cycle	prediction
$\lambda_{n_-} = 2.1001941520(11) \times 10^{-16}$ m/cycle	CODATA 2022, $\sigma = +1.71$
$\lambda_{n_-} = 2.1001941552(12) \times 10^{-16}$ m/cycle	CODATA 2018, $\sigma = -1.10$
	$\Delta_{\text{precision}} = +0.04$
	$\Delta_{\text{scaled}} = -2.67$

reduced tau Compton wavelength

43.(675)

$$\lambda_{\tau_-} = \left(\frac{l_p m_p}{m_\tau} \right) \left(1 + \frac{P_{up} i}{8} (\mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where l_p = the Planck length, m_p = the Planck mass, m_τ = the tau mass, P_{up} = the universal parabolic constant, i = the imaginary unit, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\lambda_{\tau_-} = 1.11053589548069 \dots \times 10^{-16}$ m/cycle	prediction
$\lambda_{\tau_-} = 1.110538(75) \times 10^{-16}$ m/cycle	CODATA 2022, $\sigma = -0.03$
$\lambda_{\tau_-} = 1.110538(75) \times 10^{-16}$ m/cycle	CODATA 2018, $\sigma = -0.03$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

Wien frequency displacement law constant

42.04047(85)

$$b' = \frac{1}{2\pi} \left(\frac{\nu_{\text{peak}}}{t_p T_p} \right) \left(1 + 18 \left(\frac{2\pi}{4s} \right)^2 (\mathfrak{K}_1^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where ν_{peak} = the peak emission of spectral flux per unit frequency, π = Archimedes' constant, t_p = the Planck time, T_p = the Planck temperature, s = the arc length of the unit lemniscate, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} b' &= 5.878925757\textcolor{green}{96309} \dots \times 10^{10} \text{ Hz/K} && \text{prediction} \\ b' &= 5.878925757 \times 10^{10} \text{ Hz/K} && \text{CODATA 2022, 10-digit match} \\ b' &= 5.878925757 \times 10^{10} \text{ Hz/K} && \text{CODATA 2018, 10-digit match} \\ &&& \Delta_{\text{precision}} = +0.00 \\ &&& \Delta_{\text{scaled}} = +0.00 \end{aligned}$$

$\nu_{\text{peak}} = 2.82143937212207 \dots$ is the solution to the equation $\frac{x e^x}{e^x - 1} = 3$.

Boltzmann constant in Hz/K

42.0407(24)

$$\ddot{k}_B = \frac{1}{2\pi} \left(\frac{1}{t_p T_p} \right) \left(1 + 18 \left(\frac{2\pi}{4s} \right)^2 (\mathfrak{K}_1^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where π = Archimedes' constant, t_p = the Planck time, T_p = the Planck temperature, s = the arc length of the unit lemniscate, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} \ddot{k}_B &= 2.083661912\textcolor{green}{44486} \dots \times 10^{10} \text{ Hz/K} && \text{prediction} \\ \ddot{k}_B &= 2.083661912 \times 10^{10} \text{ Hz/K} && \text{CODATA 2022, 10-digit match} \\ \ddot{k}_B &= 2.083661912 \times 10^{10} \text{ Hz/K} && \text{CODATA 2018, 10-digit match} \\ &&& \Delta_{\text{precision}} = +0.00 \\ &&& \Delta_{\text{scaled}} = +0.00 \end{aligned}$$

kelvin-hertz relationship

42.0414(24)

$$K : \text{Hz} = \frac{1}{2\pi} \left(\frac{K}{t_p T_p} \right) \left(1 + 18 \left(\frac{2\pi}{4s} \right)^2 (\mathfrak{K}_1^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where π = Archimedes' constant, K = the kelvin, t_p = the Planck time, T_p = the Planck temperature, s = the arc length of the unit lemniscate, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$K : \text{Hz} = 2.083661912\textcolor{teal}{44486} \dots \times 10^{10} \text{ Hz}$	prediction
$K : \text{Hz} = 2.083661912 \times 10^{10} \text{ Hz}$	CODATA 2022, 10-digit match
$K : \text{Hz} = 2.083661912 \times 10^{10} \text{ Hz}$	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

hertz-kelvin relationship

-42.0414(24)

$$\text{Hz} : K = 2\pi \left(\frac{t_p T_p}{s} \right) \left(1 - 18 \left(\frac{2\pi}{4s} \right)^2 (\mathfrak{K}_1^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where π = Archimedes' constant, t_p = the Planck time, T_p = the Planck temperature, s = the second, s = the arc length of the unit lemniscate, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\text{Hz} : K = 4.799243073\textcolor{teal}{02320} \dots \times 10^{-11} \text{ K}$	prediction
$\text{Hz} : K = 4.799243073 \times 10^{-11} \text{ K}$	CODATA 2022, 10-digit match
$\text{Hz} : K = 4.799243073 \times 10^{-11} \text{ K}$	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

Wien wavelength displacement law constant

-45.6561(28)

$$b = 2\pi \left(\frac{l_p T_p}{\lambda_{\text{peak}}} \right) \left(1 + P^2 \left(\mathfrak{K}_1^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2 \right) \boxtimes \right)$$

Where π = Archimedes' constant, λ_{peak} = the peak wavelength of spectral radiance per unit wavelength, l_p = the Planck length, T_p = the Planck temperature, P = the plastic constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} b &= 2.89777195379426 \dots \times 10^{-3} \text{ m} \cdot \text{K/cycle} && \text{prediction} \\ b &= 2.897771955 \times 10^{-3} \text{ m} \cdot \text{K/cycle} && \text{CODATA 2022, 9-digit match} \\ b &= 2.897771955 \times 10^{-3} \text{ m} \cdot \text{K/cycle} && \text{CODATA 2018, 9-digit match} \\ &&& \Delta_{\text{precision}} = +0.00 \\ &&& \Delta_{\text{scaled}} = +0.00 \end{aligned}$$

$$\lambda_{\text{peak}} = 4.96511423174427 \dots \text{ is the solution to the equation } \frac{x e^x}{e^x - 1} = \mathbf{5}.$$

second radiation constant

-45.6561(28)

$$c_2 = 2\pi \left(l_p T_p \right) \left(1 + P^2 \left(\mathfrak{K}_1^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2 \right) \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, T_p = the Planck temperature, P = the plastic constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} c_2 &= 1.43877687681333 \dots \times 10^{-2} \text{ m} \cdot \text{K} && \text{prediction} \\ c_2 &= 1.438776877 \times 10^{-2} \text{ m} \cdot \text{K} && \text{CODATA 2022, 10-digit match} \\ c_2 &= 1.438776877 \times 10^{-2} \text{ m} \cdot \text{K} && \text{CODATA 2018, 10-digit match} \\ &&& \Delta_{\text{precision}} = +0.00 \\ &&& \Delta_{\text{scaled}} = +0.00 \end{aligned}$$

inverse meter-kelvin relationship

-45.6543(35)

$$\frac{1}{\text{m}} : \text{K} = 2\pi \left(\frac{l_p T_p}{\text{m}} \right) \left(1 + P^2 (\mathfrak{K}_1^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, T_p = the Planck temperature, m = the meter, P = the plastic constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$1/\text{m} : \text{K} = 1.43877687\textcolor{teal}{681333} \dots \times 10^{-2} \text{ K}$	prediction
$1/\text{m} : \text{K} = 1.438776877 \times 10^{-2} \text{ K}$	CODATA 2022, 10-digit match
$1/\text{m} : \text{K} = 1.438776877 \times 10^{-2} \text{ K}$	CODATA 2018, 10-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

kelvin-inverse meter relationship

45.6543(35)

$$\text{K} : \frac{1}{\text{m}} = \frac{1}{2\pi} \left(\frac{\text{K}}{l_p T_p} \right) \left(1 - P^2 (\mathfrak{K}_1^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes \right)$$

Where π = Archimedes' constant, K = the kelvin, l_p = the Planck length, T_p = the Planck temperature, P = the plastic constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\text{K} : 1/\text{m} = 6.95034800\textcolor{teal}{805251} \dots \times 10^1 \text{ cycles/m}$	prediction
$\text{K} : 1/\text{m} = 6.950348004 \times 10^1 \text{ cycles/m}$	CODATA 2022, 9.23-digit match
$\text{K} : 1/\text{m} = 6.950348004 \times 10^1 \text{ cycles/m}$	CODATA 2018, 9.23-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

$$\dot{k}_B = \frac{1}{2\pi} \left(\frac{1}{l_p T_p} \right) \left(1 - P^2 \left(\mathfrak{K}_1^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2 \right) \boxtimes \right)$$

Where π = Archimedes' constant, l_p = the Planck length, T_p = the Planck temperature, K_{-6} = the Khinchin mean of order -6 , $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$\dot{k}_B = 6.95034800805251 \dots \times 10^1 \text{ cycles/m} \cdot \text{K}$

prediction

$\dot{k}_B = 6.950348004 \times 10^1 \text{ cycles/m} \cdot \text{K}$

CODATA 2022, 9.23-digit match

$\dot{k}_B = 6.950348004 \times 10^1 \text{ cycles/m} \cdot \text{K}$

CODATA 2018, 9.23-digit match

$\Delta_{\text{precision}} = +0.00$ $\Delta_{\text{scaled}} = +0.00$

hartree-kelvin relationship

-54.431024(11)

$$E_h : K = \mathfrak{K}_1^4 \left(\frac{T_p m_e}{m_p} \right) \left(1 + \frac{s}{\sqrt{2\pi}} \left(\mathfrak{K}_1^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2 \right) \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, T_p = the Planck temperature, m_e = the electron mass, m_p = the Planck mass, s = the arc length of the unit lemniscate, π = Archimedes' constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$E_h : K = 3.15775024803817 \dots \times 10^5 \text{ K}$	prediction
$E_h : K = 3.1577502480398(34) \times 10^5 \text{ K}$	CODATA 2022, $\sigma = -0.48$
$E_h : K = 3.1577502480407(61) \times 10^5 \text{ K}$	CODATA 2018, $\sigma = -0.41$
	$\Delta_{\text{precision}} = +0.25$
	$\Delta_{\text{scaled}} = -0.15$

kelvin-hartree relationship

54.431320(11)

$$K : E_h = \frac{1}{\mathfrak{K}_1^4} \left(\frac{K m_p}{T_p m_e} \right) \left(1 + \frac{s}{\sqrt{2\pi}} \left(\mathfrak{K}_1^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2 \right) \boxtimes \right)^{-1}$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, K = the kelvin, m_p = the Planck mass, T_p = the Planck temperature, m_e = the electron mass, s = the arc length of the unit lemniscate, π = Archimedes' constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$K : E_h = 3.16681156345811 \dots \times 10^{-6} E_h$	prediction
$K : E_h = 3.1668115634564(35) \times 10^{-6} E_h$	CODATA 2022, $\sigma = +0.50$
$K : E_h = 3.1668115634556(61) \times 10^{-6} E_h$	CODATA 2018, $\sigma = +0.41$
	$\Delta_{\text{precision}} = +0.24$
	$\Delta_{\text{scaled}} = +0.13$

electron molar mass

-283.54(31)

$$M_e = \frac{6}{e^\gamma} \kappa_1^2 \left(\frac{C \text{ kg } m_e}{q_p m_p} \right) \left(1 + 18 C_Q \left(\kappa_1^2 + \kappa_3^2 + \kappa_4^2 \right) \boxtimes \right)$$

Where e = Euler's number, γ = the Euler-Mascheroni constant, κ_1 = the 1st hyperbolic partition constant, C = the coulomb, kg = the kilogram, m_e = the electron mass, q_p = the Planck charge, m_p = the Planck mass, C_Q = the QRS constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} M_e &= 5.48579909\mathbf{796588} \dots \times 10^{-7} \text{ kg/mol} && \text{prediction} \\ M_e &= 5.4857990962(17) \times 10^{-7} \text{ kg/mol} && \text{CODATA 2022, } \sigma = +1.04 \\ M_e &= 5.4857990888(17) \times 10^{-7} \text{ kg/mol} && \text{CODATA 2018, } \sigma = +5.39 \\ &&& \Delta_{\text{precision}} = +0.00 \\ &&& \Delta_{\text{scaled}} = +4.35 \end{aligned}$$

muon molar mass

-283.50(22)

$$M_\mu = \frac{6}{e^\gamma} \kappa_1^2 \left(\frac{C \text{ kg } m_\mu}{q_p m_p} \right) \left(1 + 18 C_Q \left(\kappa_1^2 + \kappa_3^2 + \kappa_4^2 \right) \boxtimes \right)$$

Where e = Euler's number, γ = the Euler-Mascheroni constant, κ_1 = the 1st hyperbolic partition constant, C = the coulomb, kg = the kilogram, m_μ = the muon mass, q_p = the Planck charge, m_p = the Planck mass, C_Q = the QRS constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} M_\mu &= 1.13428925\mathbf{199289} \dots \times 10^{-4} \text{ kg/mol} && \text{prediction} \\ M_\mu &= 1.134289258(25) \times 10^{-4} \text{ kg/mol} && \text{CODATA 2022, } \sigma = -0.24 \\ M_\mu &= 1.134289259(25) \times 10^{-4} \text{ kg/mol} && \text{CODATA 2018, } \sigma = -0.28 \\ &&& \Delta_{\text{precision}} = +0.00 \\ &&& \Delta_{\text{scaled}} = -0.04 \end{aligned}$$

molar mass constant

-283.5449(30)

$$M_{A_{\text{mass}}} = \frac{6}{e^\gamma} \kappa_1^2 \left(\frac{C \text{ kg } A_{\text{mass}}}{q_p m_p} \right) \left(1 + 18 C_Q \left(\kappa_1^2 + \kappa_3^2 + \kappa_4^2 \right) \boxtimes \right)$$

Where e = Euler's number, γ = the Euler-Mascheroni constant, κ_1 = the 1st hyperbolic partition constant, C = the coulomb, kg = the kilogram, A_{mass} = the atomic mass constant, q_p = the Planck charge, m_p = the Planck mass, C_Q = the QRS constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} M_{A_{\text{mass}}} &= 1.00000000136176 \dots \times 10^{-3} \text{ kg/mol} && \text{prediction} \\ M_{A_{\text{mass}}} &= 1.00000000105(31) \times 10^{-3} \text{ kg/mol} && \text{CODATA 2022, } \sigma = +1.01 \\ M_{A_{\text{mass}}} &= 0.99999999965(30) \times 10^{-3} \text{ kg/mol} && \text{CODATA 2018, } \sigma = +5.71 \\ &&& \Delta_{\text{precision}} = -0.01 \\ &&& \Delta_{\text{scaled}} = +4.67 \end{aligned}$$

proton molar mass

-283.5447(30)

$$M_+ = \frac{6}{e^\gamma} \kappa_1^2 \left(\frac{C \text{ kg } m_+}{q_p m_p} \right) \left(1 + 18 C_Q \left(\kappa_1^2 + \kappa_3^2 + \kappa_4^2 \right) \boxtimes \right)$$

Where e = Euler's number, γ = the Euler-Mascheroni constant, κ_1 = the 1st hyperbolic partition constant, C = the coulomb, kg = the kilogram, m_+ = the proton mass, q_p = the Planck charge, m_p = the Planck mass, C_Q = the QRS constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} M_+ &= 1.00727646795396 \dots \times 10^{-3} \text{ kg/mol} && \text{prediction} \\ M_+ &= 1.00727646764(31) \times 10^{-3} \text{ kg/mol} && \text{CODATA 2022, } \sigma = +1.01 \\ M_+ &= 1.00727646627(31) \times 10^{-3} \text{ kg/mol} && \text{CODATA 2018, } \sigma = +5.43 \\ &&& \Delta_{\text{precision}} = +0.00 \\ &&& \Delta_{\text{scaled}} = +4.42 \end{aligned}$$

neutron molar mass

-283.5457(57)

$$M_n = \frac{6}{e^\gamma} \kappa_1^2 \left(\frac{C \text{ kg } m_n}{q_p m_p} \right) \left(1 + 18 C_Q \left(\kappa_1^2 + \kappa_3^2 + \kappa_4^2 \right) \boxtimes \right)$$

Where e = Euler's number, γ = the Euler-Mascheroni constant, κ_1 = the 1st hyperbolic partition constant, C = the coulomb, kg = the kilogram, m_n = the neutron mass, q_p = the Planck charge, m_p = the Planck mass, C_Q = the QRS constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$M_n = 1.00866491\textcolor{teal}{684647} \dots \times 10^{-3} \text{ kg/mol}$	prediction
$M_n = 1.00866491712(51) \times 10^{-3} \text{ kg/mol}$	CODATA 2022, $\sigma = -0.54$
$M_n = 1.00866491560(57) \times 10^{-3} \text{ kg/mol}$	CODATA 2018, $\sigma = +2.19$
	$\Delta_{\text{precision}} = +0.05$
	$\Delta_{\text{scaled}} = +2.67$

tau molar mass

-287.(680)

$$M_\tau = \frac{6}{e^\gamma} \kappa_1^2 \left(\frac{C \text{ kg } m_\tau}{q_p m_p} \right) \left(1 + 18 C_Q \left(\kappa_1^2 + \kappa_3^2 + \kappa_4^2 \right) \boxtimes \right)$$

Where e = Euler's number, γ = the Euler-Mascheroni constant, κ_1 = the 1st hyperbolic partition constant, C = the coulomb, kg = the kilogram, m_τ = the tau mass, q_p = the Planck charge, m_p = the Planck mass, C_Q = the QRS constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$M_\tau = 1.90754\textcolor{teal}{046781407} \dots \times 10^{-3} \text{ kg/mol}$	prediction
$M_\tau = 1.90754(13) \times 10^{-3} \text{ kg/mol}$	CODATA 2022, $\sigma = +0.00$
$M_\tau = 1.90754(13) \times 10^{-3} \text{ kg/mol}$	CODATA 2018, $\sigma = +0.00$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

deuteron molar mass

-283.5443(31)

$$M_{\text{de}} = \frac{6}{e^\gamma} \kappa_1^2 \left(\frac{C \text{ kg } m_{\text{de}}}{q_p m_p} \right) \left(1 + 18 C_Q \left(\kappa_1^2 + \kappa_3^2 + \kappa_4^2 \right) \boxtimes \right)$$

Where e = Euler's number, γ = the Euler-Mascheroni constant, κ_1 = the 1st hyperbolic partition constant, C = the coulomb, kg = the kilogram, m_{de} = the deuteron mass, q_p = the Planck charge, m_p = the Planck mass, C_Q = the QRS constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$M_{\text{de}} = 2.01355321\textcolor{green}{529080} \dots \times 10^{-3} \text{ kg/mol}$	prediction
$M_{\text{de}} = 2.01355321466(63) \times 10^{-3} \text{ kg/mol}$	CODATA 2022, $\sigma = +1.00$
$M_{\text{de}} = 2.01355321205(61) \times 10^{-3} \text{ kg/mol}$	CODATA 2018, $\sigma = +5.31$
	$\Delta_{\text{precision}} = -0.01$
	$\Delta_{\text{scaled}} = +4.28$

helion molar mass

-283.5447(30)

$$M_{\text{he}} = \frac{6}{e^\gamma} \kappa_1^2 \left(\frac{C \text{ kg } m_{\text{he}}}{q_p m_p} \right) \left(1 + 18 C_Q \left(\kappa_1^2 + \kappa_3^2 + \kappa_4^2 \right) \boxtimes \right)$$

Where e = Euler's number, γ = the Euler-Mascheroni constant, κ_1 = the 1st hyperbolic partition constant, C = the coulomb, kg = the kilogram, m_{he} = the helion mass, q_p = the Planck charge, m_p = the Planck mass, C_Q = the QRS constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$M_{\text{he}} = 3.01493225\textcolor{green}{103642} \dots \times 10^{-3} \text{ kg/mol}$	prediction
$M_{\text{he}} = 3.01493225010(94) \times 10^{-3} \text{ kg/mol}$	CODATA 2022, $\sigma = +1.00$
$M_{\text{he}} = 3.01493224613(91) \times 10^{-3} \text{ kg/mol}$	CODATA 2018, $\sigma = +5.39$
	$\Delta_{\text{precision}} = -0.01$
	$\Delta_{\text{scaled}} = +4.36$

triton molar mass

-283.5439(30)

$$M_{\text{tri}} = \frac{6}{e^\gamma} \kappa_1^2 \left(\frac{C \text{ kg } m_{\text{tri}}}{q_p m_p} \right) \left(1 + 18 C_Q \left(\kappa_1^2 + \kappa_3^2 + \kappa_4^2 \right) \boxtimes \right)$$

Where e = Euler's number, γ = the Euler-Mascheroni constant, κ_1 = the 1st hyperbolic partition constant, C = the coulomb, kg = the kilogram, m_{tri} = the triton mass, q_p = the Planck charge, m_p = the Planck mass, C_Q = the QRS constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$M_{\text{tri}} = 3.01550071982806 \dots \times 10^{-3} \text{ kg/mol}$	prediction
$M_{\text{tri}} = 3.01550071913(94) \times 10^{-3} \text{ kg/mol}$	CODATA 2022, $\sigma = +0.74$
$M_{\text{tri}} = 3.01550071517(92) \times 10^{-3} \text{ kg/mol}$	CODATA 2018, $\sigma = +5.06$
	$\Delta_{\text{precision}} = -0.01$
	$\Delta_{\text{scaled}} = +4.30$

alpha particle molar mass

-283.5454(30)

$$M_{\alpha} = \frac{6}{e^\gamma} \kappa_1^2 \left(\frac{C \text{ kg } m_{\alpha}}{q_p m_p} \right) \left(1 + 18 C_Q \left(\kappa_1^2 + \kappa_3^2 + \kappa_4^2 \right) \boxtimes \right)$$

Where e = Euler's number, γ = the Euler-Mascheroni constant, κ_1 = the 1st hyperbolic partition constant, C = the coulomb, kg = the kilogram, m_{α} = the alpha particle mass, q_p = the Planck charge, m_p = the Planck mass, C_Q = the QRS constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$M_{\alpha} = 4.00150618457836 \dots \times 10^{-3} \text{ kg/mol}$	prediction
$M_{\alpha} = 4.0015061833(12) \times 10^{-3} \text{ kg/mol}$	CODATA 2022, $\sigma = +1.07$
$M_{\alpha} = 4.0015061777(12) \times 10^{-3} \text{ kg/mol}$	CODATA 2018, $\sigma = +5.73$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +4.67$

molar mass of carbon-12

-283.544(33)

$$M_{12_C} = \frac{6}{e^\gamma} \kappa_1^2 \left(\frac{12 \text{ C kg } A_{\text{mass}}}{q_p m_p} \right) \left(1 + 18 C_Q \left(\kappa_1^2 + \kappa_3^2 + \kappa_4^2 \right) \boxtimes \right)$$

Where e = Euler's number, γ = the Euler-Mascheroni constant, κ_1 = the 1st hyperbolic partition constant, C = the coulomb, kg = the kilogram, A_{mass} = the atomic mass constant, q_p = the Planck charge, m_p = the Planck mass, C_Q = the QRS constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$M_{12_C} = 1.200000001\textcolor{teal}{63412} \dots \times 10^{-2} \text{ kg/mol} \quad \text{prediction}$$

$$M_{12_C} = 1.20000000126(36) \times 10^{-2} \text{ kg/mol} \quad \text{CODATA 2022, } \sigma = +1.04$$

$$M_{12_C} = 1.19999999958(36) \times 10^{-2} \text{ kg/mol} \quad \text{CODATA 2018, } \sigma = +5.71$$

$$\Delta_{\text{precision}} = +0.00$$

$$\Delta_{\text{scaled}} = +4.67$$

Avogadro constant

-283.5364(83)

$$N_A = \frac{6}{e^\gamma} \kappa_1^2 \left(\frac{\text{C kg}}{q_p m_p} \right) \left(1 + 18 C_Q \left(\kappa_1^2 + \kappa_3^2 + \kappa_4^2 \right) \boxtimes \right)$$

Where e = Euler's number, γ = the Euler-Mascheroni constant, κ_1 = the 1st hyperbolic partition constant, C = the coulomb, kg = the kilogram, q_p = the Planck charge, m_p = the Planck mass, C_Q = the QRS constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$N_A = 6.02214076\textcolor{teal}{443131} \dots \times 10^{23} \text{ 1/mol} \quad \text{prediction}$$

$$N_A = 6.02214076 \times 10^{23} \text{ 1/mol} \quad \text{CODATA 2022, 9.13-digit match}$$

$$N_A = 6.02214076 \times 10^{23} \text{ 1/mol} \quad \text{CODATA 2018, 9.13-digit match}$$

$$\Delta_{\text{precision}} = +0.00$$

$$\Delta_{\text{scaled}} = +0.00$$

Sackur-Tetrode constant (1K, 100 kPa)

-28.0784(13)

$$ST_0 = \frac{5}{2} + \log \left(\left(\frac{1}{2\pi} \frac{K A_{\text{mass}}}{l_p^2 T_p m_p} \right)^{3/2} \frac{K E_p}{T_p p_0} \right) \left(1 + j_{0,1} \left(\frac{2\pi}{14} \right) (\mathfrak{K}_1^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes \right)$$

Where $\log(x)$ = the natural logarithm, π = Archimedes' constant, K = kelvin, A_{mass} = the atomic mass constant, l_p = the Planck length, T_p = the Planck temperature, m_p = the Planck mass, E_p = the Planck energy, $p_0 = 100.000000000000 \dots$ kPa, $j_{0,1}$ = the 1st root of the Bessel function, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$ST_0 = -1.15170753432430 \dots$	prediction
$ST_0 = -1.15170753496(47)$	CODATA 2022, $\sigma = +1.35$
$ST_0 = -1.15170753706(45)$	CODATA 2018, $\sigma = -6.08$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = -4.66$

The Planck energy is equal to the product of the Planck length squared and the Planck mass, divided by the Planck time squared,

$$E_p = \frac{l_p^2 m_p}{t_p^2}.$$

Sackur-Tetrode constant (1K, 101.325 kPa)

-27.9776(13)

$$ST_1 = \frac{5}{2} + \log\left(\left(\frac{1}{2\pi} \frac{\text{K } A_{\text{mass}}}{l_p^2 T_p m_p}\right)^{3/2} \frac{\text{K } E_p}{T_p p_1}\right)\left(1 + \frac{1}{G_{Ga}} \left(\frac{4\pi}{14}\right) (\mathfrak{K}_1^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes\right)$$

Where $\log(x)$ = the natural logarithm, π = Archimedes' constant, A_{mass} = the atomic mass constant, K = kelvin, l_p = the Planck length, m_p = the Planck mass, T_p = the Planck temperature, E_p = the Planck energy, p_1 = 101.325000000000 kPa, G_{Ga} = Gauss's constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$ST_1 = -1.16487052045596 \dots$

prediction

$ST_1 = -1.16487052149(47)$

CODATA 2022, $\sigma = +2.20$

$ST_1 = -1.16487052358(45)$

CODATA 2018, $\sigma = -6.94$

$\Delta_{\text{precision}} = -0.02$

$\Delta_{\text{scaled}} = -4.64$

The Planck energy is equal to the product of the Planck length squared and the Planck mass, divided by the Planck time squared,

$$E_p = \frac{l_p^2 m_p}{t_p^2}.$$

$$\sigma = \frac{3}{\textcolor{red}{5}} \left(\frac{4\pi}{\Gamma(\textcolor{red}{5})} \right)^2 \left(\frac{m_p}{t_p^3 T_p^4} \right) \left(1 + (18 + 35) \left(\frac{2K}{14} \right) (\mathfrak{K}_1^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where π = Archimedes' constant, $\Gamma(x)$ = the gamma function, m_p = the Planck mass, t_p = the Planck time, T_p = the Planck temperature, K = Catalan's constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\sigma = 5.670374419\textcolor{green}{87317} \dots \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$$

prediction

$$\sigma = 5.670374419 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$$

CODATA 2022, 10-digit match

$$\sigma = 5.670374419 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$$

CODATA 2018, 10-digit match

$$\Delta_{\text{precision}} = +0.00$$

$$\Delta_{\text{scaled}} = +0.00$$

conventional value of volt-90

1.066651(50)

$$V_{90} = \text{volt} \left(1 + \frac{L_1}{32} \left(\mathfrak{K}_1^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2 \right) \boxtimes \right)$$

Where L_1 = the 1st lemniscate constant, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$V_{90} = 1.00000010665218 \dots \text{ V}$	prediction
$V_{90} = 1.00000010666 \text{ V}$	CODATA 2022, 11.11-digit match
$V_{90} = 1.00000010666 \text{ V}$	CODATA 2018, 11.11-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

conventional value of watt-90

1.955351(50)

$$W_{90} = \text{watt} \left(1 - \frac{j_{0,1}}{32} \left(\mathfrak{K}_1^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2 \right) \boxtimes \right)$$

Where $j_{0,1}$ = the 1st root of the Bessel function, $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$W_{90} = 1.00000019552285 \dots \text{ W}$	prediction
$W_{90} = 1.00000019553 \text{ W}$	CODATA 2022, 11.15-digit match
$W_{90} = 1.00000019553 \text{ W}$	CODATA 2018, 11.15-digit match
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = +0.00$

$$A_{\text{ef}\nabla} = \mathfrak{K}_1^7 \left(\frac{m_e^3}{t_p^2 q_p m_p^2} \right) \left(1 + \sqrt{35} \left(\frac{V_{\text{fe}}}{6} \right) (\mathfrak{K}_2^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, m_e = the electron mass, t_p = the Planck time, q_p = the Planck charge, m_p = the Planck mass, V_{fe} = the figure eight knot hyperbolic volume, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$A_{\text{ef}\nabla} = 9.71736242779302 \dots \times 10^{21} \text{ V/m}^2$	prediction
$A_{\text{ef}\nabla} = 9.7173624424(30) \times 10^{21} \text{ V/m}^2$	CODATA 2022, $\sigma = -4.87$
$A_{\text{ef}\nabla} = 9.7173624292(29) \times 10^{21} \text{ V/m}^2$	CODATA 2018, $\sigma = -0.49$
	$\Delta_{\text{precision}} = -0.02$
	$\Delta_{\text{scaled}} = +4.55$

$$A_{\text{ef}\nabla} = \mathfrak{K}_1^7 \left(\frac{m_e^3}{t_p^2 q_p m_p^2} \right) \left(1 - \frac{1}{\sqrt{C_{\text{CFP}}}} \left(\frac{4\pi}{18} \right) (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{\text{ef}\nabla} = \mathfrak{K}_1^7 \left(\frac{m_e^3}{t_p^2 q_p m_p^2} \right) \left(1 - \mathcal{S} \left(\frac{2\pi}{32} \right) (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{\text{ef}\nabla} = \mathfrak{K}_1^7 \left(\frac{m_e^3}{t_p^2 q_p m_p^2} \right) \left(1 + \frac{s^2}{35} \left(\frac{32}{4\pi} \right) (\mathfrak{K}_2^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{\text{current}} = \mathfrak{K}_1^5 \left(\frac{q_p m_e}{t_p m_p} \right) \left(1 - C_{\text{CFP}} \left(\frac{V_{\text{fe}}}{8} \right) (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, q_p = the Planck charge, m_e = the electron mass, t_p = the Planck time, m_p = the Planck mass, C_{CFP} = the fixed point of the hyperbolic cotangent, V_{fe} = the figure eight knot hyperbolic volume, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$A_{\text{current}} = 6.62361823748092 \dots \times 10^{-3} \text{ A}$	prediction
$A_{\text{current}} = 6.6236182375082(72) \times 10^{-3} \text{ A}$	CODATA 2022, $\sigma = -3.79$
$A_{\text{current}} = 6.623618237510(13) \times 10^{-3} \text{ A}$	CODATA 2018, $\sigma = -2.24$
	$\Delta_{\text{precision}} = +0.26$
	$\Delta_{\text{scaled}} = -0.15$

$$A_{\text{mfd}} = \mathfrak{x}_1^3 \left(\frac{m_e^2}{t_p q_p m_p} \right) \left(1 + \sqrt{8} \left(\frac{4\pi}{35} \right) (\mathfrak{x}_2^2 + \mathfrak{x}_3^2 + \mathfrak{x}_4^2) \boxtimes \right)$$

Where $\mathfrak{x}_1$ = the 1st hyperbolic partition constant, m_e = the electron mass, t_p = the Planck time, q_p = the Planck charge, m_p = the Planck mass, π = Archimedes' constant, $\mathfrak{x}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{x}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{x}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$A_{\text{mfd}} = 2.35051756770603 \dots \times 10^5 \text{ T}$	prediction
$A_{\text{mfd}} = 2.35051757077(73) \times 10^5 \text{ T}$	CODATA 2022, $\sigma = -4.20$
$A_{\text{mfd}} = 2.35051756758(71) \times 10^5 \text{ T}$	CODATA 2018, $\sigma = +0.18$
	$\Delta_{\text{precision}} = -0.01$
	$\Delta_{\text{scaled}} = +4.49$

$$A_{\text{mfd}} = \mathfrak{x}_1^3 \left(\frac{m_e^2}{t_p q_p m_p} \right) \left(1 + G_{\text{Gi}} (\mathfrak{x}_2^2 + \mathfrak{x}_3^2 + \mathfrak{x}_4^2) \boxtimes \right)$$

$$A_{\text{edm}} = \frac{1}{\mathfrak{K}_1} \left(\frac{l_p q_p m_p}{m_e} \right) \left(1 + \frac{P}{L_2} \left(\frac{4\pi}{\textcolor{red}{5}!} \right) (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, q_p = the Planck charge, m_p = the Planck mass, m_e = the electron mass, P = the plastic constant, L_2 = the 2nd lemniscate constant, π = Archimedes' constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} A_{\text{edm}} &= 8.4783536\textcolor{green}{2016674} \dots \times 10^{-30} \text{ m} \cdot \text{C} && \text{prediction} \\ A_{\text{edm}} &= 8.4783536198(13) \times 10^{-30} \text{ m} \cdot \text{C} && \text{CODATA 2022, } \sigma = +0.28 \\ A_{\text{edm}} &= 8.4783536255(13) \times 10^{-30} \text{ m} \cdot \text{C} && \text{CODATA 2018, } \sigma = -4.10 \\ &&& \Delta_{\text{precision}} = +0.00 \\ &&& \Delta_{\text{scaled}} = -4.38 \end{aligned}$$

$$A_{\text{edm}} = \frac{1}{\mathfrak{K}_1} \left(\frac{l_p q_p m_p}{m_e} \right) \left(1 + \frac{L_2}{D_{Do}} \left(\frac{4\pi}{\textcolor{red}{5}!} \right) (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{\text{edm}} = \frac{1}{\mathfrak{K}_1} \left(\frac{l_p q_p m_p}{m_e} \right) \left(1 - \frac{18}{4} \left(\frac{V_{\text{fe}}}{4\pi} \right) (\mathfrak{K}_2^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{\text{edm}} = \frac{1}{\mathfrak{K}_1} \left(\frac{l_p q_p m_p}{m_e} \right) \left(1 - \frac{8}{\textcolor{blue}{7}} \left(\frac{8}{4\pi} \right) (\mathfrak{K}_2^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{\text{edm}} = \frac{1}{\mathfrak{K}_1} \left(\frac{l_p q_p m_p}{m_e} \right) \left(1 - \frac{1}{\textcolor{blue}{7}} \left(\frac{32}{2\pi} \right) (\mathfrak{K}_2^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{\text{edm}} = \frac{1}{\mathfrak{K}_1} \left(\frac{l_p q_p m_p}{m_e} \right) \left(1 - \frac{V_{\text{fe}}}{4} \left(\frac{18}{4\pi} \right) (\mathfrak{K}_2^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{\text{edm}} = \frac{1}{\mathfrak{K}_1} \left(\frac{l_p q_p m_p}{m_e} \right) \left(1 + 4 \left(\frac{V_{\text{fe}}}{35} \right) (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{\text{mag}} = \frac{1}{\mathfrak{K}_1^2} \left(\frac{l_p^2 q_p^2 m_p^2}{m_e^3} \right) \left(1 + L_{LL} \left(\frac{4\pi}{18} \right) (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, l_p = the Planck length, q_p = the Planck charge, m_p = the Planck mass, m_e = the electron mass, L_{LL} = the Laplace limit, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} A_{\text{mag}} &= 7.8910365\textcolor{green}{6936017} \dots \times 10^{-29} \text{ J/T}^2 && \text{prediction} \\ A_{\text{mag}} &= 7.8910365794(49) \times 10^{-29} \text{ J/T}^2 && \text{CODATA 2022, } \sigma = -2.05 \\ A_{\text{mag}} &= 7.8910366008(48) \times 10^{-29} \text{ J/T}^2 && \text{CODATA 2018, } \sigma = -6.55 \\ &&& \Delta_{\text{precision}} = -0.01 \\ &&& \Delta_{\text{scaled}} = -4.45 \end{aligned}$$

$$A_{\text{mag}} = \frac{1}{\mathfrak{K}_1^2} \left(\frac{l_p^2 q_p^2 m_p^2}{m_e^3} \right) \left(1 + P \left(\frac{2\pi}{18} \right) (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{\text{mag}} = \frac{1}{\mathfrak{K}_1^2} \left(\frac{l_p^2 q_p^2 m_p^2}{m_e^3} \right) \left(1 - \frac{4}{\textcolor{blue}{7}} \left(\frac{32}{4\pi} \right) (\mathfrak{K}_2^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{\text{mag}} = \frac{1}{\mathfrak{K}_1^2} \left(\frac{l_p^2 q_p^2 m_p^2}{m_e^3} \right) \left(1 + \frac{\omega_2}{2} \left(\frac{2\pi}{18} \right) (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{\text{mag}} = \frac{1}{\mathfrak{K}_1^2} \left(\frac{l_p^2 q_p^2 m_p^2}{m_e^3} \right) \left(1 + \frac{S}{2} \left(\frac{2\pi}{\textcolor{red}{5}} \right) (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{\text{mag}} = \frac{1}{\mathfrak{K}_1^2} \left(\frac{l_p^2 q_p^2 m_p^2}{m_e^3} \right) \left(1 + \frac{1}{\sqrt{35}} \left(\frac{2\pi}{x_\infty} \right) (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{\text{mag}} = \frac{1}{\mathfrak{K}_1^2} \left(\frac{l_p^2 q_p^2 m_p^2}{m_e^3} \right) \left(1 - G_{\text{Gi}} \left(\frac{18}{4\pi} \right) (\mathfrak{K}_2^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{\text{mag}} = \frac{1}{\mathfrak{K}_1^2} \left(\frac{l_p^2 q_p^2 m_p^2}{m_e^3} \right) \left(1 - \sqrt{32 e^{1/e}} (\mathfrak{K}_2^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{\text{eqm}} = \frac{1}{\kappa_1^3} \left(\frac{l_p^2 q_p m_p^2}{m_e^2} \right) \left(1 + \frac{2}{\sqrt{3}} \left(\frac{4\pi}{32} \right) (\kappa_2^2 - \kappa_3^2 - \kappa_4^2) \boxtimes \right)$$

Where κ_1 = the 1st hyperbolic partition constant, l_p = the Planck length, q_p = the Planck charge, m_p = the Planck mass, m_e = the electron mass, π = Archimedes' constant, κ_2 = the 2nd hyperbolic partition constant, κ_3 = the 3rd hyperbolic partition constant, κ_4 = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$A_{\text{eqm}} = 4.48655151342649 \dots \times 10^{-40} \text{ m} \cdot \text{C}$	prediction
$A_{\text{eqm}} = 4.4865515185(14) \times 10^{-40} \text{ m} \cdot \text{C}$	CODATA 2022, $\sigma = -3.62$
$A_{\text{eqm}} = 4.4865515246(14) \times 10^{-40} \text{ m} \cdot \text{C}$	CODATA 2018, $\sigma = -9.87$
	$\Delta_{\text{precision}} = +0.00$
	$\Delta_{\text{scaled}} = -4.36$

$$A_{\text{eqm}} = \frac{1}{\kappa_1^3} \left(\frac{l_p^2 q_p m_p^2}{m_e^2} \right) \left(1 + \frac{7}{18} \left(\frac{8K}{2\pi} \right) (\kappa_2^2 - \kappa_3^2 - \kappa_4^2) \boxtimes \right)$$

$$A_{\text{eqm}} = \frac{1}{\kappa_1^3} \left(\frac{l_p^2 q_p m_p^2}{m_e^2} \right) \left(1 + \frac{\gamma}{2} \left(\frac{4\pi}{8} \right) (\kappa_2^2 - \kappa_3^2 - \kappa_4^2) \boxtimes \right)$$

$$A_{\text{eqm}} = \frac{1}{\kappa_1^3} \left(\frac{l_p^2 q_p m_p^2}{m_e^2} \right) \left(1 - \frac{7}{18} \left(\frac{4K}{18} \right) (\kappa_2^2 + \kappa_3^2 + \kappa_4^2) \boxtimes \right)$$

$$A_{\text{eqm}} = \frac{1}{\kappa_1^3} \left(\frac{l_p^2 q_p m_p^2}{m_e^2} \right) \left(1 - \frac{12}{j_{0,1}} \left(\frac{4\pi}{!5} \right) (\kappa_2^2 + \kappa_3^2 + \kappa_4^2) \boxtimes \right)$$

$$A_{\text{eqm}} = \frac{1}{\kappa_1^3} \left(\frac{l_p^2 q_p m_p^2}{m_e^2} \right) \left(1 - \frac{2\pi}{\sqrt{3}} \left(\frac{4\pi}{32} \right) (\kappa_2^2 + \kappa_3^2 + \kappa_4^2) \boxtimes \right)$$

$$A_{1^{\text{st}} \text{ hp}} = \frac{1}{\mathfrak{K}_1^{11}} \left(\frac{t_p^4 q_p^3 m_p^3}{l_p m_e^5} \right) \left(1 - \frac{3}{\sqrt{2}} \left(\frac{4\pi}{8} \right) (\mathfrak{K}_2^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, t_p = the Planck time, q_p = the Planck charge, m_p = the Planck mass, l_p = the Planck length, m_e = the electron mass, π = Archimedes' constant, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} A_{1^{\text{st}} \text{ hp}} &= 3.20636129566261 \dots \times 10^{-53} \text{ m}^3 \cdot \text{C}^3 / \text{J}^2 && \text{prediction} \\ A_{1^{\text{st}} \text{ hp}} &= 3.2063612996(15) \times 10^{-53} \text{ m}^3 \cdot \text{C}^3 / \text{J}^2 && \text{CODATA 2022, } \sigma = -2.62 \\ A_{1^{\text{st}} \text{ hp}} &= 3.2063613061(15) \times 10^{-53} \text{ m}^3 \cdot \text{C}^3 / \text{J}^2 && \text{CODATA 2018, } \sigma = -6.96 \\ &&& \Delta_{\text{precision}} = +0.00 \\ &&& \Delta_{\text{scaled}} = -4.33 \end{aligned}$$

$$A_{1^{\text{st}} \text{ hp}} = \frac{1}{\mathfrak{K}_1^{11}} \left(\frac{t_p^4 q_p^3 m_p^3}{l_p m_e^5} \right) \left(1 + \pi \left(\frac{V_{\text{fe}}}{6} \right) (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{1^{\text{st}} \text{ hp}} = \frac{1}{\mathfrak{K}_1^{11}} \left(\frac{t_p^4 q_p^3 m_p^3}{l_p m_e^5} \right) \left(1 + \frac{D_d}{L_{LL}} \left(\frac{4\pi}{32} \right) (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{1^{\text{st}} \text{ hp}} = \frac{1}{\mathfrak{K}_1^{11}} \left(\frac{t_p^4 q_p^3 m_p^3}{l_p m_e^5} \right) \left(1 + \frac{2}{D_{Do}} \left(\frac{4\pi}{32} \right) (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{1^{\text{st}} \text{ hp}} = \frac{1}{\mathfrak{K}_1^{11}} \left(\frac{t_p^4 q_p^3 m_p^3}{l_p m_e^5} \right) \left(1 - \sqrt{\frac{7}{5}} \left(\frac{4\pi}{14} \right) (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{1^{\text{st}} \text{ hp}} = \frac{1}{\mathfrak{K}_1^{11}} \left(\frac{t_p^4 q_p^3 m_p^3}{l_p m_e^5} \right) \left(1 + \frac{2\pi}{\sqrt{35}} (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{1^{\text{st}} \text{ hp}} = \frac{1}{\mathfrak{K}_1^{11}} \left(\frac{t_p^4 q_p^3 m_p^3}{l_p m_e^5} \right) \left(1 - D_{Do} \left(\frac{18}{4\pi} \right) (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{2^{\text{nd}} \text{ hp}} = \frac{1}{\mathfrak{K}_1^{16}} \left(\frac{t_p^6 q_p^4 m_p^4}{l_p^2 m_e^7} \right) \left(1 - \frac{12}{D_{Do}} \left(\frac{4\pi}{!5} \right) (\mathfrak{K}_2^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes \right)$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant, t_p = the Planck time, q_p = the Planck charge, m_p = the Planck mass, l_p = the Planck length, m_e = the electron mass, D_{Do} = the Dottie number, π = Archimedes' constant, $!n$ = the derangement function, $\mathfrak{K}_2$ = the 2nd hyperbolic partition constant, $\mathfrak{K}_3$ = the 3rd hyperbolic partition constant, $\mathfrak{K}_4$ = the 4th hyperbolic partition constant, and $\boxtimes$ = the hyperbolic inversion boundary.

$$\begin{aligned} A_{2^{\text{nd}} \text{ hp}} &= 6.23537999\textcolor{teal}{8029131} \dots \times 10^{-65} \text{ m}^4 \cdot \text{C}^4 / \text{J}^3 && \text{prediction} \\ A_{2^{\text{nd}} \text{ hp}} &= 6.2353799735(39) \times 10^{-65} \text{ m}^4 \cdot \text{C}^4 / \text{J}^3 && \text{CODATA 2022, } \sigma = +1.74 \\ A_{2^{\text{nd}} \text{ hp}} &= 6.2353799905(38) \times 10^{-65} \text{ m}^4 \cdot \text{C}^4 / \text{J}^3 && \text{CODATA 2018, } \sigma = -2.69 \\ &&& \Delta_{\text{precision}} = -0.01 \\ &&& \Delta_{\text{scaled}} = -4.47 \end{aligned}$$

$$A_{2^{\text{nd}} \text{ hp}} = \frac{1}{\mathfrak{K}_1^{16}} \left(\frac{t_p^6 q_p^4 m_p^4}{l_p^2 m_e^7} \right) \left(1 - P \left(\frac{!5}{4\pi} \right) (\mathfrak{K}_2^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{2^{\text{nd}} \text{ hp}} = \frac{1}{\mathfrak{K}_1^{16}} \left(\frac{t_p^6 q_p^4 m_p^4}{l_p^2 m_e^7} \right) \left(1 - \varphi \left(\frac{18}{2\pi} \right) (\mathfrak{K}_2^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{2^{\text{nd}} \text{ hp}} = \frac{1}{\mathfrak{K}_1^{16}} \left(\frac{t_p^6 q_p^4 m_p^4}{l_p^2 m_e^7} \right) \left(1 + P \left(\frac{14}{4\pi} \right) (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{2^{\text{nd}} \text{ hp}} = \frac{1}{\mathfrak{K}_1^{16}} \left(\frac{t_p^6 q_p^4 m_p^4}{l_p^2 m_e^7} \right) \left(1 + L_{LL} \left(\frac{14}{2\pi} \right) (\mathfrak{K}_2^2 - \mathfrak{K}_3^2 - \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{2^{\text{nd}} \text{ hp}} = \frac{1}{\mathfrak{K}_1^{16}} \left(\frac{t_p^6 q_p^4 m_p^4}{l_p^2 m_e^7} \right) \left(1 - \frac{16}{j_{0,1}} \left(\frac{4\pi}{18} \right) (\mathfrak{K}_2^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes \right)$$

$$A_{2^{\text{nd}} \text{ hp}} = \frac{1}{\mathfrak{K}_1^{16}} \left(\frac{t_p^6 q_p^4 m_p^4}{l_p^2 m_e^7} \right) \left(1 - \frac{18}{8} \left(\frac{2\pi}{s} \right)^4 (\mathfrak{K}_2^2 + \mathfrak{K}_3^2 + \mathfrak{K}_4^2) \boxtimes \right)$$

proton rms charge radius

8.7541(46)

$$\frac{r_n}{r_+} = K = \sum_{n=1}^{\infty} \frac{\sin\left(n \frac{\pi}{2}\right)}{n^2}$$

Where K = Catalan’s constant, $\sin(x)$ = the sine function, and π = Archimedes’ constant.

$r_+ = 8.43431614404474 \dots \times 10^{-16} \text{ m}$	prediction
$r_+ = 8.4075(64) \times 10^{-16} \text{ m}$	CODATA 2022, $\sigma = +4.18$
$r_+ = 8.414(19) \times 10^{-16} \text{ m}$	CODATA 2018, $\sigma = +1.05$
	$\Delta_{\text{precision}} = 0.47$
	$\Delta_{\text{scaled}} = 0.36$

neutron radius

$$\frac{r_n}{r_e} = \left(\frac{4\pi}{\Gamma(5)}\right)^2 = \sum_{n=1}^{\infty} \frac{\cos\left(n \frac{\pi}{3}\right)}{n^2}$$

Where r_e = the classical electron radius, π = Archimedes’ constant, $\Gamma(x)$ = the gamma function, and $\cos(x)$ = the cosine function.

$r_n = 7.72554339835845 \dots \times 10^{-16} \text{ m}$	prediction
$r_n = 8 \times 10^{-16} \text{ m}$	Povh, B.: Rith, K.: Scholz, C.: Zetsche, F. (2002)

deuteron rms charge radius

$$\frac{r_e}{r_{\text{de}}} = \sqrt{\frac{2\pi}{\log(36)}}$$

Where r_e = the classical electron radius, r_{de} = the deuteron rms charge radius, π = Archimedes' constant, and $\log(x)$ = the natural logarithm.

$r_{\text{de}} = 2.12812292371170 \dots \times 10^{-15} \text{ m}$	prediction
$r_{\text{de}} = 2.12778(27) \times 10^{-15} \text{ m}$	CODATA 2022, $\sigma = +1.26$
$r_{\text{de}} = 2.12799(74) \times 10^{-15} \text{ m}$	CODATA 2018, $\sigma = +0.18$
	$\Delta_{\text{precision}} = 0.44$
	$\Delta_{\text{scaled}} = 0.28$

alpha particle rms charge radius

$$\frac{r_n}{r_\alpha} = \sqrt{\frac{C_U}{4}} \quad C_U = \sqrt{2} \left(\frac{2\pi}{2s} \right)$$

Where r_n = the neutron radius, r_α = the alpha particle rms charge radius, and C_U = the ubiquitous constant, π = Archimedes' constant, and s = the arc length of the unit lemniscate.

$r_\alpha = 1.67865985670315 \dots \times 10^{-15} \text{ m}$	prediction
$r_\alpha = 1.6785(21) \times 10^{-15} \text{ m}$	CODATA 2022, $\sigma = +0.05$
$r_\alpha = 1.6785(21) \times 10^{-15} \text{ m}$	CODATA 2018, $\sigma = +0.05$
	$\Delta_{\text{precision}} = 0.00$
	$\Delta_{\text{scaled}} = 0.00$

fine-structure constant

$$\alpha = \mathfrak{K}_1^2$$

Where $\mathfrak{K}_1$ = the 1st hyperbolic partition constant.

$$\alpha = 7.29735257295520 \dots \times 10^{-3} \quad \text{prediction}$$
$$\alpha = 7.2973525684(14) \times 10^{-3} \quad \text{average 11-digit measure, } \sigma = +3.21$$

$\alpha = 7.29735256491(80) \times 10^{-3}$	Fan et al. 2023, $\sigma = +10.05$
$\alpha = 7.2973525643(11) \times 10^{-3}$	CODATA 2022, $\sigma = +7.82$
$\alpha = 7.2973525628(06) \times 10^{-3}$	Morel et al. 2020, $\sigma = +16.83$
$\alpha = 7.2973525693(11) \times 10^{-3}$	CODATA 2018, $\sigma = +3.27$
$\alpha = 7.2973525713(14) \times 10^{-3}$	Parker, Yu, et al. 2018, $\sigma = +1.14$
$\alpha = 7.2973525657(18) \times 10^{-3}$	Aoyama et al. 2017, $\sigma = +4.00$
$\alpha = 7.2973525664(17) \times 10^{-3}$	CODATA 2014, $\sigma = +3.82$
$\alpha = 7.2973525717(48) \times 10^{-3}$	Bouchendira 2010, $\sigma = +0.25$
$\alpha = 7.2973525698(24) \times 10^{-3}$	CODATA 2010, $\sigma = +1.29$
$\alpha = 7.2973525692(27) \times 10^{-3}$	Gabrielse, Hanneke 2008, $\sigma = +1.37$
$\alpha = 7.2973525700(52) \times 10^{-3}$	Gabrielse 2007, $\sigma = +0.56$
$\alpha = 7.2973525376(50) \times 10^{-3}$	CODATA 2006, $\sigma = +7.06$
$\alpha = 7.297352568(24) \times 10^{-3}$	CODATA 2002, $\sigma = +2.04$
$\alpha = 7.297352533(27) \times 10^{-3}$	CODATA 1998, $\sigma = +1.44$
$\alpha = 7.297352582(27) \times 10^{-3}$	Kinoshita 1998, $\sigma = -0.37$
$\alpha = 7.29735308(33) \times 10^{-3}$	CODATA 1986, $\sigma = -1.55$
$\alpha = 7.2973461(81) \times 10^{-3}$	CODATA 1973, $\sigma = +0.79$
$\alpha = 7.297351(11) \times 10^{-3}$	CODATA 1969, $\sigma = +0.09$

* Note Fan et al. 2023 is not an experimentally measured value. It is a value determined by *current theory* based on an experimentally determined value of the *electron magnetic moment*.

** CODATA values are computed by averaging other measurements; they are not independent experiments.

electron charge

$$e = \varkappa_1 q_p$$

Where $\varkappa_1$ = the 1st hyperbolic partition constant, and q_p = the Planck charge.

$e = 1.60217657405973 \dots \times 10^{-19} \text{ C}$	prediction
$e = 1.602176634 \times 10^{-19} \text{ C}$	CODATA 2022, 8-digit match
$e = 1.602176634 \times 10^{-19} \text{ C}$	CODATA 2018, 8-digit match
$e = 1.602176565(35) \times 10^{-19} \text{ C}$	CODATA 2010, $\sigma = +0.26$
	$\Delta_{\text{precision}} = 0.00$
	$\Delta_{\text{scaled}} = 0.00$

Also known as the *elementary charge*, and the *atomic unit of charge*.

W to Z mass ratio

$$\frac{m_W}{m_Z} = \log(1 + \sqrt{2})$$

Where m_W = the W boson mass, m_Z = the Z boson mass, and $\log(x)$ = the natural logarithm.

$m_W/m_Z = 0.881373587019543 \dots$	prediction
$m_W/m_Z = 0.88145(13)$	CODATA 2022, $\sigma = -0.62$
$m_W/m_Z = 0.88153(17)$	CODATA 2018, $\sigma = -0.94$
	$\Delta_{\text{precision}} = 0.12$
	$\Delta_{\text{scaled}} = 0.47$

$$\log(1 + \sqrt{2}) = P_{up} - \sqrt{2}$$

$$\frac{m_W}{m_Z} + \sqrt{2} = P_{up}$$

weak mixing angle

$$\frac{m_W}{m_Z} = \cos(\theta_W) \qquad 1 - \left(\frac{m_W}{m_Z}\right)^2 = \sin^2(\theta_W)$$

Where $\cos(x)$ = the cosine function, $\sin(x)$ = the sine function, m_W = the mass of the W boson, and m_Z = the mass of the Z boson.

$\sin^2(\theta_W) = 0.223180600010430 \dots$	prediction
$\sin^2(\theta_W) = 0.22305(23)$	CODATA 2022, $\sigma = +0.57$
$\sin^2(\theta_W) = 0.22290(30)$	CODATA 2018, $\sigma = +0.93$
	$\Delta_{\text{precision}} = 0.12$
	$\Delta_{\text{scaled}} = 0.50$